

Analysis Log

Geometry set	Included	Excluded	Main purpose
G0_BARE_AIRFRAME	Fuselage, wing, HT, VT	Nacelles, props, struts, gear	Basic airframe aerodynamics
G1_NACELLE_AIRFRAME	G0 + both nacelles/spinners	Props, struts, gear	Measure nacelle drag penalty
G2_COMPLETE_AIRFRAME	G1 + struts + fixed landing gear	Propellers	Actual clean unpowered aircraft
G3_POWERED_AIRCRAFT	G2 + both propellers/disks	—	Powered-wing and asymmetric-propulsion analysis

Mode	Geometry set	Slat	Fowler	Prop state	Purpose
M_CRZ	G2	0°	0°	excluded	Clean cruise/stability
M_TOU	G2	15°	12°	excluded/off	Unpowered take-off high lift
M_TOP	G3	15°	12°	symmetric powered disks	Powered take-off
M_LDG	G2	25°	35°	no lift credit	Landing
M_VMC	G3	15°	12°	one live, one failed/feathered	Critical OEI control

01 CompGeom

G0

G0 — Bare Airframe	Wetted area (m²)	Status
Fuselage	44.529	PASS
Main wing	59.599	PASS — merged symmetric mesh
Horizontal tail	16.074	PASS — merged symmetric mesh
Vertical tail	7.917	PASS
Total	128.118	PASS
Triangle count	8,288	Baseline

G1

Parameter	G0 — Bare Airframe	G1 — Nacelle Airframe	Change G1–G0	Assessment
Number of components	4	6	+2	PASS — two nacelles added
Number of meshes	6	8	+2	PASS
Triangle count	8,288	9,824	+1,536 (+18.5%)	Reasonable
Fuselage wetted area	44.529 m²	44.529 m²	0.000	PASS
Main-wing wetted area	59.599 m²	58.530 m²	–1.069 m²	Expected nacelle/wing trimming
Horizontal-tail wetted area	16.074 m²	16.074 m²	0.000	PASS
Vertical-tail wetted area	7.917 m²	7.917 m²	0.000	PASS
Nacelle R wetted area	—	3.718 m²	+3.718	PASS
Nacelle L wetted area	—	3.718 m²	+3.718	PASS — exact L/R symmetry
Total nacelle exposed area	—	7.436 m²	+7.436	PASS
Total aircraft wetted area	128.118 m²	134.486 m²	+6.368 m² (+4.97%)	PASS
Total theoretical area	137.128 m²	145.636 m²	+8.508 m²	Plausible
Total wetted volume	24.362 m³	25.492 m³	+1.130 m³	Plausible
Mesh warning	MainWing + HTail merged	MainWing + HTail merged	No new warning	PASS
Overall CompGeom status	PASS	PASS	—	Proceed to G2

G1 CompGeom validation: Addition of the two packaged nacelles increased total exposed wetted area from 128.118 m² to 134.486 m² (+4.97%). Each nacelle contributes 3.718 m² exposed wetted area. Main-wing wetted area decreases by 1.069 m² due to nacelle/wing intersection trimming,

giving a net installed nacelle increment of 6.368 m². Left/right nacelle areas and volumes are identical. No new mesh warnings were introduced relative to G0; only the existing symmetric MainWing and HTail mesh merges remain. G1 accepted for subsequent drag analysis.

(Current – strut leaving no space)

Metric	G0 — Bare Airframe	G1 — Nacelle Airframe	G2 — Complete Airframe	Assessment
Geoms	4	6	11	Correct progression
Meshes	6	8	16	PASS
Triangles	8,288	9,824	13,888	Appropriate growth
Fuselage (Swet)	44.529	44.529	44.483	G2 attachment trimming
Wing (Swet)	59.599	58.530	58.488	Nacelle + strut trimming
HT (Swet)	16.074	16.074	16.074	Stable
VT (Swet)	7.917	7.917	7.917	Stable
Nacelles	—	7.436	7.436	Stable
Lift struts	—	—	4.260	Installed exposed area
Main gear legs	—	—	0.972	Installed exposed area
Main wheels	—	—	1.362	Installed exposed area
Nose gear leg	—	—	0.137	Installed exposed area
Nose wheel	—	—	0.460	Installed exposed area
Total (Swet)	128.118	134.486	141.591	PASS
MainWing merge warning	Yes	Yes	Yes	Accepted
HTail merge warning	Yes	Yes	Yes	Accepted
Other warnings	None	None	None	PASS
Overall status	PASS	PASS	PASS	CompGeom gate closed

(Previous – strut leaving some space)

Parameter	G0 Bare Airframe	G1 + Nacelles	G2 Complete Airframe	G2–G1	Assessment
OpenVSP Geoms	4	6	11	+5	Correct
Meshes	6	8	16	+8	Reasonable with symmetric images
Triangles	8,288	9,824	13,376	+3,552	Reasonable
Fuselage (Swet)	44.529	44.529	44.505	–0.024	Expected small intersection trimming
Main wing (Swet)	59.599	58.530	58.530	0.000	See strut-junction check below
HT (Swet)	16.074	16.074	16.074	0.000	PASS
VT (Swet)	7.917	7.917	7.917	0.000	PASS

Parameter	G0 Bare Airframe	G1 + Nacelles	G2 Complete Airframe	G2–G1	Assessment
Nacelle R	—	3.718	3.718	0.000	PASS
Nacelle L	—	3.718	3.718	0.000	PASS
Lift struts, total	—	—	4.140	+4.140	CHECK attachment
Main gear legs, total	—	—	0.972	+0.972	PASS
Main wheels, total	—	—	1.362	+1.362	PASS
Nose-gear leg	—	—	0.137	+0.137	PASS
Nose wheel	—	—	0.460	+0.460	PASS
Total (Swet)	128.118	134.486	141.533	+7.047	PASS/CHECK
Warning	Wing + HT merged	Wing + HT merged	Wing + HT merged	No new warning	PASS

G2 CompGeom validation: The complete unpowered G1.3 aircraft comprises 11 OpenVSP Geoms and produces 141.533 m² exposed wetted area, compared with 134.486 m² for the nacelle-installed G1 configuration and 128.118 m² for the bare G0 airframe. Addition of the lift struts and fixed tricycle landing gear increases exposed wetted area by 7.047 m² (+5.24% relative to G1). Explicit added-component area totals 7.071 m²; 0.024 m² is removed through installation/intersection trimming, exactly reproducing the configuration-level increment. No new CompGeom mesh warnings are generated; the existing MainWing and HTail symmetric-mesh merges persist. G2 is accepted subject to visual confirmation of the lift-strut attachment junctions.

G2_COMPLETE_AIRFRAME — PASS. The complete unpowered Tyche R2-T1 G1.3 airframe comprises 11 OpenVSP Geoms, 16 CompGeom meshes and 13,888 triangles. Total exposed wetted area is 141.591 m². Relative to G1, installation of the permanent lift-strut and fixed-gear system increases aircraft wetted area by 7.105 m² (+5.28%). Each lift strut has 2.223 m² theoretical and 2.130 m² exposed wetted area; concurrent reductions of 0.042 m² in main-wing wetted area and 0.046 m² in fuselage wetted area confirm successful intersection and trimming at the installed attachments. No additional mesh warnings occur beyond the previously accepted MainWing and HTail symmetric-mesh merges. **G2 accepted as the complete clean-aircraft geometry for subsequent parasite-drag and VSPAERO analysis.**

Final 01 CompGeom status

Set	Purpose	Final wetted area	Status
G0_BARE_AIRFRAME	Wing-body-tail baseline	128.118 m²	PASS
G1_NACELLE_AIRFRAME	G0 + installed nacelles	134.486 m²	PASS
G2_COMPLETE_AIRFRAME	G1 + struts + fixed gear	141.591 m²	PASS
G3_POWERED_AIRCRAFT	Adds actuator-disk props	Not required for wetted-area CompGeom	N/A

02 ParasiteDrag

G0 Parasite Drag analysis log

Parameter	Fuselage	Main Wing	H-Tail	V-Tail	Total
(Swet), m ²	44.5289	59.5985	16.0743	7.9167	128.1184
(Lref), m	10.8061	1.8805	1.4028	1.5080	—
(t/c) or body ratio	6.0092	0.1701	0.1199	0.1199	—
Reynolds number	(4.011*10 ⁷)	(6.980*10 ⁶)	(5.207*10 ⁶)	(5.597*10 ⁶)	—
Laminar fraction	0%	0%	0%	0%	—
(C _f)	0.002422	0.003121	0.003267	0.003230	—
Form factor, (FF)	1.1341	1.3903	1.2521	1.2521	—
Interference (Q)	1.000	1.000	1.000	1.000	—
Flat-plate area (f), m ²	0.12233	0.25860	0.06576	0.03202	0.47871
(CD0)	0.004078	0.008620	0.002192	0.001067	0.015957
Share of G0 drag	25.55%	54.02%	13.74%	6.69%	100%
Status	PASS	PASS	PASS	PASS	PASS

G0_BARE_AIRFRAME Parasite Drag — PASS. At the controlled 130 KTAS, 2,600 m standard-atmosphere cruise condition and Sref=30.00 m², the fully turbulent OpenVSP parasite-drag build-up gives f=0.47871 m² and CD0=0.015957. Parasite Drag wetted areas reproduce the accepted G0 CompGeom total of approximately 128.118 m². The main wing contributes 54.02% of G0 parasite drag, followed by the fuselage at 25.55%, horizontal tail at 13.74% and vertical tail at 6.69%. Reynolds numbers, form factors and component drag arithmetic are internally consistent. G0 is accepted as the bare-airframe baseline for configuration-delta analysis.

G1 Parasite Drag analysis log

Parameter	Fuselage	Main Wing	H-Tail	V-Tail	Nacelle R	Nacelle L	Total
(Swet), m ²	44.529	58.530	16.074	7.917	3.718	3.718	134.486
(Lref), m	10.806	1.881	1.403	1.508	2.200	2.200	—
(t/c) or fineness ratio	6.009	0.170	0.120	0.120	3.142	3.142	—
Reynolds number	(4.011*10 ⁷)	(6.980*10 ⁶)	(5.207*10 ⁶)	(5.597*10 ⁶)	(8.165*10 ⁶)	(8.165*10 ⁶)	—
Laminar fraction	0%	0%	0%	0%	0%	0%	—
(C _f)	0.002422	0.003121	0.003267	0.003230	0.003047	0.003047	—
Form factor (FF)	1.134	1.390	1.252	1.252	1.495	1.495	—
(Q)	1.00	1.00	1.00	1.00	1.00	1.00	—

Parameter	Fuselage	Main Wing	H-Tail	V-Tail	Nacelle R	Nacelle L	Total
Flat-plate area (f), m²	0.122329	0.253967	0.065758	0.032019	0.016934	0.016934	0.507942
(C_D0)	0.004078	0.008466	0.002192	0.001067	0.000564	0.000564	0.016931
% of G1 total	24.08%	50.00%	12.95%	6.30%	3.33%	3.33%	100%
Status	PASS	PASS	PASS	PASS	PASS	PASS	PASS

G0 → G1 comparison

Metric	G0 Bare	G1 + Nacelles	Change
Total (S_wet)	128.118 m²	134.486 m²	+6.368 m²
Flat-plate area (f)	0.478708 m²	0.507942 m²	+0.029234 m²
(C_D0)	0.015957	0.016931	+0.000974
(C_D0) increase	—	—	+6.10%
Main-wing (C_D0)	0.008620	0.008466	−0.000154
Two nacelle (C_D0)	—	0.001128	+0.001128
Overall status	PASS	PASS	—

G1_NACELLE_AIRFRAME Parasite Drag — PASS. At 130 KTAS and 2,600 m standard-atmosphere cruise, addition of the two packaged nacelles raises the equivalent flat-plate area from 0.47871 to 0.50794 m² and CD0 from 0.015957 to 0.016931. The net installed nacelle penalty is therefore ΔCD0=0.000974, equivalent to a 6.10% increase relative to G0. Each nacelle contributes CD0=0.000564, while the main-wing contribution decreases by 0.000154 due to nacelle/wing trimming. Left/right nacelle results are identical, confirming model symmetry. G1 accepted for progression to the complete-airframe G2 build-up.

G2 Parasite Drag analysis log

Component	(S_wet) m²	(L_ref) m	Re	(C_f)	FF	(f) m²	(C_D0)	% total	Assessment
Fuselage	44.483	10.806	(4.01*10^7)	0.002422	1.134	0.122204	0.004073	21.46%	PASS
Main wing	58.488	1.881	(6.98*10^6)	0.003121	1.390	0.253784	0.008459	44.56%	PASS
H-tail	16.074	1.403	(5.21*10^6)	0.003267	1.252	0.065758	0.002192	11.55%	PASS
V-tail	7.917	1.508	(5.60*10^6)	0.003230	1.252	0.032019	0.001067	5.62%	PASS
Nacelle R	3.718	2.200	(8.17*10^6)	0.003047	1.495	0.016934	0.000564	2.97%	PASS
Nacelle L	3.718	2.200	(8.17*10^6)	0.003047	1.495	0.016934	0.000564	2.97%	PASS
Lift struts, pair	4.261	0.285	(1.06*10^6)	0.004278	1.367	0.024921	0.000831	4.38%	PASS
Main gear legs, pair	0.973	0.211	(7.84*10^5)	0.004518	1.622	0.007129	0.000238	1.25%	PASS/REVIEW

Component	(S_wet) m²	(L_ref) m	Re	(C_f)	FF	(f) m²	(C_D0)	% total	Assessment
Main wheels, pair	1.363	0.600	(2.23*10^6)	0.003754	4.228	0.021629	0.000721	3.80%	REVIEW
Nose gear leg	0.137	0.201	(7.45*10^5)	0.004562	1.576	0.000983	0.000033	0.17%	PASS/REVIEW
Nose wheel	0.460	0.500	(1.86*10^6)	0.003873	4.078	0.007267	0.000242	1.28%	REVIEW
TOTAL	141.591	—	—	—	—	0.569563	0.018985	100%	PASS as baseline

G0 → G1 → G2 final build-up table

Metric	G0 Bare	G1 + Nacelles	G2 Complete	Increment
(S_wet), m²	128.118	134.486	141.591	+13.473 vs G0
Flat-plate area (f), m²	0.478708	0.507942	0.569563	+0.090855
(C_D0)	0.015957	0.016931	0.018985	+0.003028 vs G0
Increase from previous set	—	+6.10%	+12.13%	—
Increase from G0	—	+6.10%	+18.98%	—
Status	PASS	PASS	PASS baseline	—

G2_COMPLETE_AIRFRAME Parasite Drag — PASS as geometry-based baseline. At 130 KTAS, 2,600 m and Sref=30.0 m², the complete G1.3 airframe produces an OpenVSP equivalent flat-plate area of 0.56956 m² and CD0=0.018985. Addition of the lift struts and fixed tricycle gear increases CD0 by 0.002054 (+12.13%) relative to G1. The main wing remains the dominant parasite-drag source at 44.56%, followed by the fuselage at 21.46%. The corrected strut installation shows consistent geometric trimming and drag accounting. The raw G2 result remains 32.9% below the report value of 0.0283; this discrepancy is retained for further investigation through explicit excrescence, cooling/interference, and bluff-body landing-gear sensitivity rather than tuning the OpenVSP model to the report.

G2 Parasite Drag analysis log (with excrescence)
Final G2 parasite-drag sensitivity table

Case	Purpose	Operational / cooling allowance	Gear treatment	Total (f) (m²)	Total (C_D0)	Increase vs S0	Difference from 0.0283
S0	Raw geometry baseline	0%	1.0×	0.569563	0.018985	—	−32.92%
S1	Low operational sensitivity	+5% of (C_D,geom)	1.0×	0.598041	0.019935	+5.00%	−29.56%
S2	Moderate operational sensitivity	+10% of (C_D,geom)	1.0×	0.626519	0.020884	+10.00%	−26.20%
S3	High operational sensitivity	+15% of (C_D,geom)	1.0×	0.654998	0.021833	+15.00%	−22.85%
G1	Gear-model sensitivity	0%	1.5× effective gear	0.588073	0.019602	+3.25%	−30.73%

Case	Purpose	Operational / cooling allowance	Gear treatment	Total (f) (m²)	Total (C_D0)	Increase vs S0	Difference from 0.0283
			drag				
G2	Strong gear sensitivity	0%	2.0× effective gear drag	0.606583	0.020219	+6.50%	−28.55%
C1	Conservative working case	+10%	1.5× gear	0.645029	0.021501	+13.25%	−24.02%
C2	Upper-bound sensitivity	+15%	2.0× gear	0.692018	0.023067	+21.50%	−18.49%
Previous report	Legacy analytical estimate	—	—	0.849000 equivalent	0.028300	+49.1% vs S0	—

Parasite-drag validation conclusion. The frozen G1.3 complete-aircraft geometry gives $CD_0=0.018985$ at 130 KTAS and 2,600 m. Explicit operational/cooling and fixed-gear sensitivities increase the result to 0.021501 for the conservative working case and 0.023067 for the deliberately adverse upper-bound case. The previous analytical value of 0.0283 is 49.1% above the geometry baseline and remains 18.5% above the upper sensitivity. No identified physical drag source supports imposing the additional penalty required to recover the previous value; 0.0283 is therefore retained as a legacy analytical comparison rather than used to tune the validation model.

03 VSPAERO Convergence

Convergence table – 16 wake iterations

Quantity	Coarse	Medium	Fine	Coarse→Medium	Medium→Fine	Criterion	Status
Span/load records	62	94	144	+51.6%	+53.2%	increasing	PASS
Total sweep runtime	36.2 s	108.7 s	322.6 s	×3.00	×2.97	increasing	PASS
CL(4°)	0.78961	0.80591	0.82051	2.02%	1.78%	≤1%	FAIL
(C_D_i(4°)	0.02409	0.02522	0.02723	4.49%	7.38%	≤2%	FAIL
(C_My(4°)	-0.01388	-0.00429	-0.01276	(	\Delta	=0.00959)	**(
(C_L_alpha, rad ⁻¹	5.6894	5.6327	5.6218	1.01%	0.19%	≤2%	PASS
(C_m_alpha, rad ⁻¹	-1.4693	-1.4637	-1.5796	0.38%	7.34%	≤2%	FAIL
Fitted induced-drag factor (k)	0.03834	0.03921	0.04009	2.22%	2.19%	≈≤2%	Borderline/ FAIL
Fitted span efficiency (e)	0.9261	0.9055	0.8857	2.28%	2.24%	≈≤2%	Borderline/ FAIL

Medium MainWing W = 33 MainWing U = 5, 17, 3, 9 HTail W = 33 HTail U = 13 VTail W = 33 VTail U = 13	Fine MainWing: W = 49 U = 7, 25, 5, 13 HTail: W = 49 U = 19 VTail: W = 49 U = 19	Extra Fine MainWing: W = 73 U = 9, 37, 7, 19 HTail: W = 73 U = 29 VTail: W = 73 U = 29	Coarse MainWing W = 33 MainWing U = 3, 11, 3, 7 HTail W = 21 HTail U = 9 VTail W = 21 VTail U = 9
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Alpha Start = -2.000 Alpha End = 8.000 Npts = 6	Beta Start = 0 Beta End = 0 Npts = 1 Mach Start = 0.203 Mach End = 0.203 Npts = 1	Manual reference = ON Sref = 30.000 m ² bref = 16.400 m cref = 1.881 m	Xref = 3.750 m Yref = 0.000 m Zref = 0.000 m	Slice Direction = X Num Slices = 10
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	ReCref Start = 6.98e6 ReCref End = 6.98e6 Npts = 1			
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Updated VSPAERO convergence table — 20 wake iterations

Quantity	Coarse	Medium	Fine	Medium → Fine	Criterion	Status
Load-distribution records	62	94	144	+53.2%	Increasing	PASS
Total sweep runtime	86.4 s	201.7 s	658.1 s	×3.26	Increasing	PASS
(C_L(4°))	0.81303	0.80949	0.81983	1.262%	≤1%	Marginal FAIL
(C_D_i(4°))	0.025029	0.025495	0.027206	6.288%	≤2%	FAIL *
(C_My(4°))	−0.01908	−0.01438	−0.01163	(	\Delta	=0.00275)
(C_L_alpha, rad ^{−1})	5.6864	5.6398	5.6254	0.256%	≤2%	PASS
Fitted induced-drag factor (k)	0.03850	0.03921	0.04000	1.994%	≤2%	PASS, just
Fitted (e=1/(\pi ARk))	0.9221	0.9056	0.8875	2.034%	≈≤2%	Borderline
Full-sweep (C_m_alpha, rad ^{−1})	−1.5227	−1.4749	−1.6001	7.83%	≤2%	FAIL

Extra Fine

Parameter	Fine	Extra-Fine
MainWing Section 1 SectTess_U	7	9
MainWing Section 2	25	37
MainWing Section 3	5	7
MainWing Section 4	13	19
MainWing Num_W	49	73
HTail SectTess_U	19	29
HTail Num_W	49	73
VTail SectTess_U	19	29
VTail Num_W	49	73
LE clustering	0.25	0.25
TE clustering	0.25	0.25
Cap Tess	3	3

Fine → Extra-Fine convergence

Quantity	Coarse	Medium	Fine	Extra-Fine	Fine → Extra-Fine	Target	Status
Span/load records	62	94	144	218	+51.4%	clear refinement	PASS
(C_L(4°))	0.81303	0.80949	0.81983	0.82316	0.405%	≤1%	PASS
(C_L_alpha), rad ⁻¹	5.6951	5.6522	5.6121	5.5671	0.809%	≤2%	PASS
(C_D_i(4°))	0.02503	0.02550	0.02721	0.02976	8.57%	≤2%	FAIL
fitted (k)	0.03680	0.03799	0.03907	0.04247	8.00%	≤2%	FAIL
fitted (e)	0.9649	0.9346	0.9087	0.8360	8.70%	≈≤2%	FAIL
VSPAERO (E) at (4°)	0.9377	0.9125	0.8771	0.8085	8.50%	≈≤2%	FAIL
(C_My (4°))	-0.01908	-0.01438	-0.01163	-0.00191	(	\Delta	=0.00972)
low-angle (C_m_alpha), rad ⁻¹	-1.3952	-1.3206	-1.3443	-1.3225	1.65%	≤2%	PASS

VSPAERO spatial-convergence assessment: A four-level VLM refinement from 62 to 218 span/load stations demonstrates convergence of the lifting solution. Fine-to-Extra-Fine differences are 0.405% in CL(4°) and 0.809% in CL α , satisfying the adopted 1%/2% criteria. Low-angle C $m\alpha$ also differs by only 1.65%. Induced drag does not exhibit equivalent convergence: CDi(4°) increases by 8.57%, and the fitted induced-drag factor increases from 0.03907 to 0.04247 (+8.00%), reducing fitted span efficiency from 0.909 to 0.836. Because surface refinement was conducted with wake node count fixed at eight, induced-drag validation remains open pending a targeted wake-discretization sensitivity. No further surface refinement is justified until this numerical dependency is isolated.

Extra-Fine: 8 vs 16 wake nodes

Quantity	Extra-Fine, 8 nodes	Extra-Fine, 16 nodes	Change	Assessment
(C_L(-2°))	0.240109	0.239679	-0.179%	PASS
(C_L(0°))	0.435436	0.434043	-0.320%	PASS
(C_L(2°))	0.629543	0.627535	-0.319%	PASS
(C_L(4°))	0.823162	0.820990	-0.264%	PASS
(C_D_i(-2°))	0.003488	0.003585	+2.77%	Sensitive at low lift
(C_D_i(0°))	0.008797	0.008912	+1.31%	PASS
(C_D_i(2°))	0.017555	0.017704	+0.85%	PASS
(C_D_i(4°))	0.029758	0.030046	+0.97%	PASS
(E(4°))	0.81059	0.79858	-1.48%	PASS / noticeable
(C_My(4°))	-0.001913	-0.004905	-0.00299 abs.	PASS by absolute criterion

Derived quantity	8 wake nodes	16 wake nodes	Change	Status
(C_L_alpha), rad ⁻¹)	5.5671	5.5503	-0.30%	PASS
(C_m_alpha), rad ⁻¹)	-1.3225	-1.3452	+1.72% magnitude	PASS
Fitted (k)	0.04247	0.04301	+1.27%	PASS
Fitted (e)	0.83598	0.82549	-1.26%	PASS

The 16-node solution itself is also wake-iteration converged

(alpha)	(ΔC_L) iter. 19→20	(ΔC_D_i)	(ΔC_My) abs.
-2°	+0.015%	+0.039%	-0.000126
0°	+0.011%	-0.092%	+0.000013
2°	-0.039%	-0.192%	+0.000226
4°	+0.095%	+0.066%	-0.000240

16 wake nodes should replace 8 for the production VSPAERO setup. Lift is spatially converged by the Fine mesh, but induced drag is still not formally surface-mesh converged between Fine and Extra-Fine. Extra-Fine/16 is the best-resolved solution we currently have.

Final 2×2 numerical sensitivity comparison

Quantity	Fine / 8	Fine / 16	Extra-Fine / 8	Extra-Fine / 16
(C_L(4°))	0.819832	0.814470	0.823162	0.820990
(C_D_i(4°))	0.027206	0.027961	0.029758	0.030046
(C_My(4°))	-0.011634	-0.003911	-0.001913	-0.004905
(C_L_alpha), rad ⁻¹)	5.6121	5.5667	5.5671	5.5503
low-angle (C_m_alpha), rad ⁻¹)	-1.3443	-1.2689	-1.3225	-1.3452
fitted (k)	0.039071	0.040751	0.042470	0.043010
fitted (e)	0.9087	0.8713	0.8360	0.8255
VSPAERO (E) at 4°	0.8771	0.8423	0.8085	0.7965

Wake-node sensitivity: Fine versus Extra-Fine

Metric	8 nodes	16 nodes	Change
(C_L(4°))	0.819832	0.814470	0.66%
(C_D_i(4°))	0.027206	0.027961	2.70%
(C_L_alpha)	5.6121	5.5667	0.82%
(k)	0.039071	0.040751	4.12%
fitted (e)	0.9087	0.8713	4.30%

Final convergence table for 03_VSPAERO_Convergence

Verification metric	Fine / 16	Extra-Fine / 16	Difference	Criterion	Disposition
(C_L(4°))	0.81447	0.82099	0.79%	$\leq 1\%$	PASS
(C_L_alpha)	5.5667	5.5503	0.30%	$\leq 2\%$	PASS
(C_D_i(4°))	0.027961	0.030046	6.94%	$\leq 2\%$	OPEN
fitted (k)	0.040751	0.043010	5.25%	$\leq 2\%$	OPEN
fitted (e)	0.8713	0.8255	5.54%	$\approx \leq 2\%$	OPEN
(C_My(4°))	-0.003911	-0.004905	0.000993 abs.	≤ 0.005	PASS
low-angle (C_m_alpha)	-1.2689	-1.3452	5.67%	$\leq 2\%$	OPEN
Wake-node sensitivity in (k)	4.12%	1.26%	—	$\leq 2\%$	XF PASS
Iteration convergence	PASS	PASS	—	stable	PASS

VSPAERO numerical convergence: Wake iteration and wake-node sensitivity were resolved at 20 iterations and 16 wake nodes. Fine-to-Extra-Fine refinement produced differences of 0.79% in $CL(4^\circ)$ and 0.30% in CL_α , satisfying the adopted lift-convergence limits. The local pitching-moment coefficient also differed by only 0.0010. Induced drag remained more mesh-sensitive, with a 5.25% Fine-to-Extra-Fine difference in fitted k. Accordingly, the Extra-Fine/16 solution is retained as the best-resolved aerodynamic solution and the residual 5–6% induced-drag sensitivity is propagated as a numerical uncertainty rather than forcing artificial convergence. The final controlled solver settings are 20 wake iterations and 16 wake nodes.

04 Clean Aero

00_G0_Clean_Lifting

1. Extracted CA-G0 polar

(α)	(C _L)	(C _{D_i})	(C _{My})	VSPAERO (E)
−4°	0.04480	0.001740	+0.17335	0.0410*
−2°	0.23968	0.003585	+0.13629	0.5690
0°	0.43405	0.008912	+0.09512	0.7505
2°	0.62753	0.017704	+0.04917	0.7897
4°	0.82098	0.030046	−0.00487	0.7965
6°	1.01467	0.045806	−0.06429	0.7980
8°	1.20781	0.065221	−0.13566	0.7941
10°	1.39983	0.087932	−0.21194	0.7912

The E value at −4° should not be interpreted physically because CL is almost zero, making efficiency-factor calculations ill-conditioned.

2. CA-G0 analysis log

ID	Verification item	Result	Criterion / expectation	Disposition
CA-G0-01	Analysis representation	MainWing + HTail + VTail, thin VLM	Intended baseline	PASS
CA-G0-02	Surface discretization	Extra-Fine; 218 load records/case	Frozen Folder-03 mesh	PASS
CA-G0-03	Wake iterations	20	Frozen numerical setting	PASS
CA-G0-04	Wake nodes	16	Frozen numerical setting	PASS
CA-G0-05	(α) sweep	−4° to +10°, 8 points	Complete intended sweep	PASS
CA-G0-06	Mach	0.203	Controlled cruise state	PASS
CA-G0-07	Reynolds number	($6.98 \cdot 10^6$)	Controlled cruise state	PASS
CA-G0-08	References	(S=30.0000034 m ² , b=16.40 m, c=1.881 m)	Matches controlled state	PASS
CA-G0-09	Moment reference	(X _{ref} =3.750 m)	Fixed reference	PASS
CA-G0-10	Max (	ΔC_L	$_{\{19\to 20\}}$	0.112%
CA-G0-11	Max (	ΔC_{D_i}	$_{\{19\to 20\}}$	0.234%
CA-G0-12	Max (	ΔC_{My}	$_{\{19\to 20\}}$	0.00113
CA-G0-13	(C _L (4°))	0.82098	Baseline recording	Accepted
CA-G0-14	(C _{D_i} (4°))	0.030046	Baseline recording	Accepted

ID	Verification item	Result	Criterion / expectation	Disposition
CA-G0-15	(C_My (4°))	-0.00487	Baseline recording	Accepted
CA-G0-16	(C_L_alpha), -2°→8°	5.546 rad⁻¹	Linear attached-flow response	PASS
CA-G0-17	(C_L-alpha) fit (R ²)	0.9999993	Near unity	PASS
CA-G0-18	(C_{L0})	0.4337	Plausible positive-lift baseline	Accepted
CA-G0-19	Zero-lift angle	-4.48°	Consistent with cambered/high-lift wing system	Accepted
CA-G0-20	Low-angle (C_m_alpha), -2°→4°	-1.345 rad⁻¹	Negative for longitudinal stability screening	PASS / provisional
CA-G0-21	Low-angle (C_m_alpha) fit (R ²)	0.9962	Strong local linearity	PASS
CA-G0-22	Approx. (C_m=0) crossing	α ≈ 3.82°	Informational only	Record
CA-G0-23	Induced-drag fit, 0°→10°	k= 0.04467	Strong parabolic relationship	PASS
CA-G0-24	(C_D_i-C_L ²) fit (R ²)	0.99992	Near unity	PASS
CA-G0-25	Fitted (e), 0°→10°	0.795	Baseline lifting-system value	Accepted / provisional
CA-G0-26	Lateral numerical residuals	(	C_S	_{\max}=0.00233), (
CA-G0-27	Solver repeatability vs prior XF/16 case	(C_L) common points within ~0.002%; (C_D_i) within ~0.012%	Repeatable	PASS
CA-G0-28	Runtime	~405–424 s/case; ~55.3 min total inferred	Informational	Record
CA-G0-29	VSPAERO CDo/CDtot	Present but not used as complete-aircraft parasite drag	PDT controls (C_{D0})	Correct treatment
CA-G0-30	Overall CA-G0 gate	Numerically stable, repeatable and usable	Required before G1	PASS — CLOSE G0

CA-G0 PASS. The Extra-Fine lifting-surface VLM model with 20 wake iterations and 16 wake nodes produced finite, repeatable solutions over $-4^\circ \leq \alpha \leq 10^\circ$. Maximum final-iteration changes were 0.112% in CL, 0.234% in CDi, and 0.00113 absolute in CMy. The lift curve is effectively linear with $CL_\alpha = 5.546 \text{ rad}^{-1}$, $CL_0 = 0.4337$, and $\alpha_{L=0} = -4.48^\circ$. The positive-lift induced-drag polar gives $k = 0.04467$ and $e \approx 0.795$. The solution reproduces the preceding convergence case to effectively numerical precision. CA-G0 is accepted as the reference against which lift-strut effects will be quantified.

01_G1_Strut_Sensitivity

1. Extracted CA-G1 polar

(α)	(C_L)	(C_{D_i})	(C_{My})	VSPAERO (E)
-4°	0.03992	0.001796	+0.16825	0.0308*

(α)	(C_L)	(C_{D_i})	(C_{My})	VSPAERO (E)
-2°	0.23710	0.003570	+0.13364	0.5583
0°	0.43388	0.008909	+0.09500	0.7508
2°	0.62981	0.017802	+0.05172	0.7926
4°	0.82519	0.030283	+0.00207	0.8005
6°	1.02157	0.046410	-0.05456	0.8012
8°	1.21774	0.066224	-0.12438	0.7985
10°	1.41107	0.089222	-0.19537	0.7966

Again, E near zero lift should not be interpreted physically.

2. Direct G0 → G1 comparison

(α)	(C_{L,G0})	(C_{L,G1})	(ΔC_L)	(C_{D_i,G0})	(C_{D_i,G1})	(ΔC_{D_i})	(ΔC_{My})
-4°	0.04480	0.03992	-0.00488	0.001740	0.001796	+0.000056	-0.00510
-2°	0.23968	0.23710	-0.00258	0.003585	0.003570	-0.000015	-0.00264
0°	0.43405	0.43388	-0.00016	0.008912	0.008909	-0.000003	-0.00012
2°	0.62753	0.62981	+0.00228	0.017704	0.017802	+0.000098	+0.00255
4°	0.82098	0.82519	+0.00421	0.030046	0.030283	+0.000237	+0.00694
6°	1.01467	1.02157	+0.00690	0.045806	0.046410	+0.000604	+0.00973
8°	1.20781	1.21774	+0.00993	0.065221	0.066224	+0.001003	+0.01128
10°	1.39983	1.41107	+0.01124	0.087932	0.089222	+0.001290	+0.01657

CA-G1 analysis log

ID	Verification item	Result	Criterion / interpretation	Disposition
CA-G1-01	Model build-up	G0 + LiftStrut_Pair only	Controlled incremental change	PASS
CA-G1-02	Load records	238 vs G0 218	+10 stations/strut	PASS
CA-G1-03	Wake iterations	20	Frozen numerical state	PASS
CA-G1-04	Wake nodes	16	Frozen numerical state	PASS
CA-G1-05	Max final-step	ΔC_L		0.196%
CA-G1-06	Max final-step	ΔC_{D_i}		0.362%
CA-G1-07	Max final-step	ΔC_{My}		0.00327
CA-G1-08	(C_L(4°))	0.82519	+0.51% vs G0	Small effect

ID	Verification item	Result	Criterion / interpretation	Disposition
CA-G1-09	(C_{D_i})(4°)	0.030283	+0.79% vs G0	Small effect
CA-G1-10	(C_{My})(4°)	+0.00207	(Δ =+0.00694)	Material effect
CA-G1-11	(C_{L_\alpha})	5.6163 rad⁻¹	+1.26% vs G0	Accept
CA-G1-12	(C_{L-\alpha}) (R ²)	0.9999992	Near unity	PASS
CA-G1-13	(C_{L0})	0.43348	-0.056% vs G0	Negligible
CA-G1-14	($\alpha_{\{L=0\}}$)	-4.42°	+0.06° vs G0	Negligible
CA-G1-15	(k)	0.044612	-0.12% vs G0	PASS
CA-G1-16	fitted (e)	0.79585	+0.12% vs G0	PASS
CA-G1-17	Drag-polar (R ²)	0.99993	Near unity	PASS
CA-G1-18	Low-angle (C_{m_\alpha})	-1.2548 rad⁻¹	6.7% smaller magnitude than G0	TRACK
CA-G1-19	(C_{m=0}) crossing	≈4.07°	+0.25° vs G0	Record
CA-G1-20	Strut L/R load symmetry	<0.75% difference at positive (α)	Good symmetry	PASS
CA-G1-21	Strut load behavior	Smooth overall; lower-end unloading visible	Expected without fuselage	Accept with caveat
CA-G1-22	Lateral residuals	Comparable to G0	No new asymmetry issue	PASS
CA-G1-23	Runtime	66.6 min	+20.4% vs G0	Record
CA-G1-24	Strut profile drag treatment	Remains in PDT	Avoid VSPAERO double-counting	Correct
CA-G1-25	Overall strut sensitivity	Small lift/drag effect; measurable pitch effect	Suitable for next build-up	PASS

CA-G1 PASS. Explicit inclusion of the lift-strut pair changes $CL(4^\circ)$ by +0.51%, $CD_i(4^\circ)$ by +0.79%, and the fitted induced-drag factor by only -0.12%, indicating negligible impact on overall induced-drag efficiency. The principal strut effect is longitudinal moment: $C_{m\alpha}$ changes from -1.345 to -1.255 rad^{-1} , a 6.7% reduction in stabilizing slope magnitude, while remaining negative. Left/right strut loading is highly symmetric and the solver remains converged. CA-G1 is therefore accepted as a valid strut-sensitivity model, with pitching-moment effects retained for subsequent hybrid/stability assessment.

Fuselage-only analysis log

ID	Verification item	Result	Assessment	Disposition
FUS-01	Thick Set	Fuselage only	Intended diagnostic	PASS

ID	Verification item	Result	Assessment	Disposition
FUS-02	Thin Set	None	Intended diagnostic	PASS
FUS-03	(\alpha)	4.0°	Correct smoke point	PASS
FUS-04	(\beta)	0°	Correct	PASS
FUS-05	Mach	0.203	Correct	PASS
FUS-06	(Re_{\bar c})	(6.98\times10^6)	Correct	PASS
FUS-07	(S_{\rm ref})	30.0000034 m²	Correct	PASS
FUS-08	(b_{\rm ref})	16.40 m	Correct	PASS
FUS-09	(c_{\rm ref})	1.881 m	Correct	PASS
FUS-10	(X_{\rm ref})	3.750 m	Correct	PASS
FUS-11	History iterations in CSV	3	Solver settled rapidly	Record
FUS-12	Final (C_L)	+0.0002014	Very small body-only lift	Stable
FUS-13	(C_L), iter. 2→3	+0.0113%	Excellent	PASS
FUS-14	Final (C_{D_i})	-0.0003029	Tiny inviscid value; do not use as aircraft drag	Accept diagnostically
FUS-15	(	\Delta C_{D_i}	_{2\to3})	(1.69\times10^{-6})
FUS-16	Final (C_{My})	-0.0017165	Small	Stable
FUS-17	(	\Delta C_{My}	_{2\to3})	0.0000443
FUS-18	Final (C_S)	(2.33\times10^{-5})	Essentially zero	PASS
FUS-19	Final (C_{Mz})	(-3.12\times10^{-5})	Essentially zero	PASS
FUS-20	Final (\log_{10} L_2) residual	-3.742	Strong reduction	PASS
FUS-21	Final (\log_{10} R_{\max})	-3.022	Strong reduction	PASS
FUS-22	NaN/divergence	None	Required	PASS
FUS-23	Overall thick-fuselage representation	Numerically stable alone	Key diagnostic	PASS

G2A MainWing + Fuselage analysis log

ID	Verification item	Result	Gate / expectation	Disposition
MW-FUS-01	Thin Set	MainWing only	Intended isolation	PASS
MW-FUS-02	Thick Set	Fuselage only	Intended isolation	PASS
MW-FUS-03	(\alpha)	4°	Controlled smoke point	PASS
MW-FUS-04	Mach	0.203	Controlled	PASS

ID	Verification item	Result	Gate / expectation	Disposition
MW-FUS-05	(Re _c)	(6.98*10^6)	Controlled	PASS
MW-FUS-06	Wake iterations	20	Frozen	PASS
MW-FUS-07	Final (C _L)	0.717560	Finite and settled	Finite
MW-FUS-08	(C _L), iter. 19→20	+0.810%	<0.5%	FAIL
MW-FUS-09	Final (C _{D_i})	0.017965	Finite and settled	Finite
MW-FUS-10	(C _{D_i}), iter. 19→20	+2.74%	<0.5%	FAIL
MW-FUS-11	Wake (C _{D_iw})	0.022784	Should be settled	—
MW-FUS-12	(C _{D_iw}), iter. 19→20	+6.18%	<0.5%	FAIL
MW-FUS-13	Final (C _{My})	−0.026172	Finite	—
MW-FUS-14	(	\Delta C _{My}	_{19\to20})	0.002903
MW-FUS-15	Final (C _S)	+0.002915	Near zero expected	Monitor
MW-FUS-16	Final (log _{10} L ₂)	+0.2815	Ideally falling/low	Not converged
MW-FUS-17	Final (log _{10} R _{max})	+1.7947	Ideally falling/low	Not converged
MW-FUS-18	NaN	None	Required	PASS
MW-FUS-19	Wing-root intersection strips	Pathological slivers present	Should not occur	FAIL
MW-FUS-20	Overall	Finite but junction-driven instability	Must pass before adding tails	FAIL

Repair 1 analysis log

ID	Check	Previous W=73	New W=49	Gate	Disposition
R1-01	MainWing Num_W	73	49	Intended change	PASS
R1-02	Fuselage Num_W	17	17	Frozen	PASS
R1-03	Fuselage section U	6 × 6 sections	6 × 6 sections	Frozen	PASS
R1-04	Final (C _L)	0.71756	0.71283	Finite	Finite
R1-05	(	\Delta C _L	_{19\to20})	0.81%	2.52%
R1-06	Final (C _{D_i})	0.017965	0.012565	Finite/settled	FAIL
R1-07	(	\Delta C _{D_i}	_{19\to20})	2.74%	27.66%
R1-08	Wake (C _{D_iw})	0.022784	0.021874	Settled	—
R1-09	(	\Delta C _{D_iw}	_{19\to20})	6.18%	14.55%
R1-10	Final (C _{My})	−0.02617	−0.02818	Finite	Finite

ID	Check	Previous W=73	New W=49	Gate	Disposition
R1-11	(	\Delta C_{My}	_{19\to20})	0.00290	0.02072
R1-12	Final (C_S)	+0.00292	-0.00085	Near zero	Endpoint improved
R1-13	(L_2) residual final	+0.281	+0.457	Falling/low	FAIL
R1-14	Max residual final	+1.795	+2.185	Falling/low	FAIL
R1-15	Root sliver chord	0.0198 m	0.03089 m	Larger is better	Improved
R1-16	Port root (c_l)	-17.94	-7.82	Non-pathological	Improved, still bad
R1-17	Starboard root (c_l)	-17.99	-10.14	Non-pathological	Improved, still bad
R1-18	Port root (c_{d_i})	-65.90	-8.73	Non-pathological	Major improvement
R1-19	Starboard root (c_{d_i})	-66.88	-13.48	Non-pathological	Major improvement
R1-20	Overall	Junction improved but solver unstable	—	Required PASS	FAIL / continue repair

Repair 2 analysis log

ID	Check	W49 / U6	W49 / U9	Criterion	Disposition
R2-01	MainWing Num_W	49	49	Frozen	PASS
R2-02	Fuselage Num_W	17	17	Frozen	PASS
R2-03	Fuselage section U	6/6/6/6/6/6	9/9/9/9/9/9	Intended change	PASS
R2-04	Final (C_L)	0.71283	0.73807	Finite + converged	Finite only
R2-05	(	\Delta C_L	_{19\to20})	2.52%	3.12%
R2-06	Final surface (C_{D_i})	+0.01257	-0.20401	Physically/numerically sensible	SEVERE FAIL
R2-07	(	\Delta C_{D_i}	_{19\to20})	27.66%	9.92%
R2-08	Final wake (C_{D_iw})	0.02187	0.02541	Settled	—
R2-09	(	\Delta C_{D_iw}	_{19\to20})	14.55%	3.82%
R2-10	Final (C_{My})	-0.02818	-0.09842	Finite + settled	FAIL
R2-11	(	\Delta C_{My}	_{19\to20})	0.02072	0.01059
R2-12	Final (C_{L,w})	0.64862	0.67369	Compare to surface lift	Large mismatch remains
R2-13	Final (\log_{10}L_2)	0.4567	0.1689	Low and settling	Improved level, still rising
R2-14	Final (\log_{10}R_{\max})	2.1851	1.6715	Low and settling	Improved level, still rising
R2-15	Root sliver chord	0.03089 m	0.03089 m	Should disappear/enlarge	No improvement
R2-16	Port root (c_l)	-7.82	-19.12	Ordinary sectional value	Much worse
R2-17	Starboard root (c_l)	-10.14	-19.06	Ordinary sectional value	Much worse

ID	Check	W49 / U6	W49 / U9	Criterion	Disposition
R2-18	Port root (c_{d_i})	-8.73	-30.18	Ordinary sectional value	Much worse
R2-19	Starboard root (c_{d_i})	-13.48	-23.93	Ordinary sectional value	Much worse
R2-20	Overall	Failed	More pathological	Must converge	FAIL — abandon U refinement

W33 analysis log

ID	Metric	W73/U6	W49/U6	W33/U6	Gate	Disposition
R3-01	MainWing Num_W	73	49	33	Intended	PASS
R3-02	Fuselage W/U	17 / 6	17 / 6	17 / 6	Frozen	PASS
R3-03	Final (C_L)	0.71756	0.71283	0.74739	finite	—
R3-04	(	\Delta C_L	_{19\to20})	0.81%	2.52%	0.0047%
R3-05	Final (C_{L,w})	0.64570	0.64862	0.67526	settled	—
R3-06	(	\Delta C_{L,w}	_{19\to20})	0.30%	0.02%	0.34%
R3-07	Final surface (C_{D_i})	+0.01796	+0.01257	-0.03409	physically sensible	FAIL
R3-08	(	\Delta C_{D_i}	_{19\to20})	2.74%	27.7%	34.7%
R3-09	Final wake (C_{D_iw})	0.02278	0.02187	0.02304	settled	—
R3-10	(	\Delta C_{D_iw}	_{19\to20})	6.18%	14.6%	0.21%
R3-11	Final (C_{My})	-0.02617	-0.02818	-0.04346	finite	—
R3-12	(	\Delta C_{My}	_{19\to20})	0.00290	0.02072	0.00247
R3-13	Final (\log_{10}L_2)	+0.281	+0.457	+0.0317	settling	Still oscillatory
R3-14	Final (\log_{10}R_{\max})	+1.795	+2.185	+1.517	settling	Still high
R3-15	Root-strip chord	0.0198 m	0.03089 m	5.32979 m	plausible local chord	SEVERE FAIL
R3-16	Root (c_l), port/starboard	-17.94/-17.99	-7.82/-10.14	-0.161/-0.162	ordinary	Coefficient improved
R3-17	Root (c_{d_i}), port/starboard	-65.9/-66.9	-8.73/-13.48	-0.091/-0.101	ordinary	Coefficient improved
R3-18	Overall	FAIL	FAIL	FAIL	valid hybrid solution	Do not use

Repair 4A analysis log

ID	Metric	W49, LE=0.25	W49, LE=0.50	Gate	Disposition
R4A-01	MainWing Num_W	49	49	Frozen	PASS
R4A-02	Fuselage Num_W	17	17	Frozen	PASS

ID	Metric	W49, LE=0.25	W49, LE=0.50	Gate	Disposition
R4A-03	Fuselage U	6/6/6/6/6/6	Same	Frozen	PASS
R4A-04	LE clustering	0.25	0.50	Intended change	PASS
R4A-05	TE clustering	0.25	0.25	Frozen	PASS
R4A-06	Final (C_L)	0.71283	0.61757	finite + settled	FAIL
R4A-07	(	\Delta C_L	_{19\to20})	2.52%	6.32%
R4A-08	Final surface (C_{D_i})	+0.01257	-0.17242	positive/sensible	SEVERE FAIL
R4A-09	(	\Delta C_{D_i}	_{19\to20})	27.7%	58.2%
R4A-10	Final wake (C_{D_{iw}})	0.02187	0.02893	settled	—
R4A-11	(	\Delta C_{D_{iw}}	_{19\to20})	14.6%	19.2%
R4A-12	Final wake (C_{Lw})	0.64862	0.59134	settled	—
R4A-13	(	\Delta C_{Lw}	_{19\to20})	0.02%	0.63%
R4A-14	Final (C_{My})	-0.02818	-0.08488	finite + settled	FAIL
R4A-15	(	\Delta C_{My}	_{19\to20})	0.02072	0.00930
R4A-16	Final (C_S)	-0.00085	+0.01191	near zero	FAIL
R4A-17	Final (\log_{10}L_2)	+0.457	+0.687	falling/low	FAIL
R4A-18	Final (\log_{10}R_{\max})	+2.185	+2.323	falling/low	FAIL
R4A-19	Root clipped chord	0.03089 m	0.03023 m	should improve	No improvement
R4A-20	Port root (c_l)	-7.82	+24.81	ordinary	Severe artifact
R4A-21	Starboard root (c_l)	-10.14	+46.97	ordinary	Severe artifact
R4A-22	Port root (c_m)	+229	-1383	ordinary	Severe artifact
R4A-23	Starboard root (c_m)	+239	-1838	ordinary	Severe artifact
R4A-24	Overall	FAIL	Much worse	valid hybrid solution	REJECT

Repair 4B analysis log

ID	Verification item	Result	Gate / interpretation	Disposition
R4B-01	MainWing Num_W	49	Frozen	PASS
R4B-02	LE clustering	0.25	Restored baseline	PASS
R4B-03	TE clustering	0.50	Intended change	PASS
R4B-04	Fuselage Num_W	17	Frozen	PASS

ID	Verification item	Result	Gate / interpretation	Disposition
R4B-05	Fuselage U	6/6/6/6/6/6	Frozen	PASS
R4B-06	Final (C_L)	0.68492	Finite	—
R4B-07	(C_L), iter. 19→20	−0.348%	<0.5%	PASS
R4B-08	Wake (C_{L,w})	0.68854	Finite	—
R4B-09	(C_{L,w}), 19→20	−0.409%	<0.5%	PASS
R4B-10	Final surface (C_{D_i})	0.16527	Physically/numerically credible	FAIL
R4B-11	(C_{D_i}), 19→20	+5.81%	<0.5%	FAIL
R4B-12	Wake (C_{D_iw})	0.025027	Finite	—
R4B-13	(C_{D_iw}), 19→20	+0.693%	<0.5%	Marginal FAIL
R4B-14	Final (C_{My})	+0.09786	Finite	—
R4B-15	(Δ C_{My})		{19to20})	0.00811
R4B-16	Final (C_S)	−0.00721	Near zero expected	Suspicious
R4B-17	Final (log_{10}L_2)	+0.0393	Lower is better	Improved late
R4B-18	Final (log_{10}R_{\max})	+1.6965	Lower is better	Still elevated
R4B-19	First root load-record chord	5.32979 m	Should resemble physical local wing chord	Severe topology artifact
R4B-20	Next normal root chord	2.28467 m	Reference	—
R4B-21	Root (c_l), port/starboard	−0.252 / −0.262	No longer enormous	Local coeff. improved
R4B-22	Root (c_{d_i}), port/starboard	−0.108 / −0.073	Small local values	Local coeff. improved
R4B-23	Surface/wake drag	0.1653 vs 0.0250	Gross inconsistency	FAIL
R4B-24	Overall	Some quantities settle, surface solution corrupted	Hybrid gate	FAIL

The complete wing-body diagnostic campaign

Case	Root/intersection behaviour	(C_L) final-step	(C_{D_i}) behaviour	Overall
W73, U6, LE/TE 0.25/0.25	19.8-mm severe sliver	0.80%	unstable	FAIL
W49, U6	30.9-mm sliver; less severe	2.59%	unstable	FAIL
W49 + fuselage U9	same sliver, much worse loads	3.03%	negative/pathological	FAIL
W33, U6	5.33-m oversized root topology	0.005%	negative/pathological	FAIL

Case	Root/intersection behaviour	(C_L) final-step	(C_{D_i}) behaviour	Overall
W49, LE 0.50	~30-mm sliver, extreme local loads	6.74%	negative/pathological	FAIL
W49, TE 0.50	5.33-m oversized root topology	0.35%	0.165 vs wake 0.025	FAIL

CA-G2 hybrid sensitivity not retained. A mixed thin-lifting-surface/thick-fuselage VSPAERO representation was investigated to quantify body interference. The isolated thick fuselage and isolated thin lifting system were each numerically stable; however, their combination produced persistent wing–fuselage intersection artifacts. Controlled tests varying wing chordwise tessellation (73, 49 and 33), fuselage longitudinal tessellation (6 and 9), and leading-/trailing-edge clustering did not produce a simultaneously converged and physically credible surface-force solution. Depending on discretization, the intersection produced either a 20–31 mm clipped root sliver with extreme sectional loads or an oversized 5.33 m root load polygon. The hybrid model was therefore rejected rather than used to introduce unquantified numerical error.

Selected clean potential-flow model should now be CA-G1

I recommend selecting: **MainWing + HTail + VTail + LiftStrut Pair** as the production clean VSPAERO representation.

G2A PANEL MainWing-only analysis log

ID	Verification item	Result	Criterion	Disposition
MW-P-01	Model	MainWing thick only	Intended isolation	PASS
MW-P-02	Thin Set	None	Intended	PASS
MW-P-03	(\alpha)	4.0°	Controlled	PASS
MW-P-04	(\beta)	0.0°	Controlled	PASS
MW-P-05	Mach	0.203	Controlled	PASS
MW-P-06	(Re_{\bar c})	(6.98\times 10^6)	Controlled	PASS
MW-P-07	(S_{\rm ref})	30.0000034 m²	Controlled	PASS
MW-P-08	(b_{\rm ref})	16.40 m	Controlled	PASS
MW-P-09	(c_{\rm ref})	1.881 m	Controlled	PASS
MW-P-10	(X_{\rm ref})	3.750 m	Controlled	PASS
MW-P-11	Wake iterations	20	Frozen	PASS
MW-P-12	Wake nodes	16	Frozen	PASS
MW-P-13	Final (C_L)	0.909413	Finite	PASS
MW-P-14	(	\Delta C_L	_{19\to 20})	0.0010%
MW-P-15	Final (C_{D_i})	0.0349590	Finite/positive	PASS
MW-P-16	(	\Delta C_{D_i}	_{19\to 20})	0.0072%
MW-P-17	Final wake (C_{D_{iw}})	0.0276277	Finite	PASS
MW-P-18	(	\Delta C_{D_{iw}}	_{19\to 20})	0.00030%

ID	Verification item	Result	Criterion	Disposition
MW-P-19	Final (C_{My})	-0.014666	Finite	PASS
MW-P-20	(	\Delta C_{My}	_{19\to20})	0.00000207
MW-P-21	Final (C_S)	(1.29\times10^{-9})	≈ 0 at ($\beta=0$)	PASS
MW-P-22	Final (C_{Mz})	(5.99\times10^{-7})	≈ 0	PASS
MW-P-23	Final (\log_{10}L_2)	-2.808	Strong decay	PASS
MW-P-24	Final (\log_{10}R_{\max})	-1.502	Strong decay	PASS
MW-P-25	L/R span-load symmetry	Essentially exact	Required	PASS
MW-P-26	Root load behaviour	Smooth and finite	No singularity	PASS
MW-P-27	Runtime	87.6 s	Informational	Record
MW-P-28	Overall	Stable, symmetric, converged	Required	PASS

CA-G2 body-interference investigation — not retained. Both the isolated thick MainWing and isolated thick fuselage produced stable, symmetric, strongly converged panel solutions. The MainWing-only panel solution at $\alpha=4^\circ$ converged to $CL=0.9094$, $CDi=0.03496$, and $CMy=-0.01467$, with final-step changes of 0.001%, 0.007%, and 2.1×10^{-6} , respectively. However, combining MainWing and fuselage produced severe local root singularities and non-converged global forces under both thin/thick hybrid and all-thick panel formulations. Controlled variations in wing chordwise tessellation, fuselage longitudinal tessellation, and LE/TE clustering did not yield a stable junction solution. The body-interference model was therefore rejected rather than allowing unresolved numerical artifacts to contaminate the validated aerodynamic database.

03_Final_Clean_Polar

00_G3_Repeatability

Repeatability analysis log

ID	Verification item	Result	Criterion	Disposition
G3-R-01	(\alpha)	4°	Required	PASS
G3-R-02	Mach	0.203	Controlled	PASS
G3-R-03	(Re_{\bar c})	(6.98\times10^6)	Controlled	PASS
G3-R-04	(S_{\rm ref})	30.0000034 m²	30.0 m²	PASS
G3-R-05	(b_{\rm ref})	16.40 m	Controlled	PASS
G3-R-06	(c_{\rm ref})	1.881 m	Controlled	PASS
G3-R-07	(X_{\rm ref})	3.750 m	Controlled	PASS
G3-R-08	Wake iterations	20	Frozen	PASS
G3-R-09	(C_L)	0.825049	CA-G1 repeatability	PASS

ID	Verification item	Result	Criterion	Disposition
G3-R-10	(C _L) difference vs G1	-0.017%	≤0.5%	PASS
G3-R-11	(C _{D_i})	0.0302665	CA-G1 repeatability	PASS
G3-R-12	(C _{D_i}) difference vs G1	-0.055%	≤1%	PASS
G3-R-13	(C _{My})	+0.002280	CA-G1 repeatability	PASS
G3-R-14	(	\Delta C _{My}	) vs G1	0.000210
G3-R-15	Final-step (C _L) change	0.208%	<0.5%	PASS
G3-R-16	Final-step (C _{D_i}) change	0.242%	<0.5%	PASS
G3-R-17	Final-step (	\Delta C _{My}	)	0.00303
G3-R-18	Lateral residual (C _S)	0.000286	Small at (\beta=0)	PASS
G3-R-19	Solver divergence / NaN	None	None allowed	PASS
G3-R-20	Total runtime	578.4 s ≈ 9.64 min	Informational	Record
G3-R-21	Configuration restoration	Reproduces CA-G1	Required	PASS
G3-R-22	Overall repeatability gate	—	Required before final sweep	PASS — CLOSE

CA-G3 repeatability PASS. The final clean-polar model reproduced the accepted CA-G1 4° solution to within 0.017% in CL, 0.055% in CDi, and 0.00021 absolute in CMy. Final wake-iteration changes remained within the adopted convergence limits. The CA-G3 configuration is therefore confirmed as numerically equivalent to the accepted CA-G1 lifting-system model and is cleared for the final -4° to 10° aerodynamic sweep.

1. Final CA-G3 VSPAERO polar

(α)	(C _L)	Surface (C _{D_i})	Wake (C _{{D_{iw}}})	(C _{My})	VSPAERO (E)
-4°	0.039914	0.001796	0.001425	+0.168266	0.0308*
-2°	0.237099	0.003570	0.002456	+0.133656	0.5583
0°	0.433879	0.008909	0.006315	+0.095006	0.7508
2°	0.629810	0.017802	0.013043	+0.051722	0.7926
4°	0.825266	0.030292	0.022624	+0.001833	0.8004
6°	1.021521	0.046402	0.035128	-0.054491	0.8012
8°	1.217688	0.066223	0.050645	-0.124256	0.7985
10°	1.411015	0.089204	0.069232	-0.194975	0.7967

8. Direct nominal/adverse polar

Using the actual discrete VSPAERO CDi rather than the fitted equation:

(α)	(C _L)	(C _D)	(C _D , _{nom})	(C _D , _{adv})	((L/D) _{nom})	((L/D) _{adv})
−4°	0.0399	0.00180	0.02330	0.02486	1.71	1.61
−2°	0.2371	0.00357	0.02507	0.02664	9.46	8.90
0°	0.4339	0.00891	0.03041	0.03198	14.27	13.57
2°	0.6298	0.01780	0.03930	0.04087	16.02	15.41
4°	0.8253	0.03029	0.05179	0.05336	15.93	15.47
6°	1.0215	0.04640	0.06790	0.06947	15.04	14.70
8°	1.2177	0.06622	0.08772	0.08929	13.88	13.64
10°	1.4110	0.08920	0.11071	0.11227	12.75	12.57

11. Final CA-G3 analysis log

ID	Verification item	Result	Criterion	Disposition
G3-01	Sweep	−4° to +10°, 8 points	Required	PASS
G3-02	Mach	0.203	Controlled	PASS
G3-03	(Re _c)	(6.98×10 ⁶)	Controlled	PASS
G3-04	Wake iterations/nodes	20 / 16	Frozen	PASS
G3-05	Max final (	Δ C _L	)	0.237%
G3-06	Max final (	Δ C _D)	)	0.363%
G3-07	Max final (	Δ C _{My}	)	0.00347
G3-08	4° repeatability (C _L)	0.026% difference	<0.5%	PASS
G3-09	4° repeatability (C _D)	0.085% difference	<1%	PASS
G3-10	(C _{L0})	0.43349	Record	Accepted
G3-11	(C _{Lα})	5.6161 rad⁻¹	Strong linearity	PASS
G3-12	Lift fit (R ²)	0.9999993	Near unity	PASS
G3-13	(α _{L=0})	−4.422°	Record	Accepted
G3-14	(C _{mα}), low-angle	−1.257 rad⁻¹	Negative	PASS / provisional
G3-15	(C _{m=0}) crossing	≈4.07°	Informational	Record
G3-16	Surface (k)	0.044608	Final working fit	Accepted
G3-17	Surface fitted (e)	0.7959	Final working value	Accepted
G3-18	Surface drag fit (R ²)	0.999927	Near unity	PASS

ID	Verification item	Result	Criterion	Disposition
G3-19	Nominal total polar	$(0.0216437+0.0446077C_L^2)$	Required	COMPLETE
G3-20	Adverse total polar	$(0.0232097+0.0446077C_L^2)$	Required	COMPLETE
G3-21	Combined adverse polar	$(0.0232097+0.0469719C_L^2)$	Robustness case	COMPLETE
G3-22	Nominal ($L/D_{\max}$)	16.09	Performance input	Accepted
G3-23	Nominal optimum (C_L)	0.6966	Performance input	Accepted
G3-24	Nominal optimum (α)	2.68°	Performance input	Accepted
G3-25	Combined-adverse ($L/D_{\max}$)	15.14	Robustness input	Accepted
G3-26	Surface vs wake drag discrepancy	Material (~34% at 4°)	Should be documented	TRACK
G3-27	Wake-(k) mesh convergence	≈1.15% Fine→XF	<2%	PASS cross-check
G3-28	G2 body model	Rejected after controlled diagnostics	No contamination of G3	PASS
G3-29	Overall final clean polar	Stable and repeatable	Required	PASS

1. Scope and controlled analysis basis

The G1.4–G1.6 campaign was opened after the original mixed thin/thick G2 clean-aerodynamic model generated nonphysical root slivers, malformed effective chords, large sectional-load spikes, surface/Trefftz disagreement, and zero-sideslip asymmetry. The objective was not to tune VSPAERO to recover a preferred report value. The objective was to determine whether a physically credible mixed thin-lifting/thick-body representation could be established for Tyche R2-T1 and, if not, to bound and document the model-form limitation.

1.1 Reference and solver controls

Control	Value
Reference area, S_{ref}	30.0 m ²
Reference span, B_{ref}	16.4 m
Reference chord, C_{ref}	1.881 m
Moment reference	$X_{ref} = 3.750$ m; $Y_{ref} = 0$; $Z_{ref} = 0$
Diagnostic flow condition	$\alpha = 4^\circ$, $\beta = 0^\circ$, $M = 0.203$, $Re_{Cref} = 6.98 \times 10^6$
Wake / iteration controls	16 wake nodes; 20 iterations; relaxation 1.0; fixed wake OFF
Production mesh state	MainWing W73, U = 9/37/7/19; HTail

	W73/U29; VTail W73/U29
Symmetry	XZ symmetry OFF; physical left/right symmetry assessed from loads and coefficients

Configuration-control note: several G1.5/G1.6 scripts retained lower tessellation defaults, but the executed diagnostic .vsp3 cases were manually restored to the established Extra-Fine values before VSPAERO. The CSV results, not the script defaults, are the authoritative solver-state evidence.

1.2 Acceptance philosophy

A run could converge numerically and still fail physically. Therefore coefficient flatness was never used as the sole acceptance criterion. A configuration was retained only when geometry topology, iterative convergence, symmetry and surface/wake sanity were jointly credible.

Gate	Working criterion
Final-step lift	$ \Delta CL < 0.5\%$
Final-step wake lift	$ \Delta CL,w < 0.5\%$
Final-step Trefftz induced drag	$ \Delta CD_{i,w} < 0.5\%$
Final-step pitching moment	$ \Delta CMy < 0.005$
Zero-sideslip symmetry	$ CS $ and $ Cn \lesssim 10^{-3}$ for isolated symmetric diagnostics
Root topology	No clipped ~0.02–0.10 m remnants, no multi-metre malformed chord, mirrored left/right geometry
Residual quality	Prefer $\log_{10}(L2) < -2$ without late growth; interpret together with physical checks
Drag interpretation	Trefftz $CD_{i,w}$ primary for mixed-model diagnosis; PDT remains source of parasite drag

2. G1.4 analysis log — MainWing root repair and first H2 failure

2.1 G1.4-H1 Mode 1 — blended WingRootFairing

The first WingRootFairing repair removed the original catastrophic wing/body root topology. The first exposed MainWing station became physically plausible (approximately $y = \pm 0.629$ m, $c = 2.279$ m), replacing the earlier near-zero / oversized root-strip behaviour. A localized first-strip load spike remained close to the fairing edge.

Result	Value	Assessment
CL / CLw	0.670999 / 0.647364	Global lift stable
CDi / CDi,w	0.015032 / 0.017530	Surface CDi noisy; wake value usable diagnostically

CMy	-0.008925	Moment numerically bounded
CS / Cn	+7.93×10 ⁻⁵ / +4.64×10 ⁻⁵	Excellent global symmetry
Root load	cl ≈ 3.47 at first strip	Localized edge artifact remains
Disposition	CONDITIONAL	Topology fixed; local fairing-edge sensitivity remains

Source: TRIAL.csv; G1.4 script.

2.2 G1.4-H1 Mode 2 — enclosed WingRootFairing

FAIRING_MODE = 2 moved the first exposed wing station farther from the body and reduced the first-strip singularity. The physical root chord was retained and global symmetry remained good. This became the working AC-02 wing/body repair.

Result	Value	Assessment
First exposed root	y ≈ ±0.6967 m; c ≈ 2.2705 m	Physical planform chord
CL / CLw	0.689660 / 0.645805	Stable global lift
CDi / CDi,w	0.010675 / 0.018459	Near-field drag remains model-sensitive
CMy	-0.019213	Final-step ΔCMy = 0.00320; pass
CS / Cn	-5.50×10 ⁻⁴ / -4.08×10 ⁻⁵	Acceptable symmetry
Disposition	RETAIN	AC-02 working repair; do not retune wing geometry

Source: H1_MODE2_MW_FUSE_FAIRING_A4.csv.

2.3 G1.4-H2 — add HTail and VTail

Adding the tails exposed a separate empennage intersection failure. The MainWing repair survived, but the HTail root collapsed to a ~0.103 m remnant followed by a ~1.679 m strip, with enormous local coefficients. The global solution was physically invalid.

Indicator	Observed	Disposition
HTail root topology	c ≈ 0.103 m → 1.679 m	FAIL
Representative HTail spike	cl ≈ 144–150; cdi ≈ -1.4×10 ⁴ ; cm ≈ -6.6×10 ⁴	FAIL
CL / CLw	0.391234 / 0.610084	Severe disagreement
CDi / CDi,w	-0.014973 / +0.016970	Nonphysical near-field drag
CMy	+0.852438	Contaminated
CS / Cn	+0.017661 / +0.006049	Zero-β symmetry fail
Disposition	REJECT	Localised to HTail/body root topology

Source: H2_Mode_1.csv.

3. G1.5 analysis log — TailRootFairing and direct tail-cone penetration

3.1 G1.5-H2 with dedicated TailRootFairing

A dedicated TailRootFairing removed the gross 0.103 m HTail chord remnant in the combined H2 geometry, but introduced a strongly asymmetric first-strip intersection state. The geometry looked better while the pressure solution remained invalid.

Indicator	Observed	Assessment
HTail chord	First exposed chord $\approx$ 1.654 m	Gross sliver removed
Local asymmetry	One side $cl \approx -46$; opposite side ≈ -0.58	Severe one-sided singularity
CL / CLw	0.599320 / 0.626331	Moderate global mismatch
CDi / CDi,w	-0.018803 / +0.018100	Near-field sign failure
CMy	+0.262228	Contaminated
CS / Cn	+0.004594 / +0.001236	Physical symmetry fail
Disposition	REJECT	TailRootFairing side-wall exit topology unresolved

Source: H2_Mode_1(1).csv; G1.5 script.

3.2 Isolated HT_TEST with TailRootFairing

The isolated HTail + thick fuselage/TailRootFairing test converged exceptionally well but remained physically wrong. This was an important demonstration that residual convergence alone cannot validate a mixed-model result.

Result	Value	Assessment
CL / CLw	-0.030232 / +0.044663	Opposite-sign lift evidence
CDi / CDi,w	-0.032355 / +0.000760	Near-field result nonphysical
CMy	+0.095085	Contaminated by root singularity
CS / Cn	-0.007400 / -0.003405	Strong asymmetry
Residuals	$\log_{10}(L2)=-3.841$; $\log_{10}(Rmax)=-2.303$	Numerically excellent
Disposition	REJECT	Converged-to-invalid solution

Source: HT_TEST.csv.

3.3 Direct tail-cone penetration: fairing OFF, TAIL_CONE_RISE = 0.060

The TailRootFairing was removed and the tail-cone crown raised by 60 mm to create direct, symmetric penetration. This isolated test was the first convincing HTail/body success: lateral coefficients collapsed toward zero, global coefficients became plausible, and residuals converged strongly.

Result	Value	Assessment
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CL / CLw	0.045955 / 0.046670	Good agreement
CDi / CDi,w	0.001298 / 0.000695	Surface/wake difference remains, but both positive
CMy	-0.138492	Physically plausible tail moment
CS / Cn	-1.38×10 ⁻⁴ / +6.35×10 ⁻⁵	PASS symmetry
Residuals	log10(L2)=-3.392; log10(Rmax)=-1.606	PASS
Disposition	PASS DIAGNOSTIC	Direct HTail penetration viable in isolation

Source: HT_Test_2.csv.

3.4 Combined H2 and H2_NO_VT with 60 mm rise

The isolated HTail repair did not survive full combination. With the MainWing/WRF system added, the complete H2 solution again became asymmetric and non-convergent; removing the VTail did not cure the failure. This ruled out the VTail as a necessary trigger and established a coupled multi-component problem.

Test	CL / CLw	CDi / CDi,w	CMy	CS / Cn	Residuals	Disposition
H2_Test_2	0.6579 / 0.6202	0.0120 / 0.0182	+0.0750	+0.00923 / +0.00294	L2 +0.784; Rmax +2.869	FAIL
H2_NO_VT_TEST	0.5413 / 0.6124	-0.00405 / 0.01717	+0.4256	-0.00267 / -0.00051	L2 +0.714; Rmax +2.618	FAIL

Sources: H2_Test_2.csv; H2_NO_VT_TEST.csv.

4. G1.6 analysis log — predictive trim-boundary repair and campaign closure

G1.6 replaced the earlier tail-fairing logic with an explicit camber-plane trim-boundary strategy. The goal was to force a constant HTail inner boundary and eliminate narrow or malformed residual strips before solver time was committed.

4.1 Initial G1.6 TailRootFairing: 0.600 m width

The new fairing established a constant trim boundary at $y = \pm 0.300$ m and restored the exact local HTail planform chord. The old sliver mechanism was closed, but the first surviving strip was only ~28% of the normal spanwise width.

Test	WRF in HT thick set?	CL / CLw	CDi / CDi,w	CS / Cn	Residuals	Disposition
HT_TEST_1	YES	0.10584 / 0.03949	0.01368 / 0.000659	+0.07662 / -0.00349	L2 -1.072; Rmax +0.366	HOLD/FAIL
HT_TEST_2	NO	0.04489 / 0.04461	-0.00485 / 0.000818	+0.00142 / +0.000616	L2 -0.721; Rmax +0.698	HOLD

The one-flag A/B comparison showed that removing WingRootFairing eliminated most of the gross pressure asymmetry, but did not restore residual convergence. The remaining 28%-width first strip therefore remained a credible numerical sensitivity.

4.2 Final AC-04 fairing geometry: 0.680 m width

The fairing was widened and shifted forward-to-full transition aft: $x_{\text{nose}} = 8.400$ m, $x_{\text{full,fwd}} = 8.600$ m, width = 0.680 m (half-width 0.340 m). The first surviving HTail strip became 0.10176 m wide, about 92.1% of a normal strip, while the first-strip chord remained exactly consistent with the analytical tail planform.

Test	Configuration	Key result	Disposition
HT_TEST_3	HTail thin; Fuselage+TRF thick; WRF OFF	yinner = 0.34000 m; c = 1.63110 m; CL=0.045154; CDiw=0.000796; CMY=-0.131935; CS=-3.58×10 ⁻⁴ ; L2=-3.696; Rmax=-2.236	PASS DIAGNOSTIC
HT_TEST_4	Same improved TRF; WRF ON	Geometry and residuals remain excellent, but CS=+0.076276; CL/CLw=0.108647/0.042133; surface CDi=0.018347 vs CDiw=0.000663	NUMERIC PASS / PHYSICAL FAIL

AC-04 was therefore closed as a geometry-topology repair. The remaining failure was no longer a malformed HTail root: it was a global thick-body/multi-body pressure-field problem.

4.3 Combined repaired H2_NO_VT

The repaired MainWing and repaired HTail were combined with Fuselage + WingRootFairing + TailRootFairing. Both root geometries remained physically correct, but the global model again failed zero-sideslip symmetry and convergence.

Metric	Result	Assessment
MainWing root	$y \approx \pm 0.6973$ m; $c \approx 2.2703$ m	AC-02 retained
HTail root	$y \approx \pm 0.3909$ m; $c = 1.63110$ m; inner edge 0.34000 m	AC-04 retained
CL / CLw	0.697529 / 0.627483	11% scale disagreement
CDi / CDi,w	0.020357 / 0.018027	Final-step CDiw variation 3.97%
CMY	+0.073604	Not reliable
CS / Cn	+0.050463 / -0.001902	Severe zero- β physical failure
Residuals	$\log_{10}(L2) = -0.284$; $\log_{10}(Rmax) = +1.349$	FAIL
Disposition	REJECT	Combined repaired mixed model not robust

Source: H2_NO_VT_TEST_1.csv and archived H2_NO_VT.vsp3.

4.4 Final forensic H2: long TailRootFairing aft closure

A final analysis-only causality test extended the TailRootFairing aft point from FS 10.940 m to FS 11.500 m, lengthening the closure taper from 0.150 m to 0.710 m. This reduced the local aft-closure pressure peak but did not recover a valid global mixed-model solution.

Metric	Result	Interpretation
Local TRF aft-closure Cp	Reduced from approximately +0.83–0.93 to $\approx +0.325$	Abrupt short closure confirmed as local defect
CL / CLw	0.703178 / 0.624349	Large disagreement remains
CDi / CDi,w	0.016150 / 0.018344	Final-step CDi and CDi,w fail convergence
CM _y	+0.054457	Final-step $ \Delta\text{CM}_y \approx 0.00727$; fail
CS / C _n	+0.054412 / -5.8×10^{-5}	Gross side force persists
Residuals	$\log_{10}(\text{L2}) = -0.278$; $\log_{10}(\text{Rmax}) = +1.215$	FAIL
Lifting-surface roots	MainWing, HTail and VTail regular	Local root topology no longer dominant
Disposition	REJECT / CAMPAIGN CLOSED	Multi-body mixed thick/thin model-form limitation

Source: H2_TEST_1.csv; Tyche_R2T1_G1_6_Analysis_H2.vspscript.

5. Consolidated G1.4–G1.6 result ledger

5.1 G1.4–G1.5 results

Test	Primary CSV	CL	CDi,w	CM _y	CS	log ₁₀ L2	Disposition
G1.4-H1-M1	TRIAL	0.6710	0.01753	−0.0089	+0.00008	−0.145	Conditional
G1.4-H1-M2	H1_MODE2	0.6897	0.01846	−0.0192	−0.00055	+0.265	Retain AC-02
G1.4-H2	H2_Mode_1	0.3912	0.01697	+0.8524	+0.01766	+0.167	Fail
G1.5-H2-TRF	H2_Mode_1(1)	0.5993	0.01810	+0.2622	+0.00459	−0.295	Fail
G1.5-HT-TRF	HT_TEST	−0.0302	0.000760	+0.0951	−0.00740	−3.841	Fail physical
G1.5-HT-RISE060	HT_Test_2	0.0460	0.000695	−0.1385	−0.00014	−3.392	Pass diagnostic
G1.5-H2-RISE060	H2_Test_2	0.6579	0.01822	+0.0750	+0.00923	+0.784	Fail
G1.5-H2-NOVT	H2_NO_VT_TEST	0.5413	0.01717	+0.4256	−0.00267	+0.714	Fail

5.2 G1.6 results

Test	Primary CSV	CL	CDi,w	CM _y	CS	log ₁₀ L2	Disposition
G1.6-HT1	HT_TEST_1	0.1058	0.000659	−0.0756	+0.07662	−1.072	Hold/Fail
G1.6-HT2	HT_TEST_2(1)	0.0449	0.000818	−0.1310	+0.00142	−0.721	Hold
G1.6-HT3	HT_TEST_3	0.0452	0.000796	−0.1319	−0.00036	−3.696	Pass diagnostic
G1.6-HT4	HT_TEST_4	0.1086	0.000663	−0.0789	+0.07628	−3.917	Fail physical

G1.6-H2-NOVT	H2_NO_VT_TEST_1	0.6975	0.01803	+0.0736	+0.05046	-0.284	Fail
G1.6-H2-LONG	H2_TEST_1	0.7032	0.01834	+0.0545	+0.05441	-0.278	Close campaign

6. Engineering conclusions from the geometry-repair campaign

- AC-02 (MainWing root) is closed. WingRootFairing Mode 2 replaced the original clipped/oversized root topology with a physical first exposed chord of approximately 2.270 m.
- AC-04 (HTail root) is closed. The final 0.680 m TailRootFairing generated a constant camber-plane trim boundary at $y = \pm 0.3400$ m and a first-strip chord of 1.63110 m matching the analytical planform.
- Mesh-density insufficiency is not the controlling explanation. The decisive G1.5/G1.6 diagnostic cases were executed at the established Extra-Fine surface discretisation.
- Numerical convergence is not sufficient evidence of physical validity. HT_TEST and HT_TEST_4 converged strongly while producing nonphysical lift/drag or zero-sideslip side force.
- The abrupt 0.150 m TailRootFairing aft closure was a real local pressure defect, but not the dominant global failure mechanism. Lengthening the taper to 0.710 m reduced the local C_p peak without eliminating the spurious CS.
- After all three lifting-surface root topologies became physically regular, the full mixed thin/thick assembly still showed large zero- β side force, poor residual convergence, and surface/wake inconsistency. The remaining error is therefore classified as a multi-body mixed-model / pressure-integration model-form limitation.
- No H3 run is justified. Further fairing dimensions, higher tessellation, more wake nodes or more iterations would have low expected validation value relative to the remaining competition schedule.

7. 04_Clean_Aero folder conclusion

The 04_Clean_Aero campaign is CLOSED for the purpose of the Tyche R2-T1 competition analysis. The production clean-aerodynamic model is the converged thin lifting-system branch (MainWing + HTail + VTail + lift struts, with propulsion off), while the G1.4–G1.6 mixed thin/thick branch is retained only as validation and model-form sensitivity evidence.

7.1 Production aerodynamic disposition

Element	Final finding	Use
CA-G0 clean lifting	Baseline thin lifting surfaces	Reference polar / no-strut baseline
CA-G1 strut sensitivity	Struts added; small global effect; production lifting set selected	CLOSED / retained
CA-G3 repeatability	$\alpha=4^\circ$: $CL=0.825049$, $CDi=0.030266$, $CM_y=+0.002280$; $CL_w=0.826879$, $CD_{i,w}=0.022618$	Repeatability PASS
G2/G1.4–G1.6 mixed body	Multiple root defects repaired, but combined multi-body solution remains physically invalid	REJECT for production coefficients
Parasite drag	OpenVSP Parasite Drag Tool / controlled G2 drag build-up	Use PDT CD_0 , not VSPAERO CD_0
Final drag build-up	$CD = CD_0, PDT + CD_i, VSPAERO$	Apply clean-lifting induced drag with documented mesh uncertainty

7.2 Ready-to-paste folder closure statement

04_Clean_Aero was closed after completing the clean lifting-surface polar, strut sensitivity, repeatability checks, and a bounded mixed thin/thick geometry-repair campaign. The converged thin lifting-system model is retained as the production aerodynamic source. MainWing and HTail root-intersection defects identified in the mixed-body branch were independently repaired and verified at Extra-Fine discretisation; however, the fully combined thick-body model continued to produce nonphysical zero-sideslip lateral force, residual oscillation and surface/Trefftz inconsistencies even after the local root topologies and the TailRootFairing aft closure were corrected. The mixed-body solution was therefore rejected under the predefined physical-validity gates and retained only as model-form sensitivity evidence. Parasite drag remains sourced from the OpenVSP Parasite Drag Tool, while induced drag, lift and lifting-surface moments are taken from the converged clean VSPAERO lifting model with the established mesh/wake uncertainty. No additional H2/H3 clean-body tuning is authorised; remaining validation effort transfers to 05_Stability.

7.3 Handoff to 05_Stability

- Freeze the G1.6 files as diagnostic evidence; do not use the analysis-only FS 11.500 m TailRootFairing as production geometry.
- Use the converged thin lifting configuration for the initial stability derivative campaign.
- Prioritise β -sweep derivatives ($CY\beta$, $CI\beta$, $Cn\beta$), followed by elevator, aileron and rudder control derivatives.
- Use the clean-polar $Cm\alpha$ / $CL\alpha$ relationship for the lifting-system neutral-point estimate, then apply a separate analytical fuselage correction rather than importing the rejected thick-body H2 result.
- Retain PDT as the parasite-drag source and avoid using mixed-model VSPAERO CDo/CDtot as complete-aircraft drag.

8. Analysis-file register

Group	Files
G1.4 scripts	Tyche_R2T1_G1_4_Analysis.vspscript
G1.5 scripts	Tyche_R2T1_G1_5_Analysis.vspscript
G1.6 scripts	Tyche_R2T1_G1_6_Analysis.vspscript; Tyche_R2T1_G1_6_Analysis_H2.vspscript
G1.4 result CSVs	TRIAL.csv; H1_MODE2_MW_FUSE_FAIRING_A4.csv; H2_Mode_1.csv
G1.5 result CSVs	H2_Mode_1(1).csv; HT_TEST.csv; HT_Test_2.csv; H2_Test_2.csv; H2_NO_VT_TEST.csv
G1.6 result CSVs	HT_TEST_1.csv; HT_TEST_2(1).csv; HT_TEST_3.csv; HT_TEST_4.csv; H2_NO_VT_TEST_1.csv; H2_TEST_1.csv
Clean-aero production evidence	CA_G0_Clean_Lifting.csv; 01_G1_Strut_Sensitivity.csv; CA_G3_Repeatability_A4.csv

05_Stability

00_G5_Baseline

The comparison against CA-G3 is effectively exact:

Quantity	CA-G3 reference	G5 B0.1	Difference	Status
(C_L)	0.825049	0.82504884	(-0.000019%)	PASS
(C_{L,w})	0.826879	0.82687868	(-0.000039%)	PASS
(C_{D_i})	0.0302665	0.03026647	(-0.00011%)	PASS
(C_{D_i,w})	0.0226181	0.02261812	(+0.000068%)	PASS
(C_{My})	+0.002280	+0.00228007	(+6.9\times10^{-8}) abs.	PASS
(C_S)	+0.000286	+0.0002864	essentially identical	PASS
(C_{Mx})	-0.000203	-0.0002031	essentially identical	PASS
(C_{Mz})	+0.0000933	+0.00009329	essentially identical	PASS

Final 00_G5_Baseline analysis log

Table 05-B0.1 — G5 baseline run definition and disposition

Test ID	File	(\alpha)	(\beta)	Thin Set	Thick Set	Purpose	Final disposition
B0.1	G5_B0_REPEAT_A4.csv	+4°	0°	STAB_THIN_PROD	STAB_THICK_NONE	Reproduce accepted CA-G3 A4 reference	PASS
B0.2	G5_B0_A0_B0.csv	0°	0°	STAB_THIN_PROD	STAB_THICK_NONE	Establish zero-(\beta) derivative anchor	PASS with CS bias note

Table 05-B0.2 — Final aerodynamic results

Test	(C_L)	(C_{L,w})	(C_{D_i})	(C_{D_i,w})	(C_{My})	(C_S)	(C_{Mx})	(C_{Mz})
CA-G3 A4 reference	0.825049	0.826879	0.0302665	0.0226181	+0.002280	+0.000286	-0.000203	+0.0000933
G5 B0.1 A4	0.8250488	0.8268787	0.0302665	0.0226181	+0.0022801	+0.0002864	-0.0002031	+0.0000933
G5 B0.2 A0/B0	0.4338828	0.4347331	0.0089078	0.0063150	+0.0949899	+0.0012737	-0.0008886	+0.0004578

Table 05-B0.3 — Iterative convergence

Test	$ \Delta CL $	$ \Delta CL,w $	$ \Delta CD_{i,w} $	$ \Delta CM_y $	log10L2	log10Rmax	Status
B0.1 A4	0.208%	0.220%	0.272%	0.003026	-0.866	+0.631	PASS
B0.2 A0/B0	0.012%	0.018%	0.171%	6.0×10^{-7}	-1.756	-0.256	PASS

Table 05-B0.4 — Gate assessment

Gate	B0.1	B0.2	Final assessment
Correct flow/reference state	PASS	PASS	PASS
CA-G3 configuration continuity	PASS — essentially exact	Consistent	PASS
Final-step lift convergence	PASS	PASS	PASS
Final-step Trefftz-drag convergence	PASS	PASS	PASS
Pitching-moment convergence	PASS	PASS	PASS
(C_{M_x}) near zero at $(\beta=0)$	PASS	PASS	PASS
(C_{M_z}) near zero at $(\beta=0)$	PASS	PASS	PASS
(C_S) near zero	PASS	(+0.001274), marginal	PASS / MONITOR
Suitable centre point for beta sweep	—	Yes	PASS
Overall 00_G5_Baseline			CLOSED / PASS

00_G5_Baseline — PASS / CLOSED. The G5 stability model reproduced the accepted CA-G3 $\alpha=4^\circ$ clean-aerodynamic reference to effectively numerical precision, confirming configuration, reference-quantity, Extra-Fine mesh and solver continuity between 04_Clean_Aero and 05_Stability. The $\alpha=0^\circ, \beta=0^\circ$ anchor also converged strongly, yielding $CL=0.433883$, $CM_y=0.094990$, $CM_x=-8.89 \times 10^{-4}$ and $CM_z=4.58 \times 10^{-4}$. A small $CS=1.274 \times 10^{-3}$ zero-sideslip bias is retained as a numerical intercept and will not be forced to zero in subsequent lateral-directional derivative fitting. The baseline is therefore accepted for progression to the manual β -sweep.

02_Beta_Sweep

(β)	(C_Y)	(C_L)	(C_n)	(C_L)	(C_m)	$(C_{D_{i,w}})$
-6°	+0.046211	+0.008382	-0.014461	0.432255	0.079014	0.008134
-4°	+0.031054	+0.005879	-0.009733	0.433493	0.086431	0.007166
-2°	+0.016085	+0.003389	-0.005064	0.434142	0.091409	0.006565

(\beta)	(C_Y)	(C_l)	(C_n)	(C_L)	(C_m)	(C_{D_i,w})
0°	+0.001274	+0.000887	−0.000458	0.433879	0.095006	0.006315
+2°	−0.013564	−0.001630	+0.004156	0.432825	0.096526	0.006433
+4°	−0.028411	−0.004143	+0.008777	0.430968	0.096246	0.006917
+6°	−0.043394	−0.006657	+0.013440	0.428320	0.094423	0.007779

01_Beta_Sweep log

Table 05-B1.1 — Run matrix and final coefficients

Test	(\beta)	(C_Y)	(C_l)	(C_n)	(C_L)	(C_m)	(C_{D_i,w})	Status
B1.1	−6°	+0.046211	+0.008382	−0.014461	0.432255	+0.079014	0.008134	PASS
B1.2	−4°	+0.031054	+0.005879	−0.009733	0.433493	+0.086431	0.007166	PASS
B1.3	−2°	+0.016085	+0.003389	−0.005064	0.434142	+0.091409	0.006565	PASS
B1.4	0°	+0.001274	+0.000887	−0.000458	0.433879	+0.095006	0.006315	PASS
B1.5	+2°	−0.013564	−0.001630	+0.004156	0.432825	+0.096526	0.006433	PASS
B1.6	+4°	−0.028411	−0.004143	+0.008777	0.430968	+0.096246	0.006917	PASS
B1.7	+6°	−0.043394	−0.006657	+0.013440	0.428320	+0.094423	0.007779	PASS

Table 05-B1.2 — Derived static lateral-directional derivatives

Quantity	Primary ±4° linear fit	(R^2)	All-seven fit	Assessment
(C_{Y_\beta})	−0.42565 rad ^{−1}	0.999997	−0.42705	PASS
(C_{l_\beta})	−0.07180 rad ^{−1}	0.999995	−0.07180	PASS
(C_{n_\beta})	+0.13247 rad ^{−1}	0.999995	+0.13295	PASS

Table 05-B1.3 — Numerical-validation gates

Gate	Result	Status
B0.2 zero-beta repeatability	(C_L) difference 0.00093%	PASS
Central-range linearity	all (R^2>0.99999)	PASS
±2° symmetry	max error 0.59%	PASS
±4° symmetry	max error 0.76%	PASS

Gate	Result	Status
$\pm 6^\circ$ symmetry	max error 0.75%	PASS
Central-difference derivative consistency	<1% variation	PASS
Final-step (C_L) convergence	all <0.05%	PASS
Final-step (C_{D_i,w}) convergence	all <0.20%	PASS
Directional static stability	(C_{n_\beta}=+0.13247)	PASS, lifting-system basis
Dihedral effect	(C_{l_\beta}=-0.07180)	PASS
Overall 01_Beta_Sweep	—	PASS / CLOSED

01_Beta_Sweep — PASS / CLOSED. A seven-point manual sideslip sweep from $\beta=-6^\circ$ to $+6^\circ$ was completed at $\alpha=0^\circ$, $M=0.203$, and $Re_c^*=6.98\times10^6$ using the validated Extra-Fine STAB_THIN_PROD configuration. The $\beta=0^\circ$ point reproduced the accepted G5 baseline to numerical precision. Using standard aircraft moment conventions ($Cl=-CM_x,Cn=-CM_z$), the central $\pm 4^\circ$ regression produced $CY_\beta=-0.42565\text{ rad}^{-1}$, $Cl_\beta=-0.07180\text{ rad}^{-1}$, and $Cn_\beta=+0.13247\text{ rad}^{-1}$, with $R^2>0.99999$ for all three fits. Independent $\pm 2^\circ$, $\pm 4^\circ$ and $\pm 6^\circ$ central differences agreed to within approximately 1%, while mirrored sideslip-response errors remained below 0.8%. All seven cases passed the final-step force/moment convergence gates. The lifting-system model therefore demonstrates stable directional response and stabilising dihedral effect over the tested range. The derivatives remain lifting-system quantities pending the separately documented analytical fuselage correction.

02_Steady_Derivatives analysis log

Table 05-S2.1 — Steady derivative cross-check

Quantity	Manual/reference	VSPAERO Steady	Difference	Disposition
(C_{Y_\beta})	-0.42565	-0.41767	1.88%	PASS
(C_{l_\beta})	-0.07180	-0.08204	14.26%	SENSITIVITY — retain manual
(C_{n_\beta})	+0.13247	+0.13007	1.81%	PASS
(C_{L_\alpha})	+5.62514 ($\pm 2^\circ$)	+5.52430	1.79%	PASS
(C_{m_\alpha})	-1.17352 ($\pm 2^\circ$)	-1.02998	12.23%	SENSITIVITY — retain finite-difference fit
(X_{NP})	4.1424 m ($\pm 2^\circ$)	4.1007 m	-0.0417 m	BOUND FOR §06

Table 05-S2.2 — Steady rate derivatives

Derivative	Value	Physical interpretation	Status
(C_{l_p})	-0.55079	roll damping	Stabilising
(C_{m_q})	-19.25475	pitch damping	Stabilising

Derivative	Value	Physical interpretation	Status
(C_{n_r})	-0.11230	yaw damping	Stabilising
(C_{n_p})	-0.02398	roll–yaw coupling	Retain
(C_{l_r})	+0.12822	yaw–roll coupling	Retain
(C_{Y_p})	-0.12061	roll-sideforce coupling	Retain
(C_{Y_r})	+0.31553	yaw-sideforce coupling	Retain

Table 05-S2.3 — Run validity

Check	Result	Status
(M=0.203)	Confirmed	PASS
(V_{\infty}=66.88) m/s	Confirmed	PASS
(S/B/C_{ref}=30/16.4/1.881)	Confirmed	PASS
(X_{ref}=3.750) m	Confirmed	PASS
Base B0.2 reproduction	essentially exact	PASS
All 7 perturbation cases completed	Yes	PASS
Final-step lift convergence	all <0.02%	PASS
Final-step (C_{D_i,w}) convergence	all <0.20%	PASS
(C_{Y_{\beta}}) manual/Steady agreement	1.9%	PASS
(C_{n_{\beta}}) manual/Steady agreement	1.8%	PASS
(C_{l_{\beta}}) manual/Steady agreement	14.3%	NOTE / manual preferred
Overall	—	PASS WITH SENSITIVITY NOTE

02_Steady_Derivatives — PASS / CLOSED WITH PERTURBATION-SENSITIVITY NOTE. VSPAERO Steady analysis was completed at $\alpha=\beta=0^\circ$, $M=0.203$, $V_\infty=66.88$ m/s and $Re_c=6.98\times10^6$ using the validated Extra-Fine lifting-system model. All seven baseline/perturbation solutions passed the coefficient-convergence gates. The built-in derivatives $C_{Y\beta}=-0.41767$ and $C_{n\beta}=+0.13007$ rad $^{-1}$ agreed with the independently derived seven-point beta-sweep values within 1.9%, validating the lateral-force and directional-stability response. The built-in $C_{l\beta}=-0.08204$ differed from the robust central-difference value -0.07180 rad $^{-1}$ by 14.3%; the discrepancy is attributed to the small 0.01° one-sided perturbation and the correspondingly small rolling-moment signal, so the multi-point manual derivative is retained as the production value. The same perturbation-size sensitivity was observed in $C_{m\alpha}$. Steady rate derivatives $C_{lp}=-0.55079$, $C_{mq}=-19.25475$, and $C_{nr}=-0.11230$ all predict stabilising aerodynamic damping. The .stab neutral point $x_{NP}=4.1007$ m is retained as a local sensitivity value pending the dedicated CG/static-margin assessment.

03_Elevator

Group	Actual surfaces	Role
0 MainWing_SS_CONT_0	two MainWing copies	Aileron or flap — identify later
1 MainWing_SS_CONT_1	two MainWing copies	Flap or aileron — identify later
2 HTail_SS_CONT_0	HTail_Surf0_SS_CONT_0 + HTail_Surf1_SS_CONT_0	Elevator
3 VTail_SS_CONT_0	VTail control surface	Rudder

Final aerodynamic results

(\delta_e)	(C_L)	(C_{L,w})	(C_{D_i})	(C_{D_i,w})	(C_m)
(-5°)	0.365765	0.367051	0.008750	0.006435	+0.292504
(0°)	0.434055	0.434971	0.008780	0.006324	+0.094596
(+5°)	0.502450	0.503002	0.011372	0.008488	-0.102068

03_Elevator analysis log

Table 05-E3.1 — Run results

Test ID	(\delta_e)	(C_L)	(C_{L,w})	(C_{D_i,w})	(C_m)	Status
E3.1	-5° TE-up	0.365765	0.367051	0.006435	+0.292504	PASS
E3.0	0°	0.434055	0.434971	0.006324	+0.094596	PASS
E3.2	+5° TE-down	0.502450	0.503002	0.008488	-0.102068	PASS

Table 05-E3.2 — Elevator effectiveness

Quantity	Negative-side estimate	Positive-side estimate	Central estimate	Linearity
Cm\delta_e	-2.26786	-2.25360	-2.26073 rad⁻¹	0.63% mismatch
CL\delta_e	+0.78255	+0.78375	+0.78315 rad⁻¹	0.15% mismatch

Table 05-E3.3 — Numerical gates

Gate	Result	Status
Neutral B0.2 repeatability	$\Delta CL=0.040\%$, (	\Delta C_m
All three 20-iteration runs completed	Yes	PASS
Final-step CL convergence	max 0.033%	PASS
Final-step CL,w convergence	max 0.044%	PASS

Gate	Result	Status
Final-step CDi,w convergence	max 0.168%	PASS
Final-step Cm convergence	max 0.000260	PASS
Elevator response linearity	R2>0.99999	PASS
Symmetric control response	longitudinal trend correct; lateral bias small	PASS / NOTE
Overall folder	—	PASS / CLOSED

03_Elevator — PASS / CLOSED. Symmetric elevator effectiveness was evaluated at $\delta e = -5^\circ, 0^\circ, +5^\circ$ using the validated Extra-Fine G5 lifting-system configuration at $\alpha = \beta = 0^\circ$. All three 20-iteration solutions passed the force, Trefftz-drag and pitching-moment convergence gates. The neutral-control solution reproduced the accepted G5 baseline within 0.040% in CL and 3.94×10^{-4} in Cm. With positive elevator defined as trailing-edge down, the central finite-difference derivatives were $C_{m\delta e} = -2.2607 \text{ rad}^{-1}$ and $CL_{\delta e} = +0.7832 \text{ rad}^{-1}$. Positive- and negative-side derivative estimates differed by only 0.63% and 0.15%, respectively, confirming a highly linear response over the tested $\pm 5^\circ$ range. Minor lateral coefficient offsets remained at the negative-elevator condition but were of order 10^{-3} and did not affect the longitudinal control conclusion.

04_Aileron

Final three-point results

(δa)	(C_Y)	(C_l)	(C_n)	(C_L)	(C_m)	(C_{D_{i,w}})
-5°	+0.000661	+0.007990	−0.000888	0.434049	+0.094826	0.006597
0°	+0.001267	+0.000882	−0.000454	0.434055	+0.094596	0.006324
$+5^\circ$	+0.001859	−0.006216	−0.000019	0.434041	+0.094706	0.006605

04_Aileron analysis log

Table 05-A4.1 — Run results

Test ID	δa	CY	Cl	Cn	CL	Cm	CDi,w	Status
A4.1	-5°	+0.000661	+0.007990	−0.000888	0.434049	+0.094826	0.006597	PASS
A4.0	0°	+0.001267	+0.000882	−0.000454	0.434055	+0.094596	0.006324	PASS
A4.2	$+5^\circ$	+0.001859	−0.006216	−0.000019	0.434041	+0.094706	0.006605	PASS

Table 05-A4.2 — Aileron effectiveness

Derivative	-5→0 estimate	0→+5 estimate	Central estimate	One-sided mismatch
Clδa	-0.081445	-0.081341	-0.081393 rad ⁻¹	0.13%
Cnδa	+0.004971	+0.004987	+0.004979 rad ⁻¹	0.31%
CYδa	+0.006954	+0.006782	+0.006868 rad ⁻¹	2.51%

Table 05-A4.3 — Validation gates

Gate	Result	Status
Neutral B0.2 continuity	ΔCL=0.040%, (	\Delta C_m
20 iterations completed	All three	PASS
Final-step CL	max 0.020%	PASS
Final-step CDi,w	max 0.249%	PASS
Roll derivative linearity	one-sided mismatch 0.13%, R2=0.9999999	PASS
Yaw derivative linearity	mismatch 0.31%	PASS
Side-force derivative linearity	mismatch 2.51%	PASS
Longitudinal contamination	negligible	PASS
Aileron adverse yaw	Present, modest	DOCUMENT
Overall folder	—	PASS / CLOSED

04_Aileron — PASS / CLOSED. Clean-configuration aileron effectiveness was evaluated at δa=−5°,0°,+5° using the validated Extra-Fine G5 lifting-system model at α=β=0°. The antisymmetric control configuration produced a strong and highly linear rolling response while total lift and pitching moment remained essentially unchanged, independently confirming correct aileron grouping. For positive aileron defined as starboard trailing-edge down and port trailing-edge up, the central derivative was Clδa=−0.08139 rad−1. The associated Cnδa=+0.004979 rad−1 gives a small adverse-yaw tendency, while CYδa=+0.006868 rad−1. Positive- and negative-side roll-derivative estimates differed by only 0.13%, and all three 20-iteration cases passed the force and Trefftz-drag convergence gates. The aileron control derivative is therefore accepted for subsequent control-authority assessment.

05_Rudder

Final three-point results

(\delta_r)	(C_Y)	(C_I)	(C_n)	(C_L)	(C_m)	(C_{D_i,w})
-5°	+0.027319	+0.003236	-0.010117	0.433364	+0.097559	0.006840
0°	+0.001267	+0.000882	-0.000454	0.434048	+0.094831	0.006324
+5°	-0.024697	-0.001538	+0.009179	0.434109	+0.095377	0.007102

Analysis log

Table 05-R5.1 — Rudder run results

Test	δr	CY	CI	Cn	CL	Cm	CDi,w	Status
R5.1	−5°	+0.027319	+0.003236	−0.010117	0.433364	+0.097559	0.006840	PASS
R5.0	0°	+0.001267	+0.000882	−0.000454	0.434048	+0.094831	0.006324	PASS
R5.2	+5°	−0.024697	−0.001538	+0.009179	0.434109	+0.095377	0.007102	PASS

Table 05-R5.2 — Rudder effectiveness

Derivative	−5→0	0→+5	Central	One-sided mismatch
Cnδr	+0.11073	+0.11039	+0.11056 rad ^{−1}	0.31%
CYδr	−0.29852	−0.29753	−0.29803 rad ^{−1}	0.33%
CIδr	−0.02697	−0.02773	−0.02735 rad ^{−1}	2.78%

Table 05-R5.3 — Validation gates

Gate	Result	Status
Neutral baseline continuity	ΔCL=0.038%, (	\Delta C_m
20 iterations completed	all three	PASS
Final-step CL convergence	max 0.016%	PASS
Final-step CL,w convergence	max 0.023%	PASS
Final-step CDi,w convergence	max 0.162%	PASS
Final-step (	\Delta C_Y	)
Final-step (	\Delta C_I	)
Final-step (	\Delta C_n	)
Cn linearity	0.31% one-sided mismatch	PASS

Gate	Result	Status
CY linearity	0.33% mismatch	PASS
Cl coupling linearity	2.78% mismatch	PASS
Overall folder	—	PASS / CLOSED

05_Rudder — PASS / CLOSED. Rudder effectiveness was evaluated at $\delta r = -5^\circ, 0^\circ, +5^\circ$ using the validated Extra-Fine G5 lifting-system configuration at $\alpha = \beta = 0^\circ$, $M = 0.203$, $Re_c = 6.98 \times 10^6$, and $V_\infty = 66.88$ m/s. All three 20-iteration solutions passed the force, Trefftz-drag and lateral-moment convergence gates. With positive rudder defined as trailing edge to starboard/right, the central derivatives were $C_{n\delta r} = +0.11056 \text{ rad}^{-1}$, $C_{Y\delta r} = -0.29803 \text{ rad}^{-1}$, and $C_{l\delta r} = -0.02735 \text{ rad}^{-1}$. Positive- and negative-side estimates differed by only 0.31%, 0.33%, and 2.78%, respectively, confirming a strongly linear local response. The rudder therefore provides a well-conditioned directional-control input with non-negligible roll coupling that must be retained in subsequent coupled OEI/VMC control-authority calculations.

Final five-point aerodynamic results

(α)	(C_L)	($C_{L,w}$)	(C_{D_i})	($C_{D_i,w}$)	(C_m)
-2°	0.237455	0.238195	0.003477	0.002460	+0.133491
-1°	0.335829	0.336586	0.005693	0.004043	+0.114606
0°	0.434050	0.434966	0.008779	0.006324	+0.094832
$+1^\circ$	0.531665	0.532832	0.012769	0.009349	+0.073838
$+2^\circ$	0.629870	0.631211	0.017631	0.013052	+0.051466

Log table

Table 05-SM6.1 — Local static-stability sweep

Test	α	CL	CL,w	Cm	CDi,w	19→20 CL	19→20 Cm	Status
SM6.1a	-2°	0.237455	0.238195	+0.133491	0.002460	0.0152%	0.000109	PASS
SM6.1b	-1°	0.335829	0.336586	+0.114606	0.004043	0.0085%	0.000030	PASS
SM6.1c	0°	0.434050	0.434966	+0.094832	0.006324	0.0094%	0.000039	PASS
SM6.1d	$+1^\circ$	0.531665	0.532832	+0.073838	0.009349	0.0480%	0.000011	PASS
SM6.1e	$+2^\circ$	0.629870	0.631211	+0.051466	0.013052	0.0312%	0.000155	PASS

Table 05-SM6.2 — Local stability derivatives

Quantity	Result	Assessment
CL0	0.433774	Accepted
CL α	+5.61880 rad⁻¹	PASS
R2(CL)	0.9999984	Excellent
Cm0	+0.093647	Accepted
Cm α	-1.17352 rad⁻¹	Statically stable
R2(Cm)	0.998852	Slight curvature; acceptable
dCm/dCL	-0.20885	Stable
xNP,LS	4.14286 m	Accepted
hN,LS	58.10% MAC	Lifting system only

Table 05-SM6.3 — Lifting-system static-margin assessment

CG condition	xCG	hCG	xNP,LS	SMLS	Status
Forward design boundary	3.600 m	29.24% MAC	4.14286 m	28.86% MAC	PASS before body correction
Current reference station*	3.750 m	37.21% MAC	4.14286 m	20.89% MAC	Reference only
Aft design boundary	3.900 m	45.19% MAC	4.14286 m	12.91% MAC	PASS provisionally / body correction critical

*Only a mathematical reference comparison; Xref=3.750 is not being asserted here as an actual loading-case CG.

Table 05-SM6.4 — Validation and closure gates

Gate	Result	Status
Correct five-point alpha sweep	-2° to +2°, 5 points	PASS
M=0.203, Re=6.98×10 ⁶	Confirmed	PASS
V ∞ =66.88 m/s	Confirmed	PASS
Xref=3.750 m	Confirmed	PASS
20 wake iterations	All five cases	PASS
CL final-step convergence	max 0.048%	PASS
CL,w convergence	max 0.048%	PASS
CDi,w convergence	max 0.187%	PASS
Cm absolute convergence	max 1.55×10 ⁻⁴	PASS
CL linearity	R2=0.999998	PASS

Gate	Result	Status
Cm linearity	R2=0.998852	PASS / NOTE
NP fit-range sensitivity	1.6 mm spread	PASS
Agreement with prior clean NP	0.44 mm	PASS
Difference from Steady NP	42.2 mm	PASS, <50 mm gate
SM6.2 Xref=3.90 audit	Not necessary	SKIP
Fuselage correction	Not yet applied	OPEN
Actual loading-cloud containment	Not yet proven here	OPEN
Overall SM6.1	—	PASS
Overall 06_CG_Static_Margin	—	HOLD / OPEN

SM6.1 — PASS. A five-point local angle-of-attack sweep from -2° to $+2^\circ$ was completed using the validated Extra-Fine G5 lifting-system VLM configuration at $M=0.203$, $Re_c=6.98 \times 10^6$, $V_\infty=66.88$ m/s and $X_{ref}=3.750$ m. All five 20-iteration solutions satisfied the force, Trefftz-drag and pitching-moment convergence gates. Linear regression gave $CL_\alpha=5.6188$ rad $^{-1}$ and $Cm_\alpha=-1.1735$ rad $^{-1}$, corresponding to a lifting-system neutral point $x_{NP,LS}=4.1429$ m, or 58.10% MAC. Independent $\pm 1^\circ$, $\pm 2^\circ$ and five-point estimates differed by only 1.6 mm, and the new neutral point agrees with the previous clean-polar estimate within 0.44 mm; consequently the optional $X_{ref}=3.90$ reference-invariance rerun is not required. The uncorrected lifting-system static margins are 28.86% MAC at FS 3.60 and 12.91% MAC at FS 3.90. The aft boundary therefore retains only 2.91% MAC of destabilising margin above the current 10%-MAC conceptual floor, making the separate fuselage/body Cm_α correction the governing next gate.

SM6.2 log table

Gate	Result	Status
(X_{ref})	3.900 m	Confirmed
Alpha points	-2° , 0° , $+2^\circ$	Confirmed
(C_L) repeatability vs SM6.1	Numerically identical	PASS
Moment translation	max error (5.0×10^{-5})	PASS
(C_{L_α})	$+5.62093$ rad $^{-1}$	PASS
(C_{m_α})	-0.72528 rad $^{-1}$	PASS
(x_{NP})	4.14271 m	PASS
Same-method NP difference	0.000467 m	PASS
NP difference as MAC	0.0248%	PASS
20 iterations	all three	PASS

Gate	Result	Status
Force/Trefftz convergence	all gates satisfied	PASS
Overall SM6.2	—	PASS / CLOSED

SM6.2 — PASS / CLOSED. The moment-reference invariance audit repeated the $\alpha = -2^\circ, 0^\circ, +2^\circ$ production lifting-system cases with Xref moved aft from 3.750 m to 3.900 m. Lift and induced-drag coefficients were unchanged, while pitching moments followed the expected reference-translation relationship to within 5.0×10^{-5} . The three-point neutral point was 4.14271 m compared with 4.14318 m using the identical three-point set at Xref = 3.750 m, a difference of only 0.47 mm or 0.025% MAC. The lifting-system neutral-point determination is therefore reference-invariant within numerical precision.

SM6.3A — MainWing-only support run

The corrected wing-only run worked properly.

The three final cases are:

α	CL	CL,w	CDi,w	Cm
-2°	0.315070	0.315273	0.003283	-0.068048
0°	0.485915	0.486460	0.007778	-0.046735
$+2^\circ$	0.656363	0.657540	0.014217	-0.030239

The lateral coefficients are essentially machine zero, which is another useful indication that the isolated symmetric wing model is behaving correctly.

SM6.3A log table

Gate	Result	Status
Configuration	MainWing only	PASS
Alpha points	$-2^\circ, 0^\circ, +2^\circ$	PASS
Controls	No VSPAERO control groups	PASS
$(C_{L_{\alpha,w}})$	4.88866 rad⁻¹	PASS
$(R^2(C_L))$	0.99999955	PASS
Wing AC	≈3.5416 m	Sanity PASS
Wing AC, % MAC	≈26.1% MAC	Sanity PASS
Simple ($d\epsilon/d\alpha$) scale	0.3471	Retain as check
Maximum final-step (C_L) change	0.00448%	PASS
Maximum final-step ($C_{D_{i,w}}$) change	0.0843%	PASS
Zero- (β) lateral symmetry	essentially machine zero	PASS

Gate	Result	Status
Overall SM6.3A	—	PASS / CLOSED

TYCHE R2-T1

SM6.3B Verification and SM6.4 CG-Envelope Closure

Corrected Munk-Multhopp audit, provisional 10%-MAC boundary, and mass-loading evidence gate

Headline quantity	Verified result
SM6.3B equation and central arithmetic	PASS - independently reproduced
Fuselage $C_{m\alpha}$ correction	+0.34028 rad ⁻¹ (+0.005939 deg ⁻¹)
Fuselage-corrected $C_{m\alpha}$	-0.83324 rad ⁻¹
Fuselage-corrected neutral point	FS 4.02894 m = 52.04% MAC
Static margin at FS 3.600 / 3.900	22.80% / 6.86% MAC
Central aft-CG station for 10% MAC	FS 3.84084 m = 42.04% MAC
Defined 20-region numerical screen	FS 3.83778 m aft limit
SM6.4 actual loading-cloud calculation	HOLD - component mass arms are absent
Overall 06_CG_Static_Margin disposition	OPEN - aft-CG/tail-sizing gate reopened

Engineering conclusion. The original 14-region Munk-Multhopp arithmetic is correct and the current FS 3.900 m aft design boundary fails the 10%-MAC gate. The prior coarse/fine sensitivity rows are replaced here because their alternative segment boundaries were not recorded. Reproducible 11-, 14-, and 20-region meshes all retain the same FAIL conclusion. SM6.4 cannot demonstrate an actual loading-envelope pass because the controlled R2-T1 report explicitly leaves the component mass-arm ledger open. Accordingly, FS 3.900 m is not retained as an acceptable aft boundary; the design gate is reopened pending a complete loading ledger and body/nacelle/propeller validation.

1. Verification scope and evidence basis

This record verifies the supplied SM6.3B document at four levels: source-equation fidelity, controlled-input consistency, independent station arithmetic, and conclusion logic. It then advances SM6.4 only as far as the available controlled evidence permits. Numerical precision is not treated as aerodynamic validation: the Munk-Multhopp result remains a conceptual fuselage correction to the reference-invariant thin lifting-system solution.

Audit item	Finding	Disposition
Governing strip integral	Matches Gudmundsson Appendix C1 discrete form	PASS
36.5 constant and units	Equation returns per degree; multiplied by 180/π	PASS
Forward upwash curves	Coefficients and x/c branch selection reproduced	PASS

Audit item	Finding	Disposition
Aft downwash relation	x_i/l_h [1 - dε/dα] reproduced	PASS
Wing-covered interval	Root LE to root TE assigned zero per method map	PASS
Frozen station interpolation	Piecewise-linear width; endpoint-average w_f	PASS
14-region central result	Integral and all downstream results reproduced	PASS
Original sensitivity rows	Alternative boundaries not stated; not reproducible	REPLACED
Complete-aircraft status	Nacelle, propeller and fuselage-lift effects omitted	HOLD

1.1 Controlled quantities

Controlled quantity	Value	Evidence
Wing reference area, S	30.000 m²	G5 / SM6.1
Reference chord, $\bar{c}$	1.881 m	G5 / SM6.1
Wing span, b	16.400 m	G1.3
Aspect ratio	8.96533	Derived
Moment reference	FS 3.750 m	SM6.1
MAC leading edge	FS 3.050 m	G1.3
Wing root LE / TE	FS 2.930079 / 5.290425 m	G1.3
HT aerodynamic centre	FS 9.420166 m	G1.3
TE-to-HT-AC distance, l_h	4.129741 m	Derived
MainWing-only $C_{L\alpha,w}$	4.88866 rad ⁻¹	SM6.3A
Lifting-system $C_{L\alpha}$	5.61880 rad ⁻¹	SM6.1
Lifting-system $C_m\alpha$	-1.17352 rad ⁻¹	SM6.1
Lifting-system neutral point	FS 4.14286 m	SM6.1 / SM6.2

2. Verified Munk-Multhopp methodology

Gudmundsson's Appendix C1 implements the Munk-Multhopp fuselage or nacelle correction by dividing the body into forward, wing-covered, and aft regions. The angle-of-attack-dependent pitching-moment slope is computed from local body width, local flow-angle gradient, and segment length. With dimensions used consistently in metres, the dimensional ratio $w_f^2\Delta x/(S\bar{c})$ is unchanged.

$$C_{m\alpha, \text{FUS,deg}} = [1 / (36.5 \text{ S } \bar{c})] \sum_i [w_{f,i}^2 (\partial\beta/\partial\alpha)_i \Delta x_i] \quad (\text{SM6.3B-1})$$

$$C_{m\alpha, \text{FUS,rad}} = (180/\pi) C_{m\alpha, \text{FUS,deg}} \quad (\text{SM6.3B-2})$$

The 36.5 form is the degree-based implementation: the slope from Equation (SM6.3B-1) is per degree. The previous document handled this conversion correctly.

2.1 Wing downwash input

$$AR = b^2/S = 16.4^2/30.0 = 8.96533 \quad (\text{SM6.3B-3})$$

$$d\varepsilon/d\alpha = 2C_{L\alpha,w}/(\pi AR) = 2(4.88866)/[\pi(8.96533)] = 0.347139 \quad (\text{SM6.3B-4})$$

This is the conceptual elliptical-wing approximation used only in the aft-region flow-angle relation; it is not applied uniformly to the forward body.

2.2 Forward and aft flow-angle gradients

For forward-body stations, x is the positive distance from the wing root leading edge to the segment midpoint. Curve 1 applies for $x/\bar{c} \geq 1$; Curve 2 applies for $x/\bar{c} < 1$.

$$(\partial\beta/\partial\alpha)_1 = 1.4758 - 0.40485(x/\bar{c}) + 0.09107(x/\bar{c})^2 \quad (\text{SM6.3B-5})$$

$$(\partial\beta/\partial\alpha)_2 = 1.74611(x/\bar{c})^{-0.4254} \quad (\text{SM6.3B-6})$$

For aft-body stations, x_i is the distance aft of the wing root trailing edge and l_h is the root-TE to horizontal-tail aerodynamic-centre distance.

$$(\partial\beta/\partial\alpha)_i = (x_i/l_h)[1 - d\varepsilon/d\alpha] \quad (\text{SM6.3B-7})$$

The root-wing-covered interval FS 2.930079 to 5.290425 m is retained as a zero-contribution row, consistent with the source implementation map. The upwash Curve 2 singularity as $x/\bar{c}$ approaches zero is integrable but creates mesh-placement sensitivity; this is explicitly audited in Section 6.

2.3 Neutral point and static margin

$$C_{m\alpha, \text{corr}} = C_{m\alpha, \text{LS}} + C_{m\alpha, \text{FUS}} \quad (\text{SM6.3B-8})$$

$$x_{\text{NP,corr}} = x_{\text{ref}} - \bar{c}(C_{m\alpha, \text{corr}}/C_{L\alpha, \text{LS}}) \quad (\text{SM6.3B-9})$$

$$SM(x_{\text{CG}}) = (x_{\text{NP,corr}} - x_{\text{CG}})/\bar{c} \quad (\text{SM6.3B-10})$$

No separate fuselage lift-curve-slope term is added in this conceptual implementation. Therefore the result is a fuselage-moment-corrected lifting-system neutral point, not a complete-aircraft CFD-equivalent neutral point.

3. Frozen 10.750 m G1.3 fuselage geometry

G1.3 station	FS x (m)	Width (m)	Height (m)
Nose point	0.0000	0.000	0.000
Nose	0.5375	0.800	0.900
Cabin front	2.3650	1.550	1.700

G1.3 station	FS x (m)	Width (m)	Height (m)
Cabin rear	5.3750	1.550	1.700
Upsweep	7.0950	1.250	1.340
Tail boom	9.1375	0.780	0.800
Tail point	10.7500	0.000	0.000

Widths are linearly interpolated between these frozen stations. Each segment uses the mean of its two endpoint widths, matching the source's strip implementation. Height is retained for geometry traceability but does not enter the stated width-based slope equation.

4. Independent reproduction of the 14-region integration

The supplied baseline uses five forward segments, one zero wing-covered interval, and eight aft segments. All values below were recomputed from the controlled inputs rather than copied from the document.

4.1 Forward and wing-covered regions

ID	x_0 - x_1 (m)	Δx	w_f	$x/\bar{c}$ or x/l_h	$\partial\beta/\partial\alpha$	Term (m³)
F1	0.0000-0.5375	0.5375	0.4000	1.4148	1.0853	0.09334
F2	0.5375-1.1467	0.6092	0.9250	1.1100	1.1386	0.59351
F3	1.1467-1.7558	0.6091	1.1750	0.7862	1.9343	1.62659
F4	1.7558-2.3650	0.6092	1.4250	0.4623	2.4243	2.99902
F5	2.3650-2.9301	0.5651	1.5500	0.1502	3.9112	5.30982
W	2.9301-5.2904	2.3603	1.5500	-	0.0000	0.00000

Forward-body integral = 10.622273 m³. The wing-covered interval contributes zero by definition in this implementation.

4.2 Aft-body regions

ID	x_0 - x_1 (m)	Δx	w_f	$x/\bar{c}$ or x/l_h	$\partial\beta/\partial\alpha$	Term (m³)
A1	5.2904-5.3750	0.0846	1.5500	0.0102	0.0067	0.00136
A2	5.3750-6.2350	0.8600	1.4750	0.1246	0.0813	0.15221
A3	6.2350-7.0950	0.8600	1.3250	0.3328	0.2173	0.32809
A4	7.0950-8.1163	1.0213	1.1325	0.5606	0.3660	0.47942
A5	8.1163-9.1375	1.0212	0.8975	0.8079	0.5275	0.43387
A6	9.1375-9.6750	0.5375	0.6500	0.9966	0.6507	0.14776

ID	x_0-x_1 (m)	Δx	w_f	$x/\bar{c}$ or x/l_h	$\partial\beta/\partial\alpha$	Term (m³)
A7	9.6750-10.2125	0.5375	0.3900	1.1268	0.7356	0.06014
A8	10.2125-10.7500	0.5375	0.1300	1.2569	0.8206	0.00745

Aft-body integral = 1.610303 m³. Total integral = 12.232575 m³. The forward body supplies 86.8% and the aft body 13.2% of the positive destabilising integral.

5. Verified central SM6.3B result

$$C_{m\alpha,FUS} = 12.232575/[36.5(30.000)(1.881)] = +0.005939 \text{ deg}^{-1} \quad (\text{SM6.3B-11})$$

$$C_{m\alpha,FUS} = +0.34028 \text{ rad}^{-1} \quad (\text{SM6.3B-12})$$

$$C_{m\alpha,corr} = -1.17352 + 0.34028 = -0.83324 \text{ rad}^{-1} \quad (\text{SM6.3B-13})$$

$$x_{NP,corr} = 3.750 - 1.881(-0.83324/5.61880) = 4.02894 \text{ m} \quad (\text{SM6.3B-14})$$

Item	FS x (m)	% MAC	Assessment
Lifting-system NP	4.14286	58.10%	Reference-invariant thin model
Fuselage-corrected NP	4.02894	52.04%	Central conceptual estimate
Forward design CG	3.60000	29.24%	SM = 22.80%
Current aft design CG	3.90000	45.19%	SM = 6.86% - FAIL
NP required for 10% at FS 3.900	4.08810	55.19%	Required
Central aft CG for exactly 10%	3.84084	42.04%	Provisional central limit

10% MAC gate decision. FS 3.900 m retains only 6.86% MAC, which is 3.14 percentage points below the 10% floor. The corrected neutral point is 0.0592 m forward of the value required to retain FS 3.900. Holding the central corrected neutral point fixed requires moving the aft boundary forward by 0.0592 m to FS 3.84084 m.

6. Corrected discretisation and model-form audit

The previous document listed 11- and 20-region sensitivity values but did not give their segment boundaries. Those rows could not be independently reproduced and are therefore replaced, not carried forward. The following audit defines every alternative mesh: U11 = four uniform forward + one wing-covered + six uniform aft; U14 = five + one + eight; U20 = eight + one + eleven. S14 is the original station-aligned baseline reproduced in Section 4.

Case	Definition	Regions	$C_{m\alpha,FUS}$	x_{NP}	SM @ 3.90	10% aft limit
U11	4 F + 1 W + 6 A, uniform by region	11	0.32977	4.03246	7.04%	3.84436
U14	5 F + 1 W + 8 A, uniform by region	14	0.33711	4.03000	6.91%	3.84190
S14	5 F + 1 W + 8 A, station-aligned	14	0.34028	4.02894	6.86%	3.84084

Case	Definition	Regions	$C_{m\alpha,FUS}$	x_{NP}	SM @ 3.90	10% aft limit
U20	8 F + 1 W + 11 A, uniform by region	20	0.34943	4.02588	6.69%	3.83778
Quadrature check	Fine continuous-width midpoint audit; not selected	-	0.38902	4.01263	5.99%	3.82453

Result robustness. Every defined mesh fails the FS 3.900 m 10%-MAC gate. Within the source's 10-to-20-region conceptual range, the most restrictive defined result is U20: x_{NP} = 4.02588 m and a 10% aft limit of FS 3.83778 m. The fine quadrature check is more restrictive because it resolves the integrable near-leading-edge upwash singularity; it is retained as a numerical warning, not substituted for the source's strip method.

6.1 Model-form boundary

Discretisation agreement does not validate the classical model for Tyche's high-wing, rounded-rectangle fuselage. Published high-wing testing has reported that the uncorrected classical slope can underpredict fuselage pitching-moment slope. A configuration-specific correction factor is not transferred here because Tyche's geometry and test basis differ. Instead, this becomes a validation and uncertainty requirement.

- Nacelle Munk-Multhopp terms are not included because their full station shapes and confirmed longitudinal datum were not in the controlled SM6.3B input set.
- Powered-propeller normal-force derivatives are not included.
- A separate fuselage lift-curve-slope contribution is not included; the denominator remains the lifting-system $C_{L\alpha}$.
- The accepted central value is therefore a conceptual fuselage-moment correction, not a certification-level complete-aircraft neutral point.

6.2 SM6.3B corrected disposition

Gate / quantity	Result	Disposition
Source equations and branch map	Verified	PASS
14-region station arithmetic	$C_{m\alpha,FUS} = +0.34028 \text{ rad}^{-1}$	PASS
Reference neutral-point calculation	FS 4.02894 m	PASS
Current FS 3.900 10% gate	SM = 6.86%	FAIL
Defined discretisation screen	SM = 6.69% to 7.04%	FAIL robust
Complete-aircraft model validation	Nacelle/prop/body-lift effects open	HOLD
Overall SM6.3B	Calculation closed; design gate not closed	CLOSED / FAIL gate

7. SM6.4 - corrected CG-envelope and loading-cloud assessment

SM6.4 must compare every actual R2-T1 loading state against the corrected aft stability boundary. This requires a component mass-arm ledger. The controlled Chapters 7-9 report states that FS 3.60-3.90 m remains a design range until a complete ledger proves every occupant, cargo, and fuel state inside it; its validation matrix separately marks CG and inertia as open. The available R2-T1 study likewise calls for a source-backed 30-line mass ledger but does not provide the longitudinal arms. Therefore an actual CG cloud cannot be reconstructed without inventing geometry.

7.1 SM6.4 governing equations

$$x_{CG,k} = [\sum_j m_{j,k} x_j] / [\sum_j m_{j,k}] \quad (\text{SM6.4-1})$$

$$h_{CG,k} = (x_{CG,k} - 3.050) / 1.881 \quad (\text{SM6.4-2})$$

$$SM_k = (4.02894 - x_{CG,k}) / 1.881 \quad (\text{SM6.4-3})$$

$$x_{CG,max,10\%} = 4.02894 - 0.10(1.881) = 3.84084 \text{ m} \quad (\text{SM6.4-4})$$

$$x_{CG,max,U20} = 4.02588 - 0.10(1.881) = 3.83778 \text{ m} \quad (\text{SM6.4-5})$$

For a later uncertainty-aware closure, the deterministic gate must be extended to include aft CG uncertainty and forward neutral-point uncertainty:

$$SM_{rob,k} = [x_{NP,corr} - \Delta x_{NP,model} - x_{CG,k} - \Delta x_{CG,k}] / \bar{c} \geq 0.10 \quad (\text{SM6.4-6})$$

Neither uncertainty allowance has yet been established, so Equation (SM6.4-6) cannot currently be closed.

7.2 Known mass cases and missing evidence

Case	OEW kg	Payload kg	Fuel kg	Total kg	CG evidence
Operating empty	1327.0	0	0	1327.0	OEW arm absent; CG unknown
505 km MTOM case	1327.0	563.2	109.8	2000.0	Payload/fuel arms absent
Full RFP arithmetic	1327.0	716.0	109.8	2152.8	Mass FAIL; not flyable
Full payload, zero fuel	1327.0	716.0	0	2043.0	Mass FAIL; not flyable
Hot-high landing limit	-	-	-	1900.0	Loading distribution absent

The mass arithmetic above is useful for mass compliance but cannot determine longitudinal CG: at least the OEW CG, each occupant/seat arm, cargo-zone arms, and fuel-tank arm variation are required. The overweight 2152.8 kg and 2043.0 kg states are excluded from aerodynamic or field-performance passes, but their component data can still inform redesign loads.

7.3 Provisional corrected stability corridor

Boundary	FS x (m)	% MAC	Static margin	Disposition
Forward design boundary	3.60000	29.24%	22.80%	Static-stability PASS; rotation/trim remains separate
U20 screening aft boundary	3.83778	41.88%	10.00% on U20	Preferred deterministic screen pending ledger
Central 14-region aft boundary	3.84084	42.04%	10.00%	Central conceptual limit
Previous aft design boundary	3.90000	45.19%	6.86%	FAIL - do not retain as acceptable

The provisional deterministic screening corridor is therefore FS 3.600 to 3.83778 m when the reproducible 20-region numerical screen is used. This is an analysis corridor, not an approved operational envelope. The central 14-region limit is FS 3.84084 m. The interval from FS 3.84084 to 3.90000 m is demonstrably below 10% MAC under the central model.

7.4 Decision logic for the future loading cloud

Future loading result	Stability meaning	Disposition
$x_{CG} < 3.600\text{ m}$	Static margin exceeds forward case	HOLD unless rotation, trim and gear loads pass
$3.600 \leq x_{CG} \leq 3.83778\text{ m}$	$\geq 10\%$ on all defined 10-to-20-region screens	Deterministic PASS; model validation still open
$3.83778 < x_{CG} \leq 3.84084\text{ m}$	Passes central S14 only	HOLD - discretisation sensitive
$3.84084 < x_{CG} \leq 3.900\text{ m}$	$< 10\%$ central static margin	FAIL - load/layout or tail change required
$x_{CG} > 3.900\text{ m}$	Outside original design range	FAIL

8. Evidence required to close SM6.4

The next controlled artifact must be a component mass-arm ledger, not another aerodynamic sweep. It should use the same fuselage-station datum and preserve source, confidence, and configuration identity for every row.

Ledger block	Minimum content	Required fields
Structure	Fuselage, wing, HT, VT, struts, control surfaces	Mass, FS arm, source, uncertainty
Propulsion	Two engines, nacelles, propellers, mounts, cooling	Left/right mass and FS arm
Hybrid/electrical	Motors, inverters, batteries, cabling, protection	Installed mass and service-removal state
Landing gear	Nose and main gear, wheels, brakes, retraction/fairings	Mass and deployed/stowed arm if relevant
Systems/interior	Avionics, flight controls, seats, cabin equipment	Installed mass and FS arm
Fuel	Each tank and unusable fuel	Tank geometry; CG versus fuel quantity
Occupants	Pilot and each of five passenger seats	86 kg design mass and seat reference point
Cargo	200 kg additional payload and baggage zones	Forward/aft loading zones and tie-down limits

8.1 Minimum loading cases

- Operating empty aircraft; pilot-only; minimum crew; and zero-fuel states.
- Each sequential passenger-loading permutation that can drive forward or aft CG, including forward-first and aft-first loading.
- Maximum forward cargo, maximum aft cargo, and combined passenger-plus-cargo states subject to the 2000 kg mass cap.
- Fuel at unusable, reserve, mission, and full-usable quantities, plus the continuous fuel-burn trajectory for each critical payload state.
- Take-off at 2000 kg, hot-high landing at 1900 kg, go-around, and dispatch-limitation cases.
- Loading/unloading transients in which passengers or cargo are removed in an order that can temporarily drive the CG outside the corridor.

8.2 SM6.4 acceptance criteria

Gate	Acceptance condition
Mass closure	Every flyable state ≤ 2000 kg; landing state $\leq$ declared 1900 kg hot-high limit
Forward containment	$x_CG \geq$ FS 3.600 m unless a new rotation/trim release proves a farther-forward state
Aft deterministic screen	Every loading point and burn path $\leq$ FS 3.83778 m
Central comparison	Every loading point also reported against FS 3.84084 m
Uncertainty	Equation SM6.4-6 passes after x_CG and model uncertainties are quantified
Traceability	Mass, arm, source, revision and loading-case ID retained for every component

9. Final dispositions and design handoff

Item	Disposition	Controlled statement
SM6.3B source-method verification	PASS	Equations, branches, unit conversion verified
SM6.3B central arithmetic	PASS	$C_{m\alpha, FUS} = +0.34028 \text{ rad}^{-1}$; $x_NP = 4.02894 \text{ m}$
SM6.3B FS 3.900 gate	FAIL	6.86% MAC < 10%
SM6.3B model completeness	HOLD	Nacelle, propeller, body-lift and validation uncertainty open
SM6.4 corrected screening boundary	DEFINED	Central 3.84084 m; U20 screen 3.83778 m
SM6.4 actual mass/loading CG cloud	HOLD / OPEN	Component longitudinal arms not available
Aft-CG/tail-sizing design gate	REOPENED	FS 3.900 cannot be retained without redesign or validation
Overall 06_CG_Static_Margin	OPEN	Close only after ledger, loading cloud, and uncertainty audit

Closure statement. SM6.3B is closed as a correctly executed central conceptual fuselage correction, but its current aft-CG design gate fails. SM6.4 advances the result into a controlled provisional corridor and establishes the exact evidence needed for a loading-cloud assessment. Because no component longitudinal arms

are available, SM6.4 is not closed as a pass or a fail of actual operational loading; it is HOLD/OPEN, and the aft-CG/tail-sizing gate is formally reopened. Until the ledger is produced, FS 3.900 m must not be described as an acceptable aft CG.

Tyche R2-T1 - 07_VMC_Control_Closure

Analysis Log, Test Evidence and Hybrid OEI Calculation Record

Tyche R2-T1 Design Study

23 August 2026

Document control and executive disposition

This document consolidates the executed evidence, calculations, decisions and remaining release items from the 07_VMC_Control_Closure campaign. It covers the low-speed unpowered baseline (VMC7.1), the corrected +12 degree Fowler configuration (VMC7.2), every retained powered actuator-disk/free-wake diagnostic (VMC7.3 through VMC7.3F), and the deterministic hybrid one-engine-inoperative (OEI) equilibrium calculation (VMC7.4).

Controlled disposition. The low-speed unpowered and corrected-flap branches are accepted for their intended purposes. The coupled powered VSPAERO free-wake branch is closed as numerically non-robust at the heavily blown low-speed condition and is not used as a production source of OEI force or moment coefficients. Counter-rotation remains a favourable architecture sensitivity because it greatly improved equal-power left/right loading symmetry; no further numerical RPM-sign change is supported when one propeller geometry has already been physically reversed.

The VMC7.4 hybrid model gives the following conceptual-design control closure:

$$V_{MC,hybrid}=38.73\,kn\,EAS$$

for the governing VMC-A loading case, the windmilling failed-propeller bound $C_{D,f}=0.45$, adverse live-propeller reaction torque, full provisional finite control travel, and no differential-thrust relief. The severe failed-propeller-drag sensitivity gives:

$$V_{MC,severe}=39.09\,kn\,EAS.$$

Both are below the internal sea-level design gate:

$$V_{MC,lim}=0.95\,V_R=39.96\,kn\,EAS.$$

The result is a **conditional analytical pass / release hold**. At the gate, the windmilling case uses approximately 93.3% of provisional aileron travel and 93.7% of provisional rudder travel. A 10% simultaneous travel-reserve screen gives 40.73 kn and therefore misses the internal gate by 0.77 kn. The unresolved additional adverse OEI powered-lift rolling coefficient must satisfy approximately:

$$|C_{l,pw})\leq 0.00250$$

at about 40 kn EAS unless control authority or the speed schedule is changed.

Controlled status summary

Campaign element	Controlled result	Disposition
VMC7.1 low-speed unpowered	Stable 20-iteration solution at 20.58 m/s	PASS / CLOSED

Campaign element	Controlled result	Disposition
VMC7.2 corrected Fowler +12 degrees	Symmetric configuration recovered; $\Delta C_L = +0.4190$ at $\alpha = 0$ degrees	PASS / CLOSED for configuration verification
Initial powered $C_T = 0.40$ input	$\eta_p = 1.667 > 1$	REJECTED as physically inadmissible
Corrected disk screen	$C_T = 0.1490$, $C_P = 0.07454$, $T = 4.213$ kN at 23.60 m/s	ACCEPTED AS BOUND
VMC7.3C through VMC7.3F	Persistent coefficient and residual oscillation	FAIL / powered branch closed
Counter-rotation sensitivity	Final $ C_l $ reduced about 81.6% relative to co-rotating RLX 0.5	PHYSICS PASS / numerical fail
VMC7.4 full-travel windmill case	38.73 kn EAS	PASS / THIN
VMC7.4 full-travel severe-drag case	39.09 kn EAS	PASS / THIN
VMC7.4 90%-travel windmill case	40.73 kn EAS	FAIL internal robustness target
Additional OEI powered-lift roll allowance	$ C_{l,pw} \leq 0.00250$	OPEN / RELEASE-CRITICAL

1. Scope, conventions and acceptance philosophy

1.1 Aerodynamic coefficient mapping

The VSPAERO CSV history fields are mapped into the conventional aircraft-axis notation used in this record as follows. The signs of rolling and yawing moment are reversed from the raw CMxtot and CMztot fields to preserve the conventional C_l and C_n signs used by the G5 derivative set.

Conventional quantity	VSPAERO history field	Mapping
Lift coefficient, C_L	CLtot	Direct
Wake lift coefficient, $C_{L,w}$	CLwtot	Direct

Conventional quantity	VSPAERO history field	Mapping
Induced drag, C_{D_i}	CDi	Direct
Wake induced drag, $C_{D_{i,w}}$	CDiw	Direct
Side-force coefficient, C_Y	CStot	Direct
Rolling-moment coefficient, C_l	CMxtot	$C_l = -C M x t o t$
Pitching-moment coefficient, C_m	CMytot	Direct
Yawing-moment coefficient, C_n	CMztot	$C_n = -C M z t o t$

The final-five relative span used for lift and drag convergence checks is:

$$S p a n_{5,\%}(X) = 100 \frac{\max(X_{N-4:N}) - \min(X_{N-4:N})}{|\bar{X}_{N-4:N}|}.$$

The campaign acceptance philosophy was:

Quantity	Preferred numerical gate
C_L , $C_{L,w}$, $C_{D_{i,w}}$ final-five relative span	less than 0.5%
C_m final-five absolute span	less than 0.005
C_Y , C_l , C_n final-five absolute span	less than 5×10^{-4} preferred
Residuals	Sustained decay without late oscillation or sign cycling

VMC7.1 and VMC7.2 were configuration-verification tests. VMC7.2 higher-alpha points were therefore judged primarily on the corrected flap implementation, symmetry and smooth aerodynamic trend; small wake-drag span exceedances were not promoted as production drag-polar evidence. VMC7.3 was intended to support powered OEI analysis and was held to the stricter production-coefficient standard.

1.2 Common reference geometry and low-speed states

Parameter	Controlled value
Wing reference area, S	30.0 m2
Span, b	16.4 m
Mean aerodynamic chord, $\bar{c}$	1.881 m
Primary VMC reference station	FS 3.70312 m for aerodynamic diagnostics
Propeller diameter, D_p	2.00 m
Propeller lateral arm, y_p	1.60 m
Sea-level density used in CSV runs	1.225 kg/m3
VMC7.1-VMC7.2 speed	20.58 m/s, Mach 0.0605, $Re = 2.65 \times 10^6$
VMC7.3 diagnostic speed	23.60 m/s, Mach 0.0693, $Re = 3.04 \times 10^6$

2. Complete campaign and file audit

The executed CSV evidence spans 22-23 August 2026. The first VMC7.2 and VMC7.3 files are intentionally retained because they record the discovery of the flap-grouping error. The (1) VMC7.1 file is byte-identical to the file without the suffix and is not a separate aerodynamic test.

Test ID	Controlled source file	Purpose / configuration	Result and use
VMC7.1	VMC7_1_LOWSPD_UNP WR_A0.csv	Alpha 0 degrees, beta 0 degrees, unpowered, low-speed baseline, 20 iterations	PASS; reference baseline
VMC7.1 duplicate	VMC7_1_LOWSPD_UNP WR_A0(1).csv	Same timestamp and byte content as VMC7.1	Archive duplicate; no new test
VMC7.2 initial	VMC7_2_TOFLAP12_ALP HA_0_4_8.csv	Unpowered alpha 0/4/8 degrees; intended Fowler +12 degrees	REJECTED; asymmetric flap grouping produced spurious roll

Test ID	Controlled source file	Purpose / configuration	Result and use
VMC7.2 corrected	VMC7_2_TOFLAP12_ALP HA_0_4_8(1).csv	Corrected symmetric Fowler +12 degrees; alpha 0/4/8 degrees	PASS for flap implementation and low-speed trend
VMC7.3A initial	VMC7_3_SYM_TOFLAP12 _ALPHA_0_4_8.csv	First powered symmetric alpha sweep	REJECTED; flap- grouping contamination and non-physical disk input basis
VMC7.3B corrected grouping	VMC7_3_SYM_TOFLAP12 _ALPHA_0_4_8(1).csv	Corrected Fowler grouping, powered alpha 0/4/8 degrees, 20 iterations	Development evidence only; powered oscillation and old $C_T = 0.40$ basis
VMC7.3C	VMC7_3C_SYM_A0_40IT. csv	Corrected C_T / C_p , co- rotating, relaxation 1.0, 40 iterations	FAIL production convergence
VMC7.3D	VMC7_3D_SYM_A0_RLX0 5.csv	Same as VMC7.3C, relaxation 0.5	Improved some spans; still FAIL
VMC7.3E	VMC7_3E_CR_SYM_A0_ RLX05.csv	Counter-rotating, relaxation 0.5	Physical symmetry improved; numerical FAIL
VMC7.3F	VMC7_3F_CR_SYM_A0_ RLX025.csv	Counter-rotating, relaxation 0.25, 40 iterations	Final numerical test; FAIL and close powered branch
VMC7.4	Deterministic hybrid calculation in this record	G5 derivatives + explicit OEI force/moment model + finite controls	CONDITIONAL ANALYTICAL PASS / RELEASE HOLD

3. VMC7.1 - low-speed unpowered baseline

3.1 Test definition

The baseline isolated the low-speed airframe before adding Fowler deflection or actuator disks.

Setting	Value
Alpha / beta	0 / 0 degrees
Fowler, aileron, elevator, rudder	0 / 0 / 0 / 0 degrees
Actuator disks	Off / zero loading
V_{∞}	20.58 m/s
Mach / Reynolds number	0.0605 / 2.65×10^6
Density	1.225 kg/m ³
Reference station	FS 3.70312 m
Wake iterations	20

3.2 Final coefficients and numerical checks

Quantity	Iteration-20 value	Final-five span	Gate	Result
C_L	0.427911	0.0898%	less than 0.5%	PASS
$C_{L,w}$	0.428777	0.0656%	less than 0.5%	PASS
$C_{D,w}$	0.006182	0.201%	less than 0.5%	PASS
C_m	+0.083816	0.000130	less than 0.005	PASS
C_Y	+0.001239	0.000204	less than 5×10^{-4}	PASS

Quantity	Iteration-20 value	Final-five span	Gate	Result
C_l	+0.000870	0.000061	less than 5×10^{-4}	PASS
C_n	-0.000447	0.000015	less than 5×10^{-4}	PASS

The final residual values were $\log_{10} L_2 = -1.8316$ and $\log_{10} R_{max} = -0.2458$. More importantly for this configuration-verification run, the aerodynamic coefficients were stable and the lateral coefficients remained small. VMC7.1 was therefore closed as the valid low-speed unpowered reference.

4. VMC7.2 - +12 degree Fowler implementation

4.1 Initial flap-grouping failure

The first VMC7.2 sweep showed a large rolling coefficient in a nominally symmetric, unpowered condition. At $\alpha = 0$ degrees the initial file gave $C_l = -0.063141$, compared with $C_l = +0.001371$ after correcting the flap grouping. The correction also reduced the side-force and yawing-moment magnitudes.

Alpha = 0 degrees quantity	Initial file	Corrected file	Change
C_L	0.427900	0.846943	Fowler lift increment recovered
C_Y	+0.011656	+0.001657	85.8% reduction in magnitude
C_l	-0.063141	+0.001371	97.8% reduction in magnitude
C_n	-0.002730	-0.000641	76.5% reduction in magnitude
C_m	+0.078101	+0.174848	Corrected flap load/moment state

The initial file is retained as a diagnostic failure, not as aerodynamic evidence. It proved that the observed large roll was created by control-surface grouping rather than by the clean aircraft.

4.2 Corrected unpowered Fowler sweep

Alpha (degrees)	C_L	$C_{L,w}$	$C_{D,w}$	C_m
0	0.846943	0.851300	0.027738	+0.174848
4	1.228337	1.234087	0.052715	+0.061235
8	1.604105	1.614415	0.087709	-0.072316

Alpha (degrees)	C_Y	C_l	C_n	Lateral-state interpretation
0	+0.001657	+0.001371	-0.000641	Small corrected residual
4	+0.000492	+0.000394	-0.000175	Approaching zero
8	-0.000689	-0.000443	+0.000189	Gentle sign reversal

Alpha (degrees)	C_L span	$C_{L,w}$ span	$C_{D,w}$ span	C_m absolute span	Assessment
0	0.070%	0.069%	0.105%	0.000265	PASS
4	0.639%	0.612%	0.750%	0.001931	Minor wake-span exceedance; trend retained
8	0.123%	0.175%	0.805%	0.000439	Minor wake-drag span exceedance; trend retained

The low-speed Fowler lift increment at alpha = 0 degrees is:

$$\Delta C_{L,F12} = C_{L,F12} - C_{L,clean} = 0.846943 - 0.427911 = 0.419032.$$

The three-point lift-curve slope over 0-8 degrees is approximately:

$$C_{L_\alpha} \approx \frac{1.604105 - 0.846943}{8\pi / 180} = 5.423 \text{ rad}^{-1}.$$

The corrected lateral coefficients remain small and reverse gently about zero across the alpha sweep. VMC7.2 was therefore closed for its intended purpose: establishing a symmetric +12 degree Fowler surrogate and a credible unpowered low-speed loading trend. It was not used to create a high-fidelity separated-flow drag polar.

5. VMC7.3 - powered actuator-disk/free-wake campaign

5.1 Physical screening of actuator-disk inputs

For propeller rotational speed n in revolutions per second, the coefficient definitions used in the campaign screen are:

$$C_T = \frac{T}{\rho n^2 D^4}, C_P = \frac{P}{\rho n^3 D^5}.$$

The advance ratio and propulsive efficiency are:

$$J = \frac{V}{nD}, \eta_p = \frac{JC_T}{C_P} = \frac{TV}{P}.$$

At 2279 RPM, $D=2.00$ m, $V=23.60$ m/s and $\rho=1.225$ kg/m³:

$$n = \frac{2279}{60} = 37.983 \text{ s}^{-1}, J = 0.31066.$$

The original $C_T=0.40$ with $C_P=0.07454$ implies:

$$T = 11.311 \text{ kN}, P = 160.12 \text{ kW}, \eta_p = 1.667.$$

Because $\eta_p > 1$, the input is physically inadmissible and all old $C_T=0.40$ results are rejected for quantitative use.

The corrected screen uses:

$$C_T = 0.1490, C_P = 0.07454,$$

which gives:

$$T = 4.213 \text{ kN}, P = 160.12 \text{ kW}, \eta_p = 0.6210.$$

This point is accepted only as a conservative idealized thrust/power bound, not as a validated production propeller map.

5.2 Early powered alpha sweeps

Source	Alpha = 0 degrees final state	Diagnostic conclusion
Initial VMC7_3...csv	$C_L = 0.88396$, $C_Y = +0.00725$, $C_I = -0.09143$, $C_m = +0.49504$	Dominated by flap-grouping error; reject
Corrected-grouping VMC7_3... (1).csv	$C_L = 1.26838$, $C_Y = -0.00704$, $C_I = +0.01540$, $C_m = +0.64453$	Flap asymmetry removed, but powered iteration remains oscillatory and old C_T basis is invalid

These runs separated the control-grouping error from the powered-flow problem. They are retained as troubleshooting evidence only.

5.3 Controlled 40-iteration diagnostic settings

Setting	VMC7.3C	VMC7.3D	VMC7.3E	VMC7.3F
Alpha / beta	0 / 0 degrees	0 / 0 degrees	0 / 0 degrees	0 / 0 degrees
Fowler / other controls	+12 / 0 degrees	+12 / 0 degrees	+12 / 0 degrees	+12 / 0 degrees
V_∞	23.60 m/s	23.60 m/s	23.60 m/s	23.60 m/s
RPM magnitude	2279	2279	2279	2279
C_T / C_P	0.1490 / 0.07454	Same	Same	Same
Rotation architecture	Co-rotating	Co-rotating	Counter-rotating	Counter-rotating
Wake nodes / iterations	16 / 40	16 / 40	16 / 40	16 / 40
Fixed wake	Off	Off	Off	Off

Setting	VMC7.3C	VMC7.3D	VMC7.3E	VMC7.3F
Relaxation	1.0	0.5	0.5	0.25

5.4 Final coefficients

Test	C_L	$C_{L,w}$	$C_{D_i,w}$	C_m
VMC7.3C	1.27814	1.41405	0.031024	+0.62514
VMC7.3D	1.28147	1.41374	0.032805	+0.61839
VMC7.3E	1.29625	1.45414	0.043366	+0.66299
VMC7.3F	1.34860	1.48845	0.045933	+0.49502

Test	$ C_Y)$	$ C_I)$	$ C_n)$	Lateral interpretation
VMC7.3C	0.010801	0.016040	0.005146	Co-rotating asymmetry
VMC7.3D	0.007525	0.016238	0.003925	Asymmetry persists
VMC7.3E	0.001742	0.002991	0.000582	Counter-rotation improvement
VMC7.3F	0.001706	0.001993	0.000592	Small instant, still oscillatory

Final values alone are not accepted as converged results. They are shown to document the direction of the architecture and relaxation sensitivities.

5.5 Final-five convergence audit

Test	C_L span	$C_{L,w}$ span	$C_{D_i,w}$ span	C_m span
VMC7.3C	3.26%	2.80%	12.48%	0.12857
VMC7.3D	1.77%	0.90%	4.42%	0.07542
VMC7.3E	4.11%	2.11%	3.82%	0.17326

Test	C_L span	$C_{L,w}$ span	$C_{D_i,w}$ span	C_m span
VMC7.3F	3.91%	2.72%	8.24%	0.15738

Test	C_Y span	C_l span	C_n span	Lateral gate result
VMC7.3C	0.022017	0.001835	0.008318	FAIL
VMC7.3D	0.004114	0.002461	0.001707	FAIL
VMC7.3E	0.005732	0.000890	0.002242	FAIL
VMC7.3F	0.005705	0.000882	0.002167	FAIL

Every controlled powered case fails multiple gates. Reducing relaxation from 1.0 to 0.5 improved some lift and wake spans but did not establish convergence. Reducing it again to 0.25 did not produce monotonic residual decay and worsened several measures.

5.6 Residual audit

Test	Final $\log_{10} L_2$	Final $\log_{10} R_{max}$	Observation
VMC7.3C	-0.2859	+1.0153	Oscillatory, no sustained decay
VMC7.3D	-0.3463	+1.3910	Relaxation did not close residuals
VMC7.3E	-0.0873	+1.3677	Counter-rotation fixed symmetry, not iteration stability
VMC7.3F	-0.4550	+1.2570	Final numerical test still oscillatory

The final five iterations of VMC7.3F show the failure directly.

Iteration	C_L	$C_{L,w}$	$C_{D_i,w}$	C_m
36	1.29886	1.45337	0.042352	0.65096
37	1.29735	1.44869	0.042604	0.65240
38	1.30070	1.45718	0.042745	0.64391
39	1.30631	1.45525	0.043748	0.62739
40	1.34860	1.48845	0.045933	0.49502

Iteration	C_Y	C_l	C_n	$\log_{10} L_2$	$\log_{10} R_{max}$
36	+0.002278	+0.002351	-0.000877	-0.206	1.421
37	+0.002343	+0.002875	-0.000964	-0.354	1.142
38	-0.003362	+0.002782	+0.001203	-0.115	1.541
39	-0.002219	+0.002669	+0.000822	-0.315	1.552
40	+0.001706	+0.001993	-0.000592	-0.455	1.257

From iteration 39 to 40:

$$\frac{1.34860 - 1.30631}{1.34860} = +3.14\%, \frac{1.48845 - 1.45525}{1.48845} = +2.23\%,$$

$$\frac{0.045933 - 0.043748}{0.045933} = +4.76\%, |\Delta C_m| = |0.49502 - 0.62739| = 0.13237.$$

The lateral coefficients also change sign rather than settling toward zero. VMC7.3F is therefore a numerical failure even though its iteration-40 lateral coefficients happen to be small.

5.7 Counter-rotation and RPM-sign conclusion

At the same relaxation setting, changing from co-rotation (VMC7.3D) to counter-rotation (VMC7.3E) changed the final lateral magnitudes as follows:

Quantity	Co-rotating RLX 0.5	Counter-rotating RLX 0.5	Magnitude reduction
$ C_Y $	0.007525	0.001742	76.8%
$ C_l $	0.016238	0.002991	81.6%
$ C_n $	0.003925	0.000582	85.2%

These ratios are descriptive architecture sensitivities, not converged production coefficients. Together with the substantially more symmetric wing-load distribution, they support opposite physical rotation directions for the two propellers.

If counter-rotation was created by reversing one propeller geometry with the OpenVSP Rev control while both disk-tab RPM magnitudes remain +2279, the numeric RPM signs should remain unchanged. Negating one RPM numerically on top of the geometric reversal would risk double-reversing the intended rotation definition. The frozen conclusion is:

Opposite physical rotation produced substantially better equal-power propwash symmetry. Numerical RPM sign alone is not used as the architecture definition, and no additional RPM-sign change is supported by the evidence.

Supplier confirmation of an opposite-rotation hardware pair remains open.

5.8 VMC7.3 closure decision

Further attempts with 60-80 iterations, relaxation 0.1, arbitrary wake-node changes or unvalidated C_T reductions were not justified. After bounded diagnostics across rotation sense and relaxation, continued tuning would select a numerical setting rather than validate the physical model. The controlled decision is:

Coupled powered VSPAERO branch closed; not production-valid at the VMC condition.

6. VMC7.4 - hybrid OEI force-and-moment equilibrium

6.1 Method statement

VMC7.4 replaces the rejected coupled powered solution with a deterministic steady OEI balance. It uses only converged G5 airframe derivatives and introduces propulsion asymmetry as explicit external forces and moments. No differential-thrust relief is credited. The left- and right-engine-loss cases are mirror images; the adverse propeller-torque sense is retained.

6.2 Actual loading cases

Case	Mass	CG station	Role
VMC-A	1894.8 kg	FS 3.70312 m	Most-aft actual departure case; primary
VMC-B	2000.0 kg	FS 3.65023 m	Most-aft MTOM case
VMC-C	1900.0 kg	FS 3.63687 m	Most-aft planned landing-mass case

The artificial VMC-U aft sensitivity at FS 3.83778 m remains an envelope audit item but is not used as the headline actual-loading result.

6.3 Converged G5 derivative block

Derivative	G5 value (rad-1)	Applied VMC7.4 value
C_{Y_β}	-0.42565	-0.42565
C_{l_β}	-0.071802	-0.071802
C_{n_β}	+0.13247	CG-transferred
$C_{Y_{\delta_a}}$	+0.006868	0.75 times derivative
$C_{l_{\delta_a}}$	-0.081393	0.75 times derivative
$C_{n_{\delta_a}}$	+0.0049793	CG-transferred, then times 0.75
$C_{Y_{\delta_r}}$	-0.29803	0.85 times derivative
$C_{l_{\delta_r}}$	-0.02735	0.85 times derivative
$C_{n_{\delta_r}}$	+0.11056	CG-transferred, then times 0.85

The near-stall control-effectiveness debits are:

$$\eta_a=0.75, \eta_r=0.85.$$

For example:

$$C_{l_{\delta_a},eff}=0.75(-0.081393)=-0.061045,$$

$$C_{n_{\delta_r},eff}(x_{ref})=0.85(0.11056)=0.093976.$$

These debits are conservative conceptual-design allowances, not claims that the local high-lift control flow has been experimentally validated.

6.4 Transfer of yaw derivatives to each CG

The G5 coefficients were generated about $x_{ref}=3.750$ m. A lateral force creates a yaw-moment change when the moment reference moves longitudinally:

$$C_{n_x}(x_{CG})=C_{n_x}(x_{ref})+\frac{x_{CG}-x_{ref}}{b}C_{Y_x}, \chi \in \{\beta, \delta_a, \delta_r\}.$$

For VMC-A sideslip stability:

$$C_{n_{\beta},A}=0.13247+\frac{3.70312-3.750}{16.4}(-0.42565)=0.133687.$$

After applying the control debits:

Case	$C_{n_{\beta}}$	$C_{n_{\delta},eff}$	$C_{n_{\delta},eff}$
VMC-A	0.133687	0.003720	0.094700
VMC-B	0.135059	0.003703	0.095517
VMC-C	0.135406	0.003699	0.095723

VMC-A consequently has the weakest directional authority of the actual cases and is expected to govern.

6.5 Live-engine thrust bound

The surviving-side power screen is:

$$P_{live}=160.13\text{ k W}.$$

With propeller disk area:

$$A_p=\frac{\pi D_p^2}{4}=3.1416\text{ m}^2,$$

the induced velocity v_i is obtained at each speed from the ideal actuator-disk relation:

$$P_{live}=2\rho A_p v_i(V+v_i)^2,$$

followed by:

$$T_{live}=2\rho A_p v_i(V+v_i).$$

The local hot-day density used for the thrust model is 1.168 kg/m³; dynamic pressure for the control coefficients is EAS-referenced. The ideal disk deliberately gives a high-thrust directional-control bound. At the governing VMC-A windmilling point the live-engine thrust is approximately 4.37 kN.

6.6 Failed-propeller drag envelope

Failed-propeller drag is represented by:

$$D_f = q A_p C_{D,f}, q = \frac{1}{2} \rho_0 V_E^2.$$

Failed-propeller state	$C_{D,f}$	Use
Feathered / low drag	0.08	Favourable bound
Intermediate / stopped	0.25	Sensitivity
Windmilling control case	0.45	Primary VMC7.4 case
Severe windmilling bound	0.60	Adverse sensitivity

At the 39.96 kn EAS gate, $q = 258.84$ Pa and the corresponding failed-propeller drags are 65.1, 203.3, 365.9 and 487.9 N. These coefficients are inherited bounding sensitivities, not R2 propeller-test data.

6.7 Propeller reaction torque

At 2279 RPM:

$$\omega = 2\pi \frac{2279}{60} = 238.66 \text{ rad s}^{-1},$$

$$Q_p = \frac{P_{live}}{\omega} = 671.0 \text{ N m}.$$

The rolling coefficient is:

$$C_{l,Q} = \frac{s_Q Q_p}{q S b},$$

where s_Q is selected so that the reaction torque adds to the adverse OEI roll requirement. At 39.96 kn EAS:

$$|C_{l,Q}| = 0.00527.$$

6.8 Simultaneous equilibrium equations

The unknowns at each candidate speed and permitted bank angle are sideslip, aileron and rudder deflection:

$$x = \begin{bmatrix} \beta & \delta_a & \delta_r \end{bmatrix}^T.$$

Side-force equilibrium is:

$$C_{Y_\beta} \beta + \eta_a C_{Y_{\delta_a}} \delta_a + \eta_r C_{Y_{\delta_r}} \delta_r + \frac{W \sin \phi}{q S} = 0.$$

Roll equilibrium is:

$$C_{l_\beta} \beta + \eta_a C_{l_{\delta_a}} \delta_a + \eta_r C_{l_{\delta_r}} \delta_r + C_{l,Q} + C_{l,pw} = 0.$$

The OEI yaw disturbance is:

$$C_{n,OEI} = \frac{(T_{live} + D_f) y_p}{q S b},$$

and yaw equilibrium is:

$$C_{n_\beta,CG} \beta + C_{n_{\delta_a},CG} \delta_a + C_{n_{\delta_r},CG} \delta_r + C_{n,OEI} = 0.$$

The combined matrix equation is:

$$\begin{bmatrix} C_{Y_\beta} & \eta_a C_{Y_{\delta_a}} & \eta_r C_{Y_{\delta_r}} \\ C_{l_\beta} & \eta_a C_{l_{\delta_a}} & \eta_r C_{l_{\delta_r}} \\ C_{n_\beta,CG} & C_{n_{\delta_a},CG} & C_{n_{\delta_r},CG} \end{bmatrix} \begin{bmatrix} \beta \\ \delta_a \\ \delta_r \end{bmatrix} = - \begin{bmatrix} W \sin \phi / (q S) \\ C_{l,Q} + C_{l,pw} \\ C_{n,OEI} \end{bmatrix}.$$

6.9 Finite limits and feasibility definition

The provisional physical stops are:

$$|\delta_r| \leq 35^\circ, |\delta_a| \leq 22^\circ.$$

The additional robustness screen retains 10% travel reserve:

$$|\delta_r| \leq 31.5^\circ, |\delta_a| \leq 19.8^\circ.$$

The other limits are:

$$|\beta| \leq 5^\circ, |\phi| \leq 5^\circ.$$

For a given speed, define the minimum normalized utilization over permitted bank angle as:

$$U(V) = \min_{|\phi| \leq 5^\circ} \max \left(\frac{|\beta|}{5^\circ}, \frac{|\delta_a|}{\delta_{a,lim}}, \frac{|\delta_r|}{\delta_{r,lim}} \right).$$

The hybrid minimum control speed is the lowest speed with a feasible equilibrium:

$$V_{MC} = \min \{ V : U(V) \leq 1 \}.$$

6.10 Computational sequence

1. Sweep EAS from 35 to 44 kn, then refine around the feasibility crossing.
2. Calculate EAS-referenced dynamic pressure.
3. Convert EAS to local TAS for the ideal live-thrust calculation.
4. Solve the momentum relation for induced velocity and live thrust.
5. Calculate failed-propeller drag, OEI yaw coefficient and reaction-torque rolling coefficient.
6. Sweep permitted bank toward the live engine.
7. Solve the 3 by 3 equilibrium for β , δ_a and δ_r .
8. Reject solutions exceeding sideslip, bank or finite control limits.
9. Mirror the disturbance for the opposite failed side and retain the adverse torque sense.
10. Take the largest VMC across VMC-A/B/C and the bounded failed-propeller states.

7. VMC7.4 results

7.1 Full provisional physical travel

CG case	$C_{D,f}=0.08$	0.25	0.45 windmill	0.60 severe
VMC-A	37.83 kn	38.22 kn	38.73 kn	39.09 kn
VMC-B	37.69 kn	38.08 kn	38.56 kn	38.96 kn
VMC-C	37.67 kn	38.03 kn	38.54 kn	38.95 kn

VMC-A governs every failed-propeller-drag state. Relative to the internal 39.96 kn gate, the primary windmilling result retains 1.23 kn and the severe-drag result retains 0.87 kn.

7.2 Governing VMC-A windmilling equilibrium

At $V_{MC}=38.729$ kn EAS and $C_{D,f}=0.45$:

Variable	Equilibrium magnitude
Live-engine thrust	4.37 kN
Failed-propeller drag	0.344 kN
Reaction-torque coefficient, $ C_{l,Q})$	0.00561
Bank toward live engine, $ \phi)$	4.00 degrees
Sideslip, $ \beta)$	2.95 degrees
Aileron, $ \delta_a)$	22.00 degrees
Rudder, $ \delta_r)$	34.84 degrees

Aileron reaches its finite stop before the 5 degree bank or 5 degree sideslip limits. Rudder is also nearly saturated. The governing result is therefore a simultaneous roll-yaw VMC rather than a rudder-only yaw calculation.

7.3 Ten-percent simultaneous travel reserve

CG case	$C_{D,f}=0.08$	0.25	0.45 windmill	0.60 severe
VMC-A	39.62 kn	40.14 kn	40.73 kn	41.15 kn

CG case	$C_{D,f}=0.08$	0.25	0.45 windmill	0.60 severe
VMC-B	39.52 kn	39.95 kn	40.53 kn	40.99 kn
VMC-C	39.47 kn	39.95 kn	40.52 kn	40.99 kn

The windmilling 90%-travel result exceeds the 39.96 kn internal gate by 0.77 kn. This is not a failure of physical minimum control speed; it shows that the present conservative model does not retain a full 10% simultaneous control-travel reserve at the gate.

7.4 Control use at the 39.96 kn gate

For VMC-A, $C_{D,f}=0.45$ and adverse shaft torque:

$$|\delta_a) \approx 20.52^\circ = 93.3\% \text{ of } 22 \text{ degrees},$$

$$|\delta_r) \approx 32.78^\circ = 93.7\% \text{ of } 35 \text{ degrees}.$$

Remaining geometric travel is approximately 6.7% aileron and 6.3% rudder. Under the severe $C_{D,f}=0.60$ case, both controls rise to roughly 95-96% travel. The appropriate engineering description is **pass at physical stops, but with thin simultaneous control reserve**.

7.5 Unresolved powered-lift rolling moment

The rejected powered solver cannot supply a trustworthy one-sided powered-lift rolling coefficient. VMC7.4 therefore sets an allowable release threshold instead of manufacturing a value from the total powered lift increment.

At 39.96 kn EAS, the maximum tolerable total adverse roll disturbance is approximately:

$$|C_{l,dist,max}) \approx 0.00777.$$

Reaction torque already consumes:

$$|C_{l,Q}) \approx 0.00527.$$

The remaining allowance is:

$$|C_{l,pw})_{max} \approx 0.00777 - 0.00527 = 0.00250.$$

With $q S b = 127349.8 \text{ N m}$ at the gate, this corresponds to:

$$L_{pw,max} = q S b C_{l,pw} \approx 318 \text{ N m}.$$

If expressed only as an equivalent one-side lift decrement at $y_p = 1.60 \text{ m}$:

$$\Delta C_{L,side,max} = C_{l,pw} \frac{b}{y_p} = 0.00250 \frac{16.4}{1.60} = 0.0256.$$

This is a release requirement, not a prediction of the actual asymmetric lift.

Added adverse $ C_{l,pw} $	Calculated VMC-A, $C_{D,f} = 0.45$
0	38.73 kn
0.001	39.20 kn
0.002	39.70 kn
0.0025	approximately 39.96 kn
0.003	40.21 kn
0.005	41.29 kn
0.010	44.49 kn

8. Decision ledger and release items

Evidence item	Result	Disposition / required action
Low-speed clean-airframe numerical stability	Coefficients stable at 20.58 m/s	PASS
Fowler grouping	Initial asymmetry identified and corrected	PASS / preserve rejected file in audit trail
Corrected +12 degree Fowler state	Symmetric low-speed trend established	PASS for VMC configuration basis
Original disk $C_T = 0.40$	$\eta_p = 1.667$	REJECT

Evidence item	Result	Disposition / required action
Corrected C_T/C_P bound	4.213 kN and 160.12 kW at 23.60 m/s	PASS AS BOUND
Co-rotating powered wake	Large lateral asymmetry plus oscillation	REJECT as production evidence
Counter-rotating powered wake	Large symmetry improvement	FAVOURABLE architecture sensitivity
Relaxation 1.0 / 0.5 / 0.25	No robust convergence after 40 iterations	CLOSE powered branch
RPM-sign question	Geometric reversal already establishes physical counter-rotation	Do not negate one RPM again; supplier confirmation open
G5 beta, rudder and aileron derivatives	Converged and linear in the validated range	PASS
Actual VMC-A/B/C CG transformation	Completed	PASS
Ideal live-thrust and reaction-torque bounds	Included	PASS AS BOUNDS
Failed-propeller $C_D=0.08$ to 0.60	Sensitivity envelope only	HOLD - obtain supplier forward/feather/windmill map
R2 rudder/aileron stops	plus or minus 35 / 22 degrees provisionally inherited	HOLD - freeze mechanical geometry
VMC-A windmill, full travel	38.73 kn	PASS / THIN
VMC-A severe drag, full travel	39.09 kn	PASS / THIN
VMC-A windmill, 90% travel	40.73 kn	FAIL internal robustness target
Additional adverse powered-lift roll	Must remain at or below about 0.00250 at the gate	OPEN / RELEASE-CRITICAL

8.1 Required evidence before final VMC release

1. Freeze the R2 mechanical rudder and aileron stops and demonstrate control-system load capability at the required simultaneous deflections.
2. Obtain or defensibly bound the actual propeller forward-thrust, feathered, stopped and windmilling drag maps over the VMC advance-ratio range.
3. Establish the OEI asymmetric powered-lift rolling coefficient using a higher-fidelity method or test and demonstrate $|C_{l,pw}| \leq 0.00250$ at approximately 40 kn EAS, or revise control authority / operating speeds.
4. Confirm that the intended supplier architecture supports opposite physical propeller rotation without relying on an ambiguous numerical RPM sign convention.
5. Repeat the final left- and right-side loss calculations with frozen propeller maps, control stops, masses and CGs.

9. Controlled conclusion

VMC7.1 proved that the low-speed unpowered numerical model is stable. VMC7.2 identified and corrected the Fowler grouping error and established a symmetric +12 degree low-speed state. VMC7.3 then separated two different powered-flow issues: co-rotation created a real equal-power lateral asymmetry, while counter-rotation largely removed that asymmetry; however, neither counter-rotation nor relaxation reductions from 1.0 to 0.5 and 0.25 produced a converged coupled actuator-disk/free-wake solution. The powered branch is therefore closed as a documented model-form/numerical limitation.

VMC7.4 provides the defensible conceptual-design replacement. Using converged G5 derivatives, actual VMC-A/B/C CGs, explicit ideal live-thrust and failed-propeller drag bounds, adverse reaction torque, finite controls, 5 degree bank and sideslip limits, and no differential-thrust credit, VMC-A governs. The windmilling full-travel result is 38.73 kn EAS and the severe-drag result is 39.09 kn EAS, both below the 39.96 kn internal gate. Because the 90%-travel windmilling result is 40.73 kn and only $C_{l,pw} \approx 0.00250$ additional adverse rolling coefficient remains at the gate, the campaign closes as:

CONDITIONAL ANALYTICAL PASS / RELEASE HOLD

The current report may state the hybrid values only with the failed-propeller-drag, control-stop, reaction-torque and unresolved powered-lift-roll qualifications preserved.

07_VMC_Control_Closure

VMC7.4 replaced the non-convergent coupled powered VSPAERO solution with a three-equation steady OEI force-and-moment balance using the converged G5 lateral/control derivatives, CG-specific yaw-moment transfer, an ideal-momentum high-thrust bound, bounded failed-propeller drag and finite conventional-control travel. For the aftmost actual loading case VMC-A, with a 15% rudder-effectiveness debit, 25% aileron-effectiveness debit, provisional $\pm 35^\circ$ rudder/ $\pm 22^\circ$ aileron travel and the adverse 671 N·m live-channel reaction torque, the windmilling $CD,f=0.45$ screen gives VMC=38.73

kn EAS and the severe CD,f=0.60 sensitivity gives 39.09 kn. Both are below the 39.96 kn 0.95VR sea-level gate. The closure is nevertheless conditional: a 90%-travel robustness screen gives 40.73 kn for the windmilling case, while the remaining allowable adverse powered-lift rolling coefficient at the 39.96 kn boundary is only Cl,pw≈0.00250. Exact R2 control stops, supplier forward/feather/windmill propeller maps and the asymmetric powered-lift rolling moment therefore remain release-critical validation items.

Forward-CG Control and Gear Closure

Analysis Log and Conclusions through FG8.3

Folder: 05_Stability/08_Forward_CG_Control_Gear_Closure

Configuration: Tyche R2-T1 / G5 thin lifting-system production model

Status at issue: FG8.1 PASS | FG8.2 PASS | FG8.3 PASS | FG8.4 analytical gear-load closure pending

Purpose. This controlled log records the forward-CG longitudinal-control validation completed before the final analytical landing-gear load closure. It consolidates the executed VSPAERO cases, convergence checks, equations, derived rotation/trim results, model limitations, and release decisions so the findings can be transferred directly into the final design report.

1. Scope and Controlled Analysis Basis

Folder 08 was introduced after the SM6.4 loading analysis identified forward operational CG states that were more forward than the original nominal control boundary. The folder was intentionally limited to take-off rotation, landing trim/flare, and balked-landing flap-transition control. Powered free-wake VSPAERO was excluded because the VMC7.3 campaign had already established it as non-robust at the relevant low-speed, strongly blown condition.

Item	Controlled value / decision
Aerodynamic model	G5 thin lifting-system production model; MainWing + HTail + VTail + LiftStrut_Pair
Thick-body model	Excluded; mixed thin/thick production use rejected in prior validation
Propulsion in VSPAERO	OFF for all FG8 cases
Take-off forward state	m = 1608.8 kg; xCG = FS 3.37374 m
Landing / go-around forward state	m = 1519.8 kg; xCG = FS 3.35170 m
Take-off Fowler surrogate	+12 deg, symmetric trailing-edge-down grouping
Landing Fowler surrogate	+35 deg, used only for longitudinal control screening
Go-around transition	35 deg -> 20 deg -> 12 deg Fowler, quasi-steady

Item	Controlled value / decision
Elevator gains	+1 / -1 mirrored pair retained
Common numerical settings	VLM; 20 wake iterations; 16 wake nodes; fixed wake OFF; relaxation 1.0

1.1 Configuration-control notes

- The first FG8.1 positive-elevator files were initially named with “M” despite containing positive GUI commands. The log therefore records the actual GUI command sign rather than relying on the filename token.
- The aerodynamic response established that negative GUI elevator command produces the required nose-up pitching-moment direction in the current G5 control grouping. Final-report wording should therefore state physical control sense (nose-up / TE-up) and avoid relying only on the OpenVSP sign.
- The early FG8.1 positive-command CSVs recorded $\rho = 1.225 \text{ kg/m}^3$ while the intended hot-high condition was $\rho = 1.079 \text{ kg/m}^3$. Because propulsion was OFF and Reynolds number was imposed explicitly, their nondimensional coefficients were retained. All dimensional rotation calculations were performed with $\rho = 1.079 \text{ kg/m}^3$, and the later negative-command runs used the corrected density.

2. FG8.0 Forward-CG Static Gear Baseline

Before aerodynamic rotation/flare testing, the tricycle-gear static reaction baseline was established using the controlled nose- and main-gear stations $x_N = 0.750 \text{ m}$ and $x_M = 4.290 \text{ m}$.

$$R_N = W (x_M - x_{CG}) / (x_M - x_N) \quad (2.1)$$

$$R_{M, total} = W - R_N \quad (2.2)$$

Condition	Mass (kg)	xCG (m)	Nose fraction	RN (kN)	Main total (kN)
Forward take-off	1608.8	3.37374	25.88%	4.084	11.693
Forward landing	1519.8	3.35170	26.51%	3.950	10.954

Status. Static reaction baseline established. Dynamic/limit/ultimate landing-gear load closure remains FG8.4 and is intentionally not included as a completed result in this issue.

3. FG8.1 Hot-High Forward-CG Take-off Rotation

FG8.1 evaluated whether the actual forward take-off loading state can unload the nose wheel at the controlled hot-high rotation condition without credit for powered lift, propeller slipstream, or thrust-line pitching moment. This is deliberately conservative.

Parameter	Value
Mass	1608.8 kg
xCG / moment reference	FS 3.37374 m
Vinf	22.16 m/s
rho	1.079 kg/m^3 for dimensionalization
Mach	0.0634
Re(cbar)	2.41E6
Alpha / Beta	0 deg / 0 deg
Fowler	+12 deg
Aileron / rudder	0 deg / 0 deg
Propulsion	OFF

3.1 Rotation moment balance

$$C_{m,MG} = C_{m,CG} + C_L (x_M - x_{CG}) / c_{\bar{t}} \quad (3.1)$$

$$M_{aero,MG} = q S c_{\bar{t}} - C_{m,MG} \quad (3.2)$$

$$R_N = [W (x_M - x_{CG}) - M_{aero,MG}] / (x_M - x_N) \quad (3.3)$$

The rotation condition is satisfied when the calculated nose-wheel reaction reaches or crosses zero:

$$R_N \leq 0 \quad (3.4)$$

3.2 Executed elevator sweep and sign audit

GUI elevator command	CL	Cm about CG	Interpretation / use
0 deg	0.84704	+0.02659	Neutral baseline
+5 deg	0.91488	-0.18134	Opposite (nose-down moment) direction
+10 deg	0.98171	-0.38562	Opposite direction
+15 deg	1.04744	-0.58513	Opposite direction

GUI elevator command	CL	Cm about CG	Interpretation / use
+20 deg	1.11211	-0.78034	Opposite direction
-15 deg	0.64114	+0.66859	Nose-up authority; rotation crossing vicinity
-20 deg	0.58037	+0.86128	Nose-up authority; converged positive margin

3.3 Rotation result

Case	Cm,MG	Maero,MG (kN m)	RN (kN)	Convergence disposition
0 deg	0.4392	6.566	+2.229	Baseline
-15 deg	0.9809	14.664	-0.059	Authority demonstrated; final-step convergence note
-20 deg	1.1440	17.103	-0.748	PASS; longitudinal gates satisfied

The minimum rotation demand is therefore approximately 15 deg in the demonstrated nose-up physical direction. Because the -15 deg case is noisier than the -20 deg case, the log deliberately avoids presenting a spurious high-precision minimum. The -20 deg point provides the clean positive-authority demonstration.

$$|\delta_e, \text{max, required}| \geq |\delta_e, \text{rot}| / 0.90 \quad (3.5)$$

Using an approximately 15 deg rotation demand, a 10% geometric control-travel reserve implies a minimum design requirement of approximately 17 deg in the nose-up elevator direction.

FG8.1 convergence item	-15 deg	-20 deg	Gate / interpretation
CL 19->20	0.702%	0.199%	<0.5%; -15 NOTE, -20 PASS
CLw 19->20	0.697%	0.064%	<0.5%; -15 NOTE, -20 PASS
CDiw 19->20	0.775%	0.102%	<0.5%; -15 NOTE, -20 PASS
\Delta Cm 19->20	0.01432	0.00280	<0.005; -15 NOTE, -20 PASS

FG8.1 conclusion. PASS. Forward-CG hot-high take-off rotation authority is demonstrated without powered-lift, propwash, or thrust-line moment credit. Approximately 15 deg nose-up elevator is required; the converged 20 deg nose-up case provides a clear positive nose-wheel-unload margin. High-deflection lateral coefficients are not used as lateral-directional validation evidence.

4. FG8.2 Forward-CG Landing Trim and Flare Authority

FG8.2 first established that the +35 deg Fowler surrogate remained sufficiently stable for longitudinal control screening, then used neutral and -10 deg elevator alpha sweeps to bracket the actual forward-CG landing trim state.

4.1 +35 deg Fowler sanity check

Quantity	Final value	19->20 change	Disposition
CL	1.59808	0.064%	PASS
CLw	1.63859	0.267%	PASS
CDiw	0.11955	0.513%	MARGINAL; not used as drag evidence
Cm	+0.06752	0.000264 absolute	PASS
CY / Cl / Cn	+0.00317 / +0.00206 / -0.00113	-	Numerical-bias note; not used laterally

Scope boundary. The +35 deg Fowler VLM state is accepted only as a longitudinal control-authority surrogate. It is not validation of the actual double-slotted Fowler/slat CLmax, landing drag polar, or certification-level stall behaviour.

4.2 Mass-specific forward landing reference state

$$C_{L, req} = W / (q S) \tag{4.1}$$

Parameter	Value
Mass	1519.8 kg
xCG	FS 3.35170 m
Vinf	21.72 m/s
rho	1.079 kg/m^3
Mach	0.0621

Parameter	Value
Re(cbar)	2.35E6
q	254.51 Pa
Required CL	1.95198
Fowler	+35 deg
Propulsion	OFF

4.3 Executed trim sweeps

Elevator	Alpha	CL	Cm
0 deg	0 deg	1.59819	+0.06743
0 deg	4 deg	1.96032	-0.14137
0 deg	8 deg	2.30896	-0.36888
-10 deg	0 deg	1.46331	+0.49134
-10 deg	4 deg	1.82470	+0.28110
-10 deg	8 deg	2.17648	+0.05205

4.4 Interpolated approach trim

At the required lift coefficient, the neutral-elevator sweep gives approximately $\alpha = 3.91$ deg and $C_m = -0.13656$, whereas the -10 deg sweep gives approximately $\alpha = 5.45$ deg and $C_m = +0.19822$. The two states therefore bracket the zero-moment trim point.

$$\delta_{e,trim} \approx -4.08 \text{ deg (OpenVSP GUI command)} \quad (4.2)$$

$$\alpha_{trim} \approx 4.54 \text{ deg} \quad (4.3)$$

The tested -10 deg state therefore retains approximately 5.9 deg of additional demonstrated nose-up elevator movement beyond steady trim. At $CL = 1.95198$, its additional positive pitching moment is approximately 2.85 kN m.

FG8.2 result	Value
Required CL	1.95198
Neutral-elevator C_m at required CL	-0.13656

FG8.2 result	Value
-10 deg elevator Cm at required CL	+0.19822
Estimated steady trim elevator	~4.08 deg nose-up
Estimated trim alpha	~4.54 deg
Tested additional elevator beyond trim	~5.92 deg
Additional nose-up moment at -10 deg	~2.85 kN m

FG8.2 conclusion. PASS. The actual forward landing case trims within a small nose-up elevator demand, and the tested -10 deg state demonstrates substantial additional nose-up authority for conceptual flare screening. The result is a quasi-steady longitudinal control closure, not a dynamic flare or high-lift CLmax validation.

5. FG8.3 Balked-Landing / Go-Around Flap-Transition Screen

FG8.3 completed the planned VSPAERO portion of Folder 08. The test held the forward landing/go-around state fixed and changed only Fowler configuration to establish the pitching-moment trend during a quasi-steady 35 deg -> 20 deg -> 12 deg flap transition. Propulsion remained OFF in accordance with the powered-VSPAERO model-form closure.

5.1 Neutral-elevator transition data

Alpha	CL @35 deg	Cm @35 deg	CL @20 deg	Cm @20 deg	CL @12 deg	Cm @12 deg
0 deg	1.59819	+0.06743	1.11818	+0.03078	0.84699	+0.01670
4 deg	1.96032	-0.14137	1.49415	-0.16386	1.22843	-0.16885
8 deg	2.30896	-0.36888	1.86278	-0.37616	1.60411	-0.37271

5.2 Convergence and physical trend

Configuration	Alpha	CL 19->20	Delta Cm 19->20	CDiw 19->20	Disposition
20 deg	0 deg	0.053%	0.00156	0.045%	PASS
20 deg	4 deg	0.063%	0.00091	0.635%	Longitudinal PASS; wake-drag note
20 deg	8 deg	0.064%	0.00058	0.267%	PASS
12 deg	0 deg	0.042%	0.00005	0.029%	PASS

Configuration	Alpha	CL 19->20	Delta Cm 19->20	CDiw 19->20	Disposition
12 deg	4 deg	0.039%	0.00018	0.051%	PASS
12 deg	8 deg	0.021%	0.00009	0.036%	PASS

Across all alpha points the lift decreases monotonically as the flap is retracted, while the pitching-moment progression remains smooth. There is no configuration-induced pitch break or reversal. The single marginal wake-drag point is not used for performance evidence.

5.3 Approximate trim-continuity estimate

For the controlled log, the flap-transition pitch increment is evaluated at the established 35 deg landing trim attitude (alpha approximately 4.54 deg). A constant-required-lift elevator interpolation from FG8.2 gives an effective local trim sensitivity of approximately 0.0335 Cm per degree of nose-up elevator. This is more conservative than the preliminary same-alpha estimate and is used here for consistency.

$$\Delta \delta_e \approx - \Delta C_m / (dC_m / d\delta_e) \quad (5.1)$$

Fowler state	Cm at alpha~4.54 deg	Delta Cm vs 35 deg	Approx. additional nose-up elevator	Approx. trim command
35 deg	-0.17184	0	0	4.08 deg
20 deg	-0.19230	-0.02046	~0.61 deg	~4.69 deg
12 deg	-0.19615	-0.02431	~0.73 deg	~4.81 deg

Interpretation. These trim commands are quasi-steady configuration-transition estimates, not a powered six-degree-of-freedom go-around trajectory. The physical go-around simultaneously adds symmetric power and acceleration, reducing the required lift coefficient as speed increases. FG8.3 only demonstrates that flap retraction itself does not create an adverse/reversing pitch-control demand.

FG8.3 conclusion. PASS. The 35 deg -> 20 deg -> 12 deg transition shows monotonic lift reduction, smooth pitching-moment evolution, no pitch-control reversal, and only a small additional nose-up elevator increment relative to the already demonstrated authority. No additional VSPAERO runs are required for Folder 08.

6. Folder 08 Status Before FG8.4 Analytical Gear-Load Closure

Test block	Question answered	Result	Release status
FG8.0	Static forward-CG gear reactions established?	Yes	BASELINE COMPLETE; dynamic load closure pending
FG8.1	Can the actual forward take-off case rotate at hot-high	Yes	PASS / CLOSED

Test block	Question answered	Result	Release status
	VR?		
FG8.2 sanity	Is +35 deg Fowler usable for longitudinal control screening?	Yes, with drag/lateral limitations	PASS / NOTE
FG8.2 trim/flare	Can the actual forward landing case trim and retain nose-up authority?	Yes	PASS / CLOSED
FG8.3	Does flap retraction introduce a pitch reversal or excessive trim demand?	No	PASS / CLOSED
FG8.4	Do updated forward-CG gear limit/ultimate reactions close?	Not yet executed	NEXT / FINAL ANALYTICAL TASK

6.1 Consolidated engineering conclusion

The forward-CG control problem introduced by the SM6.4 loading cloud is now substantially closed. The actual forward take-off case demonstrates nose-wheel unloading at the hot-high rotation condition using the unpowered +12 deg Fowler surrogate, while the actual forward landing case trims with only about 4.1 deg of nose-up elevator and retains substantial additional nose-up authority for flare. The quasi-steady 35 deg -> 20 deg -> 12 deg flap-transition screen shows smooth longitudinal behaviour and no pitch-control reversal. These results support retention of the current tail/elevator architecture without a late geometry redesign.

6.2 Model boundaries to carry into the final report

- No powered free-wake VSPAERO result is used as production evidence in Folder 08. Symmetric thrust-line and powered-lift effects are intentionally not credited in FG8.1-FG8.3.
- The Fowler geometries are OpenVSP control-surface surrogates. They are used for longitudinal control increments only, not to validate the actual slat/double-slot CLmax or landing drag polar.
- High-deflection lateral coefficients in FG8.1/FG8.2 are not used for lateral-directional evidence; the validated G5 beta/rudder/aileron campaigns remain the controlled source for those derivatives.
- FG8.3 is a quasi-steady configuration-transition screen. Propeller transient, flap-rate dynamics, ground effect, pilot response, and full six-degree-of-freedom go-around trajectory remain outside this competition-level closure.
- Final landing-gear limit and ultimate reaction substantiation remains the only open Folder 08 item and will be completed analytically in FG8.4.

6.3 Recommended report-ready summary

Forward-CG control closure: At the actual forward take-off loading state (1608.8 kg, FS 3.37374 m), hot-high rotation authority was demonstrated without powered-lift or thrust-line moment credit, with approximately 15 deg nose-up elevator required and a converged 20 deg case providing

positive nose-wheel-unload margin. At the actual forward landing/go-around state (1519.8 kg, FS 3.35170 m), the +35 deg Fowler control surrogate trimmed at approximately 4.1 deg nose-up elevator and retained substantial additional flare authority. A quasi-steady 35 deg -> 20 deg -> 12 deg flap-transition screen showed smooth lift and pitching-moment evolution with no pitch-control reversal and only a small additional elevator trim requirement. The current tail/elevator architecture therefore passes the conceptual forward-CG rotation, landing-trim/flare, and flap-transition control gates; landing-gear limit/ultimate reaction closure remains the final analytical item.

Appendix A. Executed CSV Evidence Register

Test	File(s)	Use in this log
FG8.1 baseline / positive commands	FG8_1_HH_ROT_DE0.csv; FG8_1_HH_ROT_DE_M5.csv; ..._M10.csv; ..._M15.csv; ..._M20.csv	Elevator sign audit and opposite-direction response
FG8.1 nose-up authority	FG8_1_HH_ROT_DE_M15(1).csv; FG8_1_HH_ROT_DE_M20(1).csv	Main-wheel rotation closure
FG8.2 sanity	FG8_2_LDG35_SANITY_A0_DE0.csv	+35 deg Fowler numerical health
FG8.2 approach trim	FG8_2_LDG35_DE0_A0_4_8.csv	Neutral-elevator lift/pitch surface
FG8.2 flare authority	FG8_2_LDG35_DE_NEG10_A0_4_8.csv	Nose-up elevator trim bracket and reserve
FG8.3 transition 20 deg	FG8_3_GA20_DE0_A0_4_8.csv	Intermediate flap-transition state
FG8.3 transition 12 deg	FG8_3_GA12_DE0_A0_4_8.csv	Released take-off/go-around flap state

Tyche R2-T1

Forward-CG Control, Gear and Final Validation Closure

Final Combined Analysis Log: FG8.0–FG8.4, VMC Atmosphere Reconciliation and Uncertainty Register

Folder: 05_Stability/08_Forward_CG_Control_Gear_Closure

Configuration: Tyche R2-T1 / G5 thin lifting-system production model

Status at issue: FG8.1 PASS | FG8.2 PASS | FG8.3 PASS | FG8.4 CLOSED | VMC atmosphere consistency repaired | uncertainty register complete

Purpose. This final controlled record supersedes the earlier FG8-through-FG8.3 issue. It consolidates the executed forward-CG control campaign, completes the analytical landing-gear load closure, recomputes VMC7.4 on explicit atmosphere-consistent bases, and assembles the final uncertainty register needed for report integration. No new simulation is introduced after FG8.3; FG8.4, the VMC consistency repair and the uncertainty register are analytical synthesis tasks.

1. Scope and Controlled Analysis Basis

Folder 08 was introduced after the SM6.4 loading analysis identified forward operational CG states that were more forward than the original nominal control boundary. The folder was intentionally limited to take-off rotation, landing trim/flare, and bailed-landing flap-transition control. Powered free-wake VSPAERO was excluded because the VMC7.3 campaign had already established it as non-robust at the relevant low-speed, strongly blown condition.

Item	Controlled value / decision
Aerodynamic model	G5 thin lifting-system production model; MainWing + HTail + VTail + LiftStrut_Pair
Thick-body model	Excluded; mixed thin/thick production use rejected in prior validation
Propulsion in VSPAERO	OFF for all FG8 cases
Take-off forward state	m = 1608.8 kg; xCG = FS 3.37374 m
Landing / go-around forward state	m = 1519.8 kg; xCG = FS 3.35170 m
Take-off Fowler surrogate	+12 deg, symmetric trailing-edge-down grouping
Landing Fowler surrogate	+35 deg, used only for longitudinal control screening
Go-around transition	35 deg -> 20 deg -> 12 deg Fowler, quasi-steady
Elevator gains	+1 / -1 mirrored pair retained
Common numerical settings	VLM; 20 wake iterations; 16 wake nodes; fixed wake OFF; relaxation 1.0

1.1 Configuration-control notes

- The first FG8.1 positive-elevator files were initially named with “M” despite containing positive GUI commands. The log therefore records the actual GUI command sign rather than relying on the filename token.
- The aerodynamic response established that negative GUI elevator command produces the required nose-up pitching-moment direction in the current G5 control grouping. Final-report wording should therefore state physical control sense (nose-up / TE-up) and avoid relying only on the OpenVSP sign.
- The early FG8.1 positive-command CSVs recorded $\rho = 1.225 \text{ kg/m}^3$ while the intended hot-high condition was $\rho = 1.079 \text{ kg/m}^3$. Because propulsion was OFF and Reynolds number was imposed explicitly, their nondimensional coefficients were retained. All dimensional rotation calculations were performed with $\rho = 1.079 \text{ kg/m}^3$, and the later negative-command runs used the corrected density.

2. FG8.0 Forward-CG Static Gear Baseline

Before aerodynamic rotation/flare testing, the tricycle-gear static reaction baseline was established using the controlled nose- and main-gear stations $x_N = 0.750$ m and $x_M = 4.290$ m.

$$R_N = W (x_M - x_{CG}) / (x_M - x_N) \tag{2.1}$$

$$R_{M,total} = W - R_N \tag{2.2}$$

Condition	Mass (kg)	xCG (m)	Nose fraction	RN (kN)	Main total (kN)
Forward take-off	1608.8	3.37374	25.88%	4.084	11.693
Forward landing	1519.8	3.35170	26.51%	3.950	10.954

Status. Static reaction baseline established. The dynamic/limit/ultimate landing-gear load closure is completed analytically in FG8.4 (Section 7 of this final issue).

3. FG8.1 Hot-High Forward-CG Take-off Rotation

FG8.1 evaluated whether the actual forward take-off loading state can unload the nose wheel at the controlled hot-high rotation condition without credit for powered lift, propeller slipstream, or thrust-line pitching moment. This is deliberately conservative.

Parameter	Value
Mass	1608.8 kg
xCG / moment reference	FS 3.37374 m
Vinf	22.16 m/s
rho	1.079 kg/m^3 for dimensionalization
Mach	0.0634
Re(cbar)	2.41E6
Alpha / Beta	0 deg / 0 deg
Fowler	+12 deg
Aileron / rudder	0 deg / 0 deg
Propulsion	OFF

3.1 Rotation moment balance

$$C_{m,MG} = C_{m,CG} + C_{L} (x_M - x_{CG}) / c_{\bar{c}} \quad (3.1)$$

$$M_{aero,MG} = q S c_{\bar{c}} - C_{m,MG} \quad (3.2)$$

$$R_N = [W (x_M - x_{CG}) - M_{aero,MG}] / (x_M - x_N) \quad (3.3)$$

The rotation condition is satisfied when the calculated nose-wheel reaction reaches or crosses zero:

$$R_N \leq 0 \quad (3.4)$$

3.2 Executed elevator sweep and sign audit

GUI elevator command	CL	Cm about CG	Interpretation / use
0 deg	0.84704	+0.02659	Neutral baseline
+5 deg	0.91488	-0.18134	Opposite (nose-down moment) direction
+10 deg	0.98171	-0.38562	Opposite direction
+15 deg	1.04744	-0.58513	Opposite direction
+20 deg	1.11211	-0.78034	Opposite direction
-15 deg	0.64114	+0.66859	Nose-up authority; rotation crossing vicinity
-20 deg	0.58037	+0.86128	Nose-up authority; converged positive margin

3.3 Rotation result

Case	Cm,MG	Maero,MG (kN m)	RN (kN)	Convergence disposition
0 deg	0.4392	6.566	+2.229	Baseline
-15 deg	0.9809	14.664	-0.059	Authority demonstrated; final-step convergence note

Case	Cm,MG	Maero,MG (kN m)	RN (kN)	Convergence disposition
-20 deg	1.1440	17.103	-0.748	PASS; longitudinal gates satisfied

The minimum rotation demand is therefore approximately 15 deg in the demonstrated nose-up physical direction. Because the -15 deg case is noisier than the -20 deg case, the log deliberately avoids presenting a spurious high-precision minimum. The -20 deg point provides the clean positive-authority demonstration.

$$|\delta_{e,max,required}| \geq |\delta_{e,rot}| / 0.90 \quad (3.5)$$

Using an approximately 15 deg rotation demand, a 10% geometric control-travel reserve implies a minimum design requirement of approximately 17 deg in the nose-up elevator direction.

FG8.1 convergence item	-15 deg	-20 deg	Gate / interpretation
CL 19->20	0.702%	0.199%	<0.5%; -15 NOTE, -20 PASS
CLw 19->20	0.697%	0.064%	<0.5%; -15 NOTE, -20 PASS
CDiw 19->20	0.775%	0.102%	<0.5%; -15 NOTE, -20 PASS
\Delta Cm 19->20	0.01432	0.00280	<0.005; -15 NOTE, -20 PASS

FG8.1 conclusion. PASS. Forward-CG hot-high take-off rotation authority is demonstrated without powered-lift, propwash, or thrust-line moment credit. Approximately 15 deg nose-up elevator is required; the converged 20 deg nose-up case provides a clear positive nose-wheel-unload margin. High-deflection lateral coefficients are not used as lateral-directional validation evidence.

4. FG8.2 Forward-CG Landing Trim and Flare Authority

FG8.2 first established that the +35 deg Fowler surrogate remained sufficiently stable for longitudinal control screening, then used neutral and -10 deg elevator alpha sweeps to bracket the actual forward-CG landing trim state.

4.1 +35 deg Fowler sanity check

Quantity	Final value	19->20 change	Disposition
CL	1.59808	0.064%	PASS
CLw	1.63859	0.267%	PASS

Quantity	Final value	19->20 change	Disposition
CDiw	0.11955	0.513%	MARGINAL; not used as drag evidence
Cm	+0.06752	0.000264 absolute	PASS
CY / Cl / Cn	+0.00317 / +0.00206 / -0.00113	-	Numerical-bias note; not used laterally

Scope boundary. The +35 deg Fowler VLM state is accepted only as a longitudinal control-authority surrogate. It is not validation of the actual double-slotted Fowler/slat CLmax, landing drag polar, or certification-level stall behaviour.

4.2 Mass-specific forward landing reference state

$$C_{L,req} = W / (q S) \quad (4.1)$$

Parameter	Value
Mass	1519.8 kg
xCG	FS 3.35170 m
Vinf	21.72 m/s
rho	1.079 kg/m^3
Mach	0.0621
Re(cbar)	2.35E6
q	254.51 Pa
Required CL	1.95198
Fowler	+35 deg
Propulsion	OFF

4.3 Executed trim sweeps

Elevator	Alpha	CL	Cm
0 deg	0 deg	1.59819	+0.06743

Elevator	Alpha	CL	Cm
0 deg	4 deg	1.96032	-0.14137
0 deg	8 deg	2.30896	-0.36888
-10 deg	0 deg	1.46331	+0.49134
-10 deg	4 deg	1.82470	+0.28110
-10 deg	8 deg	2.17648	+0.05205

4.4 Interpolated approach trim

At the required lift coefficient, the neutral-elevator sweep gives approximately $\alpha = 3.91$ deg and $C_m = -0.13656$, whereas the -10 deg sweep gives approximately $\alpha = 5.45$ deg and $C_m = +0.19822$. The two states therefore bracket the zero-moment trim point.

$$\delta_{e,trim} \approx -4.08 \text{ deg (OpenVSP GUI command)} \quad (4.2)$$

$$\alpha_{trim} \approx 4.54 \text{ deg} \quad (4.3)$$

The tested -10 deg state therefore retains approximately 5.9 deg of additional demonstrated nose-up elevator movement beyond steady trim. At $CL = 1.95198$, its additional positive pitching moment is approximately 2.85 kN m.

FG8.2 result	Value
Required CL	1.95198
Neutral-elevator C_m at required CL	-0.13656
-10 deg elevator C_m at required CL	+0.19822
Estimated steady trim elevator	~4.08 deg nose-up
Estimated trim α	~4.54 deg
Tested additional elevator beyond trim	~5.92 deg
Additional nose-up moment at -10 deg	~2.85 kN m

FG8.2 conclusion. PASS. The actual forward landing case trims within a small nose-up elevator demand, and the tested -10 deg state demonstrates substantial additional nose-up authority for conceptual flare screening. The result is a quasi-steady longitudinal control closure, not a dynamic flare or high-lift CL_{max} validation.

5. FG8.3 Balked-Landing / Go-Around Flap-Transition Screen

FG8.3 completed the planned VSPAERO portion of Folder 08. The test held the forward landing/go-around state fixed and changed only Fowler configuration to establish the pitching-moment trend during a quasi-steady 35 deg -> 20 deg -> 12 deg flap transition. Propulsion remained OFF in accordance with the powered-VSPAERO model-form closure.

5.1 Neutral-elevator transition data

Alpha	CL @35 deg	Cm @35 deg	CL @20 deg	Cm @20 deg	CL @12 deg	Cm @12 deg
0 deg	1.59819	+0.06743	1.11818	+0.03078	0.84699	+0.01670
4 deg	1.96032	-0.14137	1.49415	-0.16386	1.22843	-0.16885
8 deg	2.30896	-0.36888	1.86278	-0.37616	1.60411	-0.37271

5.2 Convergence and physical trend

Configuration	Alpha	CL 19->20	Delta Cm 19->20	CDiw 19->20	Disposition
20 deg	0 deg	0.053%	0.00156	0.045%	PASS
20 deg	4 deg	0.063%	0.00091	0.635%	Longitudinal PASS; wake-drag note
20 deg	8 deg	0.064%	0.00058	0.267%	PASS
12 deg	0 deg	0.042%	0.00005	0.029%	PASS
12 deg	4 deg	0.039%	0.00018	0.051%	PASS
12 deg	8 deg	0.021%	0.00009	0.036%	PASS

Across all alpha points the lift decreases monotonically as the flap is retracted, while the pitching-moment progression remains smooth. There is no configuration-induced pitch break or reversal. The single marginal wake-drag point is not used for performance evidence.

5.3 Approximate trim-continuity estimate

For the controlled log, the flap-transition pitch increment is evaluated at the established 35 deg landing trim attitude (alpha approximately 4.54 deg). A constant-required-lift elevator interpolation from FG8.2 gives an effective local trim sensitivity of approximately 0.0335 Cm per degree of nose-up elevator. This is more conservative than the preliminary same-alpha estimate and is used here for consistency.

$$\Delta \delta_e \approx - \Delta C_m / (dC_m / d \delta_e) \quad (5.1)$$

Fowler state	Cm at alpha~4.54 deg	Delta Cm vs 35 deg	Approx. additional nose-up elevator	Approx. trim command
35 deg	-0.17184	0	0	4.08 deg
20 deg	-0.19230	-0.02046	~0.61 deg	~4.69 deg
12 deg	-0.19615	-0.02431	~0.73 deg	~4.81 deg

Interpretation. These trim commands are quasi-steady configuration-transition estimates, not a powered six-degree-of-freedom go-around trajectory. The physical go-around simultaneously adds symmetric power and acceleration, reducing the required lift coefficient as speed increases. FG8.3 only demonstrates that flap retraction itself does not create an adverse/reversing pitch-control demand.

FG8.3 conclusion. PASS. The 35 deg -> 20 deg -> 12 deg transition shows monotonic lift reduction, smooth pitching-moment evolution, no pitch-control reversal, and only a small additional nose-up elevator increment relative to the already demonstrated authority. No additional VSPAERO runs are required for Folder 08.

6. Interim Folder 08 Status before FG8.4 Analytical Gear-Load Closure

Test block	Question answered	Result	Release status
FG8.0	Static forward-CG gear reactions established?	Yes; static baseline carried into FG8.4	CLOSED
FG8.1	Can the actual forward take-off case rotate at hot-high VR?	Yes	PASS / CLOSED
FG8.2 sanity	Is +35 deg Fowler usable for longitudinal control screening?	Yes, with drag/lateral limitations	PASS / NOTE
FG8.2 trim/flare	Can the actual forward landing case trim and retain nose-up authority?	Yes	PASS / CLOSED
FG8.3	Does flap retraction introduce a pitch reversal or excessive trim demand?	No	PASS / CLOSED
FG8.4	Do updated forward-CG gear limit/ultimate reactions close?	Yes; main-gear loads, nose-gear requirement, tip-back and overturn screened	PASS / CLOSED with nose-gear hardware and ground-handling qualification

6.1 Consolidated engineering conclusion

The forward-CG longitudinal-control problem introduced by the SM6.4 loading cloud was closed through FG8.1-FG8.3; FG8.4 below completes the landing-gear analytical closure. The actual forward take-off case demonstrates nose-wheel unloading at the hot-high rotation condition using the unpowered +12 deg Fowler surrogate, while the actual forward landing case trims with only about 4.1 deg of nose-up elevator and retains substantial additional nose-up authority for flare. The quasi-steady 35 deg -> 20 deg -> 12 deg flap-transition screen shows smooth longitudinal behaviour and no pitch-control reversal. These results support retention of the current tail/elevator architecture without a late geometry redesign.

6.2 Model boundaries to carry into the final report

- No powered free-wake VSPAERO result is used as production evidence in Folder 08. Symmetric thrust-line and powered-lift effects are intentionally not credited in FG8.1-FG8.3.
- The Fowler geometries are OpenVSP control-surface surrogates. They are used for longitudinal control increments only, not to validate the actual slat/double-slot CLmax or landing drag polar.
- High-deflection lateral coefficients in FG8.1/FG8.2 are not used for lateral-directional evidence; the validated G5 beta/rudder/aileron campaigns remain the controlled source for those derivatives.
- FG8.3 is a quasi-steady configuration-transition screen. Propeller transient, flap-rate dynamics, ground effect, pilot response, and full six-degree-of-freedom go-around trajectory remain outside this competition-level closure.
- Final landing-gear limit and ultimate reaction substantiation is completed analytically in FG8.4 below.

6.3 Recommended report-ready summary

Forward-CG control closure: At the actual forward take-off loading state (1608.8 kg, FS 3.37374 m), hot-high rotation authority was demonstrated without powered-lift or thrust-line moment credit, with approximately 15 deg nose-up elevator required and a converged 20 deg case providing positive nose-wheel-unload margin. At the actual forward landing/go-around state (1519.8 kg, FS 3.35170 m), the +35 deg Fowler control surrogate trimmed at approximately 4.1 deg nose-up elevator and retained substantial additional flare authority. A quasi-steady 35 deg -> 20 deg -> 12 deg flap-transition screen showed smooth lift and pitching-moment evolution with no pitch-control reversal and only a small additional elevator trim requirement. The current tail/elevator architecture therefore passes the conceptual forward-CG rotation, landing-trim/flare, and flap-transition control gates; FG8.4 below completes the associated landing-gear reaction/load closure without requiring a geometry change.

7. FG8.4 Analytical Landing-Gear Reaction and Load Closure

FG8.4 completes Folder 08 analytically. No additional OpenVSP/VSPAERO run is required. The calculation retains the frozen tricycle-gear stations and applies the actual forward take-off/landing loading states established in SM6.4. The objective is to verify static reactions, vertical design loads, tip-back/overturn geometry and the minimum nose-gear structural requirement without reopening the aircraft layout.

7.1 Controlled gear geometry and loading states

Parameter	Controlled value
Nose-gear station, xN	FS 0.750 m
Main-gear station, xM	FS 4.290 m
Wheelbase	3.540 m
Main-gear track	3.900 m
CG height used for geometry screen	1.45 m
Forward take-off state	1608.8 kg; FS 3.37374 m
Forward landing state	1519.8 kg; FS 3.35170 m
Actual nominal aft state	1894.8 kg; FS 3.70312 m
U20 10%-SM screening boundary	FS 3.83778 m

7.2 Static reaction closure

$$R_N = W \frac{x_M - x_{CG}}{x_M - x_N} \quad (7.1)$$

$$R_{M,total} = W - R_N \quad (7.2)$$

$$R_{M,each} = \frac{R_{M,total}}{2} \quad (7.3)$$

Condition	Mass (kg)	xCG (m)	Nose fraction	RN (kN)	Main total (kN)	Each main (kN)
Forward take-off	1608.8	3.37374	25.88%	4.084	11.693	5.847
Forward landing	1519.8	3.35170	26.51%	3.950	10.954	5.477

Interpretation. The updated loading cloud raises the static nose-load fraction above the old preliminary 10–20% placement guideline, but the absolute static nose reaction remains only about 3.95–4.08 kN. A late main-gear relocation is therefore not justified; forward-CG ground handling, steering effort, shimmy and tyre loading are retained as ground-test items rather than redesign triggers.

7.3 Landing-impact load factor and main-gear demand

$$n_e = 1 + \frac{V_z^2}{2g(\eta_s s_s + \eta_t s_t)} \quad (7.4)$$

Using $V_z = 5.0$ m/s, oleo stroke 0.36 m, tyre deflection 0.13 m, strut efficiency 0.80 and tyre efficiency 0.58 gives $n_e = 4.508$, rounded to the released 4.51 g limit basis. The 1.5 ultimate factor gives $n_u = 6.761 \approx 6.76$ g.

$$P_{M,limit} = \frac{n_e W_{LDG}}{2} \quad (7.5)$$

$$P_{M,ultimate} = 1.5 P_{M,limit} \quad (7.6)$$

Load item	Actual FG8.4 demand	Existing design basis	Assessment
Each main, limit	33.61 kN	44.2 kN	PASS; capacity/demand ≈ 1.315
Each main, ultimate	50.41 kN	66.3 kN	PASS; capacity/demand ≈ 1.315
Worst forward nose, limit screen	18.42 kN	Supplier rating not frozen	Hardware requirement
Worst forward nose, ultimate screen	27.63 kN	Supplier rating not frozen	Hardware requirement

The governing main-gear comparison conservatively assigns the complete 4.51 g vertical reaction to the two main wheels at the actual 1519.8 kg landing state. This is more adverse for the mains than preserving the static nose/main split. The existing 44.2/66.3 kN per-main sizing basis therefore remains adequate.

7.4 Tip-back and lateral-overturn geometry

$$\theta_{TB} = \tan^{-1} \left(\frac{x_M - x_{CG}}{h_{CG}} \right) \quad (7.7)$$

$$d_{\perp} = \frac{y_M (x_{CG} - x_N)}{\sqrt{(x_M - x_N)^2 + y_M^2}} \quad (7.8)$$

$$\theta_{OT} = \tan^{-1} \left(\frac{d_{\perp}}{h_{CG}} \right) \quad (7.9)$$

CG state	FS xCG (m)	Tip-back angle	Overturn angle	Disposition
Forward landing	3.35170	32.91°	40.88°	Forward ground-handling reference
Forward take-off	3.37374	32.29°	41.12°	Forward ground-handling reference

CG state	FS xCG (m)	Tip-back angle	Overturn angle	Disposition
Actual nominal aft	3.70312	22.04°	44.50°	PASS; improved vs old aft boundary
U20 aft screening	3.83778	17.32°	45.78°	PASS screening boundary
Obsolete FS 3.900 comparison	3.90000	15.05°	46.35°	Reference only; not operational limit

FG8.4 conclusion. PASS with ground-handling/hardware qualification. The existing main-gear vertical-load basis remains conservative for the actual forward landing state, and aft-CG tip-back improves once the obsolete FS 3.900 boundary is removed. The gear geometry is retained. A minimum nose-gear vertical requirement of approximately 18.5 kN limit and 27.7 kN ultimate is carried to hardware selection, while the 25.9–26.5% light-forward static nose fraction remains a handling-validation note.

8. VMC7.4 Atmosphere-Consistency Recalculation

The earlier VMC7.4 analytical closure used EAS-referenced aerodynamic dynamic pressure but a live-thrust calculation at an incompletely traced density of 1.168 kg/m³. The equilibrium formulation was sound, but the atmosphere bookkeeping was not sufficiently explicit for final report release. This section recomputes the same model on two fully defined atmosphere bases: sea-level standard ($\rho = 1.225 \text{ kg/m}^3$) as the release screen and 610 m ISA+20°C ($\rho = 1.079 \text{ kg/m}^3$) as an operating-condition sensitivity. The aerodynamic force equations remain EAS-based, while the ideal live-propeller thrust uses the corresponding local TAS and density.

8.1 Atmosphere-consistent speed and thrust definitions

$$q = \frac{1}{2} \rho_0 V_E^2 \quad (8.1)$$

$$V_{TAS} = V_E \sqrt{\frac{\rho_0}{\rho}} \quad (8.2)$$

$$P = 2 \rho A_p v_i (V_{TAS} + v_i)^2 \quad (8.3)$$

$$T_{live} = 2 \rho A_p v_i (V_{TAS} + v_i) \quad (8.4)$$

$$D_f = q A_p C_{D,f} \quad (8.5)$$

$$C_{n,OEI} = \frac{(T_{live} + D_f) y_p}{q S b} \quad (8.6)$$

$$Q = \frac{P}{\omega}, C_{l,Q} = \frac{Q}{q S b} \quad (8.7)$$

$$A u = -d, u = [\beta, \delta_a, \delta_r]^T \quad (8.8)$$

Input	Controlled value / rule
Reference density for EAS	$\rho_0 = 1.225 \text{ kg/m}^3$
Release atmosphere	Sea-level standard, $\rho = 1.225 \text{ kg/m}^3$
Sensitivity atmosphere	610 m ISA+20°C, $\rho = 1.079 \text{ kg/m}^3$
Wing / span	$S = 30.0 \text{ m}^2$; $b = 16.4 \text{ m}$
Propeller diameter / arm	2.00 m / 1.60 m
Live shaft-power bound	160.13 kW at 2279 RPM

Input	Controlled value / rule
Failed-propeller drag states	CD,f = 0.08, 0.25, 0.45, 0.60
Control effectiveness debit	75% aileron; 85% rudder
Physical travel / sideslip / bank	$\pm 22^\circ$ aileron; $\pm 35^\circ$ rudder; $ \beta \leq 5^\circ$; $ \phi \leq 5^\circ$
Internal VMC gate	VMC $\leq$ 0.95 VR = 39.96 kn EAS

8.2 Sea-level-standard release results

Case	CD,f	Failed-prop state	VMC EAS (kn)	Tlive (kN)	Dfail (kN)	Bank (deg)	$ \beta $ (deg)	$ \delta a $ (deg)	$ \delta r $ (deg)
VMC-A	0.08	Feathered	38.06	4.472	0.059	-3.85	2.72	21.97	35.00
VMC-A	0.25	Intermediate	38.47	4.458	0.188	-3.95	2.81	21.97	35.00
VMC-A	0.45	Windmilling	38.97	4.441	0.348	-4.07	2.97	22.00	34.94
VMC-A	0.60	Severe	39.37	4.428	0.474	-4.17	3.07	22.00	34.92
VMC-B	0.08	Feathered	37.92	4.477	0.059	-3.61	2.69	21.98	35.00
VMC-B	0.25	Intermediate	38.34	4.463	0.187	-3.71	2.77	21.95	35.00
VMC-B	0.45	Windmilling	38.84	4.446	0.346	-3.82	2.96	22.00	34.88
VMC-B	0.60	Severe	39.21	4.433	0.470	-3.91	3.02	22.00	34.95
VMC-C	0.08	Feathered	37.89	4.478	0.058	-3.80	2.68	21.98	35.00
VMC-C	0.25	Intermediate	38.30	4.464	0.187	-3.89	2.77	21.97	35.00
VMC-C	0.45	Windmilling	38.80	4.447	0.345	-4.02	2.95	22.00	34.90
VMC-C	0.60	Severe	39.19	4.434	0.469	-4.11	3.03	22.00	34.91

VMC-A remains governing. On the fully consistent sea-level-standard basis, the primary windmilling case is 38.97 kn EAS and the severe-drag case is 39.37 kn EAS. Relative to the 39.96 kn internal gate, the margins are +0.99 kn and +0.59 kn, respectively. These values supersede the earlier 38.73/39.09 kn headline values that used the ambiguous 1.168 kg/m³ thrust-density basis.

8.3 Hot-high atmosphere sensitivity

Case	CD,f	State	VMC EAS (kn)	VMC TAS (kn)	Tlive (kN)	Dfail (kN)	$ \delta a $ (deg)	$ \delta r $ (deg)
VMC-A	0.45	Windmilling	38.28	40.78	4.254	0.336	21.96	35.00
VMC-A	0.60	Severe	38.64	41.17	4.242	0.456	22.00	34.99

Case	CD,f	State	VMC EAS (kn)	VMC TAS (kn)	Tlive (kN)	Dfail (kN)	δa (deg)	δr (deg)
VMC-B	0.45	Windmilling	38.14	40.64	4.258	0.333	21.95	35.00
VMC-B	0.60	Severe	38.52	41.04	4.246	0.453	21.96	35.00
VMC-C	0.45	Windmilling	38.09	40.59	4.260	0.332	21.98	35.00
VMC-C	0.60	Severe	38.47	40.99	4.247	0.452	22.00	34.96

At 610 m ISA+20°C the lower local density reduces the ideal live-engine thrust at a given EAS, so the EAS VMC is lower than the sea-level-standard release result. VMC-A windmilling becomes 38.28 kn EAS (40.78 kn TAS), about 0.70 kn below the sea-level-standard value. Sea-level standard is therefore the more adverse EAS control-speed screen for this ideal-power model.

8.4 Control-reserve and unresolved powered-lift-roll sensitivity

Condition	VMC-A result	Gate comparison	Disposition
Full travel, windmilling, sea-level standard	38.97 kn EAS	0.99 kn below 39.96 kn	PASS / THIN
Full travel, severe drag, sea-level standard	39.37 kn EAS	0.59 kn below 39.96 kn	PASS / THIN
90% simultaneous travel, windmilling, sea-level standard	40.97 kn EAS	1.01 kn above 39.96 kn	FAIL internal robustness target
Full travel, windmilling, 610 m ISA+20	38.28 kn EAS	0.70 kn below SL windmill result	Sensitivity only

At the 39.96 kn EAS gate, the atmosphere-consistent sea-level windmilling solution uses approximately 20.88° aileron and 33.12° rudder. The remaining admissible additional adverse OEI powered-lift rolling coefficient before the finite controls saturate is about 0.00206 for CD,f = 0.45 and only 0.00122 for the severe CD,f = 0.60 case. These replace the earlier ~0.00250 allowance and make the unresolved one-sided powered-lift roll an even clearer release-critical item.

VMC atmosphere-consistency conclusion. The analytical VMC architecture remains conditionally acceptable: physical-stop VMC clears the internal gate, but a full 10% simultaneous control-travel reserve does not. The final report should quote 38.97 kn EAS windmilling and 39.37 kn EAS severe-drag on the sea-level-standard basis, state 40.97 kn for the 90%-travel robustness screen, and preserve the open propeller-map, control-stop and asymmetric powered-lift-roll qualifications. No further powered VSPAERO run is authorised.

9. Final Validation and Uncertainty Register

This register is a synthesis task only. It does not create new aerodynamic or performance evidence. Its purpose is to expose the dominant numerical, model-form, configuration and hardware uncertainties that must be carried explicitly into the competition report instead of being hidden behind single-point values.

Domain	Uncertainty / validation item	Controlled evidence / range	Design effect	Final treatment
Aerodynamics	Parasite-drag model form	PDT geometry 0.018985; operational 0.021501; upper 0.023067; performance basis 0.0283	Performance basis is 22.7% above executed upper build-up	Keep 0.0283 as deliberately conservative performance basis; do not call it validated
Aerodynamics	Wing lift-curve slope cross-check	Analytical 4.837 rad ⁻¹ vs MainWing VSPAERO 4.88866 rad ⁻¹	+1.07% difference	PASS validation; do not compare 4.837 with whole lifting-system 5.6188
Aerodynamics	Induced-drag mesh/wake sensitivity	Prior clean-aero campaign recorded ~5.25% k, ~5.54% e and ~1.26% wake-node sensitivity	Affects induced-drag interpretation more than lift	Use Trefftz-plane CD _{i,w} ; carry mesh/wake uncertainty rather than retune solver
Aerodynamics	Mixed thin/thick body model	Root topology repaired but complete mixed model remained physically invalid	Body-interference coefficients unavailable from production VSPAERO model	REJECT mixed-body coefficients; retain thin lifting-system production model and separate PDT/Munk-Multhopp treatment
Stability / CG	Munk-Multhopp discretisation	U11 NP 4.03246 m; U20 NP 4.02588 m; structured spread 6.58 mm; fine-midpoint warning reaches 4.01263 m	Moves 10%-MAC aft boundary	Use conservative U20 boundary FS 3.83778; retain fine case as warning, not release value
Stability / CG	Component-arm uncertainty	Nominal aft FS 3.70312 / 17.16% SM; independent 95% FS 3.76080 / 14.09%; all-at-bounds FS 3.91324 / 5.99%	Operational loading release depends on real CAD centroids	Nominal/95% pass; deterministic all-at-bounds remains HOLD; prohibit loading beyond FS 3.83778 pending CAD update
VMC	Atmosphere bookkeeping	Windmill VMC-A 38.97 kn SL vs 38.28 kn at 610 m ISA+20	0.70 kn atmosphere sensitivity; SL governs EAS screen	Use explicit SL release atmosphere and local-TAS conversion; delete ambiguous p=1.168 result
VMC	Failed-propeller drag	SL VMC-A 38.06 kn at CD,f=0.08 to 39.37 kn at CD,f=0.60	1.31 kn span	Windmill 0.45 primary; 0.60 severe sensitivity; supplier map still required
VMC	Finite-control reserve	38.97 kn full-travel windmill vs 40.97 kn at 90% travel	≈2.00 kn penalty; 90% case misses gate by 1.01 kn	Physical-stop PASS / thin; robustness FAIL remains visible
VMC	OEI asymmetric powered-lift roll	Allowable extra adverse Cl,pw ≈0.00206 windmill;	Release-critical because full controls are already	Higher-fidelity/test evidence required; do not

Domain	Uncertainty / validation item	Controlled evidence / range	Design effect	Final treatment
		≈ 0.00122 severe at 39.96 kn	heavily used	infer from rejected powered solver
VMC	Propeller hardware / rotation architecture	Counter-rotation strongly improved symmetry; exact supplier map/hardware not frozen	Can change critical-engine and OEI roll/yaw balance	Keep counter-rotation as favourable architecture sensitivity; supplier confirmation open
Forward-CG control	Rotation minimum	$\sim 15^\circ$ nose-up estimate; 20° nose-up case converged and clearly unloads nose	Exact minimum limited by noisier -15° point	Report $\approx 15^\circ$ TE-up requirement and $\geq 17^\circ$ design travel with 10% reserve; no more solver refinement
Forward-CG control	High-lift control surrogate	35° Fowler VLM usable for C_m/CL increments but not actual slat/double-slot stall physics	Landing $CL_{max}/drag$ not validated by FG8	Use only for longitudinal trim/flare authority; retain high-lift validation limitation
Landing gear	Forward static nose fraction	25.9–26.5% vs old 10–20% preliminary placement guide; absolute RN only 3.95–4.08 kN	Ground-handling rather than primary structural risk	Retain gear position; validate steering/shimmy/tyre/braking; nose gear $\geq 18.5/27.7$ kN limit/ultimate
Performance	High-lift / field-distance sensitivity	Report uses $CL_{max}, LDG = 3.30$; hot-high adverse TO margin ≈ 21 m and landing margin ≈ 1 m	Landing gate is highly sensitive to unvalidated high-lift/braking/reverse assumptions	Flag as report-critical validation boundary; do not invent new solver evidence in final sprint

Uncertainty-register conclusion. The highest remaining release sensitivities are not clean-airframe numerical convergence; they are high-lift/landing correlation, propeller forward/failed-state maps, finite control-stop hardware, one-sided powered-lift roll, CAD-quality component arms, and low-mass forward-CG ground handling. The final report should concentrate its caveats and future-validation statements on those items. Additional clean-airframe VSPAERO iterations would not materially reduce the dominant uncertainty.

10. Final Combined Closure Record and Engineering Freeze

10.1 Four-task completion ledger

Task	Executed result	Final disposition
1. FG8.3 go-around transition	35°→20°→12° Fowler trend smooth; no pitch-control reversal; trim increment small	PASS / CLOSED
2. Landing-gear reaction / tip-back / nose-load closure	Main-gear demand 33.61/50.41 kN limit/ultimate vs 44.2/66.3 kN basis; tip-back improved; nose hardware requirement defined	PASS / CLOSED with hardware/handling qualification
3. VMC7.4 atmosphere-consistency recomputation	SL windmill 38.97 kn; severe 39.37 kn; 90% travel 40.97 kn; HH windmill 38.28 kn EAS	CONDITIONAL PASS; atmosphere bookkeeping repaired
4. Uncertainty register	Cross-domain numerical/model-form/hardware uncertainties consolidated and prioritised	COMPLETE; no simulation required

10.2 Controlled report-replacement register

Report item to reconcile	Superseded / unsafe statement	Controlled final statement for report integration
Geometry release	G1.1 / 9.70 m fuselage where still present	G1.3 analysis geometry; fuselage length 10.750 m
Wing lift-curve slope	Treating 5.6188 rad ⁻¹ as the wing-only comparison	Wing-only VSPAERO 4.88866 rad ⁻¹ vs analytical 4.837 rad ⁻¹ ; +1.07% agreement; 5.6188 is complete lifting-system slope
Parasite drag	CD0 = 0.0283 presented as directly validated	PDT geometry 0.018985; operational 0.021501; upper 0.023067; retain 0.0283 only as conservative performance basis
Neutral point / aft CG	Old NP near FS 4.25 and CG range FS 3.60– 3.90	Corrected NP FS 4.02894; nominal cloud FS 3.34618–3.70312; U20 10%-MAC aft screen FS 3.83778
VMC headline	38.73/39.09 kn based on ambiguous $p=1.168$ thrust state	Sea-level-standard atmosphere-consistent VMC-A: 38.97 kn windmill; 39.37 kn severe; 40.97 kn at 90% travel
VMC powered-lift-roll allowance	≈0.00250 at 39.96 kn	≈0.00206 windmill; ≈0.00122 severe on corrected SL basis
Forward rotation	Forward-CG elevator/rotation evidence open	FG8.1 PASS: ≈15° TE-up required; converged 20° case gives positive unload;

Report item to reconcile	Superseded / unsafe statement	Controlled final statement for report integration
		≥17° TE-up design travel for 10% reserve
Landing trim / flare	Forward-CG flare evidence open	FG8.2 PASS: ≈4.08° TE-up steady trim; >5.9° additional tested TE-up movement at -10° case
Balked landing	No executed flap-transition evidence	FG8.3 PASS: quasi-steady 35°→20°→12° transition is smooth with no pitch-control reversal
Gear nose-load envelope	11–19.5% static nose-load statement	Actual light-forward cases 25.9–26.5%, RN = 3.95–4.08 kN; retain geometry, carry handling validation
Gear vertical loads	Main-gear basis quoted without actual forward-case comparison	Actual landing demand 33.61/50.41 kN per main vs existing 44.2/66.3 kN design basis; PASS
Validation promises	Unexecuted AVL/X-Plane/external analyses presented as if required for this issue	Replace with executed evidence register and explicit future/certification validation boundaries

10.3 Final engineering freeze decision

The four scheduled residual analysis tasks are complete. No further OpenVSP/VSPAERO, fairing, wake, relaxation, mesh, powered-flow or optional simulator work is authorised before competition submission unless report integration exposes a direct arithmetic/configuration error. Remaining technical items are supplier/hardware or higher-fidelity validation tasks, not reasons to reopen the conceptual geometry during the final report sprint.

Final combined disposition. Folder 08 is CLOSED for competition-level forward-CG control and landing-gear analysis. VMC7.4 is retained as a conditional analytical closure on the corrected sea-level-standard atmosphere basis. The aircraft retains adequate demonstrated forward-CG longitudinal control and main-gear load capability; the principal open release risks are the thin VMC control reserve, failed-propeller/propeller-map fidelity, asymmetric powered-lift roll, nose-gear hardware/ground-handling substantiation, high-lift/landing correlation, and CAD-quality mass-arm confirmation. The engineering campaign should now transition fully to report reconciliation, uncertainty disclosure and submission QA.

Appendix B. Analytical Evidence and Provenance Added in this Final Issue

Evidence item	Source / calculation basis	Use
FG8.4 gear closure	SM6.4 forward/aft CG states; xN=0.750 m; xM=4.290 m; 4.51 g / 6.76 g released gear basis	Static reactions, main/nose design loads, tip-back and overturn
VMC atmosphere repair	Converged G5 derivatives; VMC-A/B/C; P=160.13 kW; y=1.60 m; CD,f bounds; finite controls; explicit p=1.225 and 1.079 cases	Supersedes ambiguous p=1.168 VMC headline values
Parasite-drag uncertainty	G2_COMPLETE_AIRFRAME_ParasiteD	Geometry/operational/upper PDT band

Evidence item	Source / calculation basis	Use
	rag.csv; G2_C1_Operational_ParasiteDrag.csv; G2_C2_UpperBound_ParasiteDrag.csv	and conservative report treatment
CG / neutral-point uncertainty	Tyche_R2T1_SM6_3B_SM6_4_Integrated_CG_Closure_Pre_VMC record	U11/U14/S14/U20 NP sensitivity; loading-cloud and arm uncertainty
Powered-flow model boundary	VMC7_3C/D/E/F CSVs and 07_VMC_Control_Closure analysis log	Documents non-convergent powered free-wake branch and counter-rotation sensitivity
Folder 08 VSPAERO evidence	FG8_1, FG8_2 and FG8_3 CSV files listed in Appendix A	Forward rotation, landing trim/flare and go-around transition

Appendix A. Executed CSV Evidence Register

Test	File(s)	Use in this log
FG8.1 baseline / positive commands	FG8_1_HH_ROT_DE0.csv; FG8_1_HH_ROT_DE_M5.csv; ..._M10. csv; ..._M15.csv; ..._M20.csv	Elevator sign audit and opposite- direction response
FG8.1 nose-up authority	FG8_1_HH_ROT_DE_M15(1).csv; FG8_1_HH_ROT_DE_M20(1).csv	Main-wheel rotation closure
FG8.2 sanity	FG8_2_LDG35_SANITY_A0_DE0.csv	+35 deg Fowler numerical health
FG8.2 approach trim	FG8_2_LDG35_DE0_A0_4_8.csv	Neutral-elevator lift/pitch surface
FG8.2 flare authority	FG8_2_LDG35_DE_NEG10_A0_4_8.csv	Nose-up elevator trim bracket and reserve
FG8.3 transition 20 deg	FG8_3_GA20_DE0_A0_4_8.csv	Intermediate flap-transition state
FG8.3 transition 12 deg	FG8_3_GA12_DE0_A0_4_8.csv	Released take-off/go-around flap state